Guide to Output Tables

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Introduction

Food security as a *multidimensional* phenomenon covers four dimensions: food availability, food access, food utilization, and vulnerability to food insecurity. The ADePT-Food Security Module produces a suite of food security indicators that encompasses some of these dimensions. To better understand food consumption patterns and better target groups of a population in a situation of food insecurity, indicators are disaggregated by the following:

- Socioeconomic, demographic, and geographical characteristics of the household such as region, household size, gender, level of education, and occupation of the head of the household (28 Excel tables)
- Groups of food commodities[1] (22 Excel tables)
- Food commodity (15 Excel tables)

These indicators include statistics on dietary energy, value of food expenditures, and cost of the diet, as well as statistics related to the composition of the food available in terms of macro- and micronutrients, amino acids, and vitamins. The purpose of this chapter is to present each output table produced by ADePT-FSM and to provide a brief interpretation of the indicators that are displayed in the tables. The same indicator may appear in different tables depending at which level of disaggregation (by population group, food commodity group, or food commodity) it is shown. To avoid repeating the interpretation of the indicator after each table, all indicators are described in the glossary of indicators that follows the section presenting the output tables. Indicators are presented in alphabetical order to assist the reader when using the glossary.

Output Tables

Food Consumption (Dietary Energy, Macronutrients, and Monetary Values)

Disaggregated by Population Group: Tables 1.1 to 1.14

Table 1.1: Prevalence of Undernourishment Using Mainly Survey Data This table shows estimates of the prevalence of undernourishment[2] (PoU), using the methodology of the Food and Agriculture Organization (FAO), along with all the parameters used for its computation (i.e., dietary energy consumption, the minimum dietary energy requirement, coefficient of variation, and skewness). The PoU is computed at national, regional, and urban/rural levels under the assumption that the survey sample is representative for such geographic domains. If this is the case, the total number of people undernourished calculated as the sum from each region or from urban/rural is expected to be close to the one at the national level.

The minimum and average dietary energy requirements at the national[3] level are those used to estimate the Millennium Development Goal (MDG)

Table 1.1: Prevalence of Undernourishment Using Mainly Survey Data

	Population ('000s)	*Average dietary energy consumption (kcal/person/day)*	*Coefficient of variation of dietary energy consumption (%)*	*Skewness of dietary energy consumption*	*Minimum dietary energy requirement (kcal/person/day)*	*Prevalence of undernourishment (%)*	*Average dietary energy requirement (kcal/person/day)*	*Depth of food deficit (kcal/person/day)*
Total	31,906.9	2,199.5	29.07	0.90	1,694.0	23.8	2,108.0	146.1
Area								
Capital city	1,845.3	2,063.6	39.13	1.23	1,784.6	N/A	2,258.2	N/A
Other urban areas	4,406.6	2,179.3	32.18	1.00	1,728.5	N/A	2,168.3	N/A
Rural areas	25,655.0	2,212.7	29.64	0.92	1,681.6	23.1	2,086.8	138.8
Region								
Region1	1,778.4	2,401.1	23.94	0.73	1,665.9	6.9	2,061.6	37.4
Region2	2,031.7	1,947.9	28.59	0.88	1,697.4	38.8	2,108.8	261.3
Region3	1,178.8	2,053.4	31.86	0.99	1,712.5	37.1	2,136.2	256.5
Region4	1,647.1	2,162.7	29.20	0.90	1,723.7	28.2	2,148.5	182.3
Region5	1,652.8	2,269.9	30.21	0.93	1,718.8	23.7	2,151.1	149.5
Region6	761.4	1,997.6	33.75	1.05	1,712.3	N/A	2,143.4	N/A

1.9 indicator and depth of food deficit, respectively, published with *The State of Food Insecurity in the world* (SOFI). The statistics of dietary energy consumption, coefficient of variation, and skewness are derived from the survey data.

When a N/A value appears, this means the value of skewness is higher than 1. A skewness higher than 1 indicates that there is a large number of food quantity outliers (or an excessive variability in the consumption/acquisition distribution), and so a more careful food consumption data analysis has to be done.

Table 1.2: Prevalence of Undernourishment Using Mainly External Sources
This table shows estimates of the prevalence of undernourishment[4], using the FAO methodology, along with all the parameters used for its computation. The PoU is computed at the national, regional, and urban/rural levels under the assumption that the survey sample is representative for such geographic domains. If this is the case, the total number of people undernourished calculated as the sum from each region or from urban/rural areas is expected to be close to the one at the national level.

When a N/A value appears, this means the value of skewness is higher than 1. A skewness higher than 1 indicates that there is a large number

Table 1.2: Prevalence of Undernourishment Using Mainly External Sources

	Population ('000s)	*Dietary energy supply adjusted for losses (kcal/person/day)*	*Coefficient of variation of dietary energy consumption (%)*	*Skewness of dietary energy consumption*	*Minimum dietary energy requirement (kcal/person/day)*	*Prevalence of undernourishment (%)*	*Average dietary energy requirement (kcal/person/day)*	*Depth of food deficit (kcal/person/day)*
MDG 1.9 indicator (SOFI)		1,970.5	32.17	0.99	1,694.0	40.7	2,108.0	285.7
Total	31,906.9	1,970.5	29.07	0.90	1,694.0	37.6	2,108.0	252.6
Area								
Capital city	1,845.3	1,848.7	39.13	1.23	1,784.6	N/A	2,258.2	N/A
Other urban areas	4,406.6	1,952.4	32.18	1.00	1,728.5	N/A	2,168.3	N/A
Rural areas	25,655.0	1,982.4	29.64	0.92	1,681.6	36.6	2,086.8	240.9
Region								
Region1	1,778.4	2,151.1	23.94	0.73	1,665.9	16.8	2,061.6	95.2
Region2	2,031.7	1,745.1	28.59	0.88	1,697.4	53.3	2,108.8	391.7
Region3	1,178.8	1,839.6	31.86	0.99	1,712.5	49.5	2,136.2	376.1
Region4	1,647.1	1,937.5	29.20	0.90	1,723.7	42.1	2,148.5	298.3
Region5	1,652.8	2,033.5	30.21	0.93	1,718.8	36.8	2,151.1	255.5
Region6	761.4	1,789.6	33.75	1.05	1,712.3	N/A	2,143.4	N/A

of food quantity outliers (or an excessive variability in the consumption/acquisition distribution), and so a more careful food consumption data analysis has to be done.

The first row of the table shows the MDG 1.9 indicator as published in the latest edition of SOFI corresponding to the same year of the survey and the parameters used (published in the Statistics Division of the FAO website[5]). These are the parameters:

- Average dietary energy supply
- Losses that occurred at the retail level
- Coefficient of variation of dietary energy consumption
- Skewness of dietary energy consumption
- Average dietary energy requirement
- Minimum dietary energy requirement

To estimate the statistics at the subnational level (regions and urban/rural areas):

- The coefficient of variation and skewness are derived from the survey data.
- The region- and area-specific dietary energy supplies adjusted for losses are estimated following the relationship of the dietary energy consumption in the subpopulation group with respect to the national calories.
- The region- and area-specific minimum and average dietary energy requirements are estimated following the relationship of the requirements of the subpopulation group with respect to the calorie requirements as from SOFI.

Example for area of residence: First, the software computes the dietary energy consumption (DEC) at the national, urban, and rural levels using national household survey (NHS) data. Then it calculates the ratio between (1) urban and national DEC and (2) rural and national DEC; the ratios are applied to the national dietary energy supply (DES), adjusted for losses to compute the DES at the urban and rural levels. The software estimates the national DES adjusted for losses from the two exogenous parameters, which are the DES as from the food balance sheets (FBS) and the share of losses at the retail level.

Second, the software computes the minimum dietary energy requirement (MDER) at the national, urban, and rural levels using the NHS data. Then it calculates the ratio between (1) urban and national MDER and (2) rural and national MDER. The ratios are applied to the minimum national dietary energy requirement (as in SOFI and introduced as an exogenous parameter) to compute the MDER at the urban and rural levels. The difference in value between the MDER as in SOFI and the one computed from the NHS data is due to a different structure of the population used by sex and age groups. Also, because the reference height values by sex and age classes for the calculation of the MDER are taken from other sources (e.g., demographic health surveys), differences between heights recorded from the survey and the reference height values can lead to differences in the MDER.

The table below shows an example of how SOFI parameters are calculated at the subnational level using information from the survey.

	Dietary energy consumption (DEC) derived from the survey (kcal/person/day)	*Ratio of subnational DEC/national DEC*	*Dietary energy supply adjusted for losses (kcal/person/day)*	*Minimum dietary energy requirement (MDER) derived from the survey (kcal/person/day)*	*Ratio of subnational MDER/national MDER*	*Minimum dietary energy requirement (kcal/person/day)*
National	2084	1.000	2360 (*)	1824	1.000	1830 (*)
Urban	2130	1.022	2412	1805	0.992	1815
Rural	2035	0.976	2305	1834	1.008	1844

Note: (*) used to estimate MDG 1.9.

Table 1.3: Selected Food Consumption Statistics by Population Groups This table shows some food consumption statistics expressed in dietary energy and monetary values, as well as the average unit value cost of 1,000 kcal and the minimum dietary energy required of a representative individual in the population group. The minimum energy requirement is computed using the age/sex structure of the population as derived from the survey. See glossary and chapter 2 for more details on the calculation of the minimum dietary energy consumption.

Table 1.4: Selected Food Consumption Statistics of Population Groups by Income Deciles This table presents average values for food and total consumption as well as income disaggregated by income deciles. The first decile refers to the poorest group of the population while the tenth refers to the wealthiest one. As poor populations have lower values of income and consumption as compared to rich ones, the values of the statistics shown in this table are expected to increase from the first to the last income group.

Table 1.3: Selected Food Consumption Statistics by Population Groups

	Number of sampled households	*Average household size*	*Average dietary energy consumption (kcal/person/day)*	*Minimum dietary energy requirement (kcal/person/day)*	*Average food consumption in monetary value (LCU/person/day)*	*Average dietary energy unit value (LCU/1000 kcals)*	*Average total consumption in monetary value (LCU/person/day)*
Total	22175	4.9	2199	1691	220.97	100.46	352.71
Quintiles of income							
Lowest quintile	2714	6.6	1596	1640	101.64	63.68	142.49
2	3236	5.6	2043	1676	166.18	81.33	241.69
3	4110	5.0	2250	1690	218.23	97.00	333.62
4	5002	4.2	2604	1719	299.82	115.12	480.80
Highest quintile	7113	3.4	3051	1782	448.89	147.13	812.53
Area							
Capital city	1225	4.3	2064	1782	395.73	191.77	726.83
Other urban areas	13382	4.5	2179	1726	291.85	133.92	494.62
Rural areas	7568	5.1	2213	1679	196.22	88.68	301.42
Household size							
One person	2503	1.0	3667	2022	529.04	144.28	924.86
Between 2 and 3 people	5676	2.6	2667	1784	307.82	115.40	501.16
Between 4 and 5 people	6226	4.5	2253	1672	237.00	105.19	380.89
Between 6 and 7 people	4141	6.4	2097	1656	198.68	94.76	309.84
More than 7	3629	10.3	1951	1677	170.34	87.30	266.79
Gender of the household head							
Male	16751	5.2	2207	1700	219.40	99.41	351.14
Female	5424	4.0	2167	1655	227.75	105.10	359.49
Age of the household head							
Less than 35	7133	3.9	2310	1635	240.19	103.99	388.01
Between 35 and 45	6728	5.3	2155	1681	222.74	103.37	360.31
Between 46 and 60	5433	5.8	2206	1745	212.94	96.55	340.93
More than 60	2881	5.0	2089	1708	198.75	95.13	296.83

Table 1.4: Selected Food Consumption Statistics of Population Groups by Income Deciles

	Number of sampled households	*Average household size*	*Estimated population*	*Average dietary energy consumption (kcal/person/day)*	*Average dietary energy unit value (LCU/1000kcals)*	*Average food consumption in monetary value (LCU/person/day)*	*Average total consumption in monetary value (LCU/person/day)*	*Average income (LCU/ person/day)*
Total								
1	1344	7.2	4624421	1439	59.03	84.96	115.49	122.00
2	1370	5.9	3851732	1784	68.18	121.66	174.91	188.26
3	1634	5.9	3825344	2050	75.83	155.46	223.84	240.31
4	1602	5.2	3384098	2035	87.59	178.29	261.86	294.60
5	1858	5.1	3293406	2235	90.56	202.39	306.42	355.73
6	2252	4.9	3130354	2265	103.69	234.89	362.24	436.89
7	2311	4.4	2858068	2557	107.09	273.83	437.51	539.38
8	2691	4.0	2569153	2657	123.72	328.73	528.95	698.55
9	3064	3.6	2335255	2905	134.94	391.95	674.51	1001.65
10	4049	3.2	2035056	3219	159.74	514.23	970.90	2927.19
Area								
Capital city								
1	7	3.7	18622	778	151.03	117.54	126.09	127.09
2	29	6.5	67083	983	121.82	119.69	174.57	185.05
3	36	5.9	79524	1109	140.47	155.79	236.00	243.00
4	37	7.3	78240	1345	146.93	197.66	290.41	300.12
5	65	6.0	198849	1316	152.92	201.27	344.05	357.19
6	86	5.6	184519	1560	162.38	253.34	414.75	439.84
7	132	4.7	201667	1744	167.35	291.78	499.91	540.86
8	174	4.4	279204	2050	185.18	379.69	659.03	720.97
9	277	4.0	361336	2422	196.12	474.99	859.09	1030.48
10	382	3.1	376271	3150	226.89	714.69	1449.77	2999.53
Other urban areas								
1	471	6.7	243508	1244	70.35	87.52	119.72	126.87
2	592	6.4	282113	1413	92.55	130.81	180.39	189.23
3	792	5.8	325095	1569	96.73	151.80	221.52	239.95
4	825	5.7	328223	1699	104.03	176.79	267.95	296.18
5	1033	5.3	377364	1829	110.06	201.27	319.24	356.85
6	1394	5.2	550613	1879	124.83	234.53	371.24	436.92
7	1460	4.6	538197	2290	119.43	273.46	459.15	544.31
8	1773	4.4	636163	2528	133.78	338.14	567.13	708.65
9	2119	3.5	515264	2679	157.24	421.25	724.13	1000.35
10	2923	3.1	610035	3095	177.98	550.86	1038.77	3228.57

Particularly in this table, the number of sampled households used to produce the estimates by income deciles has to be analyzed to assess the reliability of the estimates. For instance, in table 1.4 the estimates of the first income decile group in the capital city are considered unreliable due to the low number of households used to derive the estimates (7 households). In general, a statistic obtained with data from fewer than 30 households is considered unreliable.

Table 1.5: Shares of Food Consumption by Food Sources (in Dietary Energy) Households acquire food in different ways, the main ones being through purchases, own production, gifts/aid, bartering, and in-kind

Table 1.5: Shares of Food Consumption by Food Sources (in Dietary Energy)

	Number of sampled households	*Share of purchased food in total food consumption (%)*	*Share of own produced food in total food consumption (%)*	*Share of food consumed away from home in total food consumption (%)*	*Share of food from other sources in total food consumption (%)*
Total	22175	52.56	40.01	3.38	4.06
Quintiles of income					
Lowest quintile	2714	40.31	53.34	1.82	4.54
2	3236	42.01	51.83	2.11	4.06
3	4110	51.34	41.80	2.62	4.25
4	5002	62.10	30.64	3.37	3.88
Highest quintile	7113	67.84	21.44	7.19	3.54
Area					
Capital city	1225	84.15	0.45	13.27	2.13
Other urban areas	13382	79.69	12.58	4.69	3.04
Rural areas	7568	45.85	47.31	2.49	4.36
Household size					
One person	2503	56.62	19.60	18.00	5.78
Between 2 and 3 people	5676	55.02	36.36	3.52	5.10
Between 4 and 5 people	6226	53.90	39.13	3.16	3.81
Between 6 and 7 people	4141	52.26	40.60	2.83	4.31
More than 7	3629	49.63	44.72	2.39	3.26
Gender of the household head					
Male	16751	52.66	39.88	3.56	3.89
Female	5424	52.10	40.59	2.55	4.76
Age of the household head					
Less than 35	7133	53.61	37.69	4.59	4.11
Between 35 and 45	6728	55.88	37.31	3.23	3.58
Between 46 and 60	5433	50.43	42.97	2.69	3.91
More than 60	2881	47.48	44.51	2.72	5.29

payment. In addition, household members consume food at sit-down and fast food restaurants and from street vendors. For the purpose of the analysis, food sources are classified in four main categories according to the type of acquisition: (1) purchase (excluding food consumed away from home), (2) own production, (3) consumed away from home, and (4) others (including gifts/aid, in-kind payment, etc.).

This table shows the proportion of total dietary energy provided by each of the four food sources. This information is useful, for instance, to assess how much households rely on the following:

- Food purchases (illustrating potential vulnerability to food price shocks)
- Own production (illustrating potential vulnerability to natural shocks such as drought or flood)

Table 1.6: Shares of Food Consumption by Food Sources (in Dietary Energy) by Income Deciles This table shows the proportion of total dietary energy provided by each of the four food sources: purchases to be consumed inside the home, own production, consumption away from home, and other sources combined. The data are disaggregated by income decile groups.

Particularly in this table, the number of sampled households used to produce the estimates at income decile levels has to be analyzed to assess the reliability of the estimates. For instance, in table 1.6 the estimates of the first income decile group in the capital city are considered unreliable due to the low number of households used to derive the estimates (7 households). In general, a statistic obtained with data from fewer than 30 households is considered unreliable.

Table 1.7: Shares of Food Consumption by Food Sources (in Monetary Value) This table shows the proportion of total food expenditure that each of the four food sources (purchases to be consumed inside the home, own production, consumption away from home, and other sources combined) represents. The share of money spent to purchase food in total household food expenditure is an indirect measure of a household's vulnerability to market food crises. In general, the higher the proportion of food consumed from purchases, the higher the risk of households being affected by food price shocks. As well, the percentage of food from own production gives an idea of the dependency a household has on its own agricultural outcome. Households that practice farming are highly

Table 1.6: Shares of Food Consumption by Food Sources (in Dietary Energy) by Income Deciles

	Number of sampled households	*Average income (LCU/person/day)*	*Share of purchased food in total food consumption (%)*	*Share of own produced food in total food consumption (%)*	*Share of food consumed away from home in total food consumption (%)*	*Share of food from other sources in total food consumption (%)*
Total						
1	1344	122.00	40.44	52.96	1.69	4.91
2	1370	188.26	40.17	53.71	1.93	4.19
3	1634	240.31	40.30	53.70	1.89	4.11
4	1602	294.60	43.96	49.69	2.35	3.99
5	1858	355.73	50.55	43.32	2.26	3.88
6	2252	436.89	52.16	40.22	3.00	4.63
7	2311	539.38	60.39	33.26	2.48	3.88
8	2691	698.55	63.94	27.84	4.33	3.89
9	3064	1001.65	66.28	25.27	4.71	3.74
10	4049	2927.19	69.45	17.47	9.75	3.33
Area						
Capital city						
1	7	127.09	86.64	0.57	4.41	8.37
2	29	185.05	87.37	4.62	3.34	4.67
3	36	243.00	87.23	1.06	8.65	3.06
4	37	300.12	88.49	0.91	5.63	4.97
5	65	357.19	93.55	0.30	4.83	1.33
6	86	439.84	90.83	0.36	7.13	1.68
7	132	540.86	91.20	0.06	7.77	0.97
8	174	720.97	85.37	0.06	12.36	2.20
9	277	1030.48	85.82	0.14	12.51	1.53
10	382	2999.53	75.71	0.72	20.93	2.64
Other urban areas						
1	471	126.87	76.95	17.30	1.87	3.88
2	592	189.23	76.75	16.94	2.05	4.26
3	792	239.95	72.18	21.67	2.16	4.00
4	825	296.18	74.26	17.58	2.95	5.21
5	1033	356.85	76.54	18.56	2.09	2.81
6	1394	436.92	80.10	13.72	2.48	3.69
7	1460	544.31	81.90	11.99	3.26	2.85
8	1773	708.65	80.51	12.52	4.86	2.11
9	2119	1000.35	82.08	10.42	4.55	2.95
10	2923	3228.57	81.40	6.19	9.96	2.45
Rural areas						
1	866	121.71	38.59	54.78	1.68	4.95
2	749	188.25	37.41	56.50	1.91	4.18
3	806	240.28	37.48	56.60	1.79	4.13
4	740	294.28	40.49	53.39	2.25	3.87
5	760	355.47	45.99	47.74	2.17	4.10
6	772	436.65	45.22	46.96	2.88	4.94
7	719	537.98	53.87	39.88	1.98	4.28
8	744	690.88	55.56	36.57	3.16	4.72
9	668	994.97	57.68	34.66	3.25	4.40
10	744	2725.93	60.83	29.31	5.81	4.04

Table 1.7: Shares of Food Consumption by Food Sources (in Monetary Value)

	Number of sampled households	*Share of food consumption in total income (%) (Engel ratio)*	*Share of purchased food in total food consumption (%)*	*Share of own produced food in total food consumption (%)*	*Share of food consumed away from home in total food consumption (%)*	*Share of food from other sources in total food consumption (%)*
Total	22175	40.57	65.38	26.38	4.24	4.01
Quintiles of income						
Lowest quintile	2714	66.82	50.15	42.89	1.84	5.12
2	3236	62.52	53.50	40.11	2.19	4.20
3	4110	55.21	62.04	30.88	2.76	4.33
4	5002	48.77	71.56	20.92	3.79	3.73
Highest quintile	7113	23.65	76.58	12.06	7.96	3.40
Area						
Capital city	1225	36.13	83.70	0.45	13.25	2.59
Other urban areas	13382	33.29	85.16	6.96	5.00	2.88
Rural areas	7568	43.80	57.67	35.10	2.73	4.50
Household size						
One person	2503	30.01	62.30	12.56	20.58	4.56
Between 2 and 3 people	5676	38.97	68.19	23.30	4.15	4.36
Between 4 and 5 people	6226	41.27	67.47	25.13	3.71	3.69
Between 6 and 7 people	4141	41.35	64.87	27.19	3.39	4.55
More than 7	3629	43.07	61.79	31.93	2.79	3.49
Gender of the household head						
Male	16751	39.46	65.27	26.48	4.51	3.74
Female	5424	46.00	65.85	25.94	3.08	5.13
Age of the household head						
Less than 35	7133	35.90	66.61	23.04	6.12	4.22
Between 35 and 45	6728	40.20	68.76	23.79	3.95	3.49
Between 46 and 60	5433	43.09	63.00	30.05	3.18	3.77
More than 60	2881	49.40	59.53	32.10	3.09	5.28

exposed to natural shocks, so they are particularly vulnerable in disaster-prone countries where recurrent natural shocks damage household agricultural production.

Table 1.7 also shows the Engel ratio defined as the percentage of total income dedicated to acquire food. Engel's law states that the proportion of income spent on food decreases when income increases. This does not mean that food expenditure decreases as income increases; on the contrary, while food intake has an upper limit due to biological factors, food expenditure does not. However, the relative importance of food expenditure tends to be greater among the poor households since they focus their acquisition on primary need goods (thus limiting the expenses on the other items). The share of food expenditure tends to be lower among the wealthier households because they spend a greater proportion of their income on nonfood items. However, it should be noted that "while the share of food expenditure in total expenditure may be a good starting-point for assessing vulnerability, it is not sufficient within a given economic environment and the same food expenditure share would not necessarily represent the same level of vulnerability across different economic environments" (Schmidhuber 2003).

Table 1.8: Shares of Food Consumption by Food Sources (in Monetary Value) by Income Deciles This table shows the proportion of total food expenditure that each of the four food sources represents (purchases to be consumed

Table 1.8: Shares of Food Consumption by Food Sources (in Monetary Value) by Income Deciles

	Number of sampled households	*Average income (LCU/ person/ day)*	*Share of food consumption in total income (%) (Engel ratio)*	*Share of purchased food in total food consumption (%)*	*Share of own produced food in total food consumption (%)*	*Share of food consumed away from home in total food consumption (%)*	*Share of food from other sources in total food consumption (%)*
Total							
1	1344	122.00	69.63	49.85	42.61	1.91	5.63
2	1370	188.26	64.62	50.41	43.12	1.79	4.69
3	1634	240.31	64.69	51.68	41.89	2.08	4.34
4	1602	294.60	60.52	55.29	38.35	2.30	4.07
5	1858	355.73	56.89	60.70	32.58	2.51	4.21
6	2252	436.89	53.77	63.26	29.33	2.98	4.43
7	2311	539.38	50.77	70.26	23.02	2.83	3.90
8	2691	698.55	47.06	72.77	18.97	4.69	3.57
9	3064	1001.65	39.13	75.65	15.17	5.50	3.68
10	4049	2927.19	17.57	77.40	9.35	10.11	3.15
Area							
Capital city							
1	7	127.09	92.48	88.09	1.23	3.63	7.05
2	29	185.05	64.68	89.21	3.61	3.41	3.77
3	36	243.00	64.11	86.64	1.20	9.26	2.90
4	37	300.12	65.86	89.66	1.71	5.77	2.86
5	65	357.19	56.35	93.24	0.24	5.03	1.48
6	86	439.84	57.60	90.56	0.38	7.00	2.05
7	132	540.86	53.95	90.38	0.09	8.34	1.19
8	174	720.97	52.66	86.12	0.13	12.00	1.75
9	277	1030.48	46.09	82.61	0.18	12.75	4.45
10	382	2999.53	23.83	78.70	0.66	18.44	2.21

inside the home, own production, consumption away from home, and other sources combined). The data are disaggregated by income decile groups.

Particularly in this table, the number of sampled households used to produce the estimates at income decile levels has to be analyzed to assess the reliability of the estimates. For instance, in table 1.8 the estimates of the first income decile group in the capital city are considered unreliable due to the low number of households used to derive the estimates (7 households). In general, a statistic obtained with data from fewer than 30 households is considered unreliable.

Table 1.9: Food Consumption in Dietary Energy, Monetary, and Nutrient Content by Population Groups The human body requires energy for different purposes including metabolic processes, muscular activity, growth, and

Table 1.9: Food Consumption in Dietary Energy, Monetary, and Nutrient Content by Population Groups

	Average dietary energy consumption (kcal/person/day)	*Average food consumption in monetary value (LCU/person/day)*	*Average protein consumption (g/person/day)*	*Average fat consumption (g/person/day)*	*Average carbohydrates consumption (g/person/day)*
Total	2199	220.97	57.0	38.8	359.1
Quintiles of income					
Lowest quintile	1596	101.64	41.7	21.1	280.3
2	2043	166.18	53.8	32.0	340.9
3	2250	218.23	57.9	39.2	367.8
4	2604	299.82	65.9	50.3	412.3
Highest quintile	3051	448.89	79.8	69.5	462.9
Area					
Capital city	2064	395.73	53.4	52.1	328.0
Other urban areas	2179	291.85	55.8	48.9	347.2
Rural areas	2213	196.22	57.5	36.1	363.4
Household size					
One person	3667	529.04	95.7	76.4	512.7
Between 2 and 3 people	2667	307.82	69.1	51.4	413.9
Between 4 and 5 people	2253	237.00	58.8	40.5	364.0
Between 6 and 7 people	2097	198.68	53.2	35.0	349.6
More than 7	1951	170.34	51.2	32.8	330.1
Gender of the household head					
Male	2207	219.40	57.1	38.6	359.5
Female	2167	227.75	56.8	39.7	357.4
Age of the household head					
Less than 35	2310	240.19	60.2	42.0	375.5
Between 35 and 45	2155	222.74	55.5	38.1	352.1
Between 46 and 60	2206	212.94	57.2	38.8	360.0
More than 60	2089	198.75	54.6	34.5	343.4

synthesis of new tissues. Humans obtain the required energy through the intake of energy-yielding macronutrients from food consumption. These macronutrients are protein, fats, total carbohydrates (including fiber), and alcohol. Each of them contributes to the total dietary energy but in different proportions. Because available carbohydrates are estimated as the difference between total carbohydrates and fiber combined with the use of the Atwater[6] factors, the energy densities of the nutrients comprise the following:

- Four calories per gram of protein
- Nine calories per gram of fats
- Four calories per gram of available carbohydrates
- Two calories per gram of fiber
- Seven calories per gram of alcohol

This table shows protein, fats, and carbohydrates consumption expressed in grams per person per day. Note that the carbohydrates values, reported in table 1.9 (last column), are those corresponding to available carbohydrates. On average worldwide, people consume more carbohydrates per day than protein or fats. It is expected that macronutrient consumption increases with income, since food consumption is positively correlated with income. However, the pattern of the increase in macronutrient consumption varies among population groups because households with higher income can afford a more diverse diet (e.g., more protein from meat) than those with lower income.

Table 1.10: Nutrient Contribution to Dietary Energy Consumption This table shows the proportion of dietary energy provided by each macronutrient. The proportion of calories from protein and fats are estimated as their respective consumption in grams times 4 and 9, respectively. Then the calories from total carbohydrates and alcohol are estimated as the difference between total dietary energy consumption and the calories coming from protein and fats.

The concept of a balanced diet is applied in more than one of the ADePT-FSM output tables. A joint WHO/FAO group of experts established guidelines for a "balanced diet," described in terms of the proportions of total dietary energy provided by diverse sources of energy (WHO 2003). These guidelines are related to the effects of chronic

Table 1.10: Nutrient Contribution to Dietary Energy Consumption

	Average dietary energy consumption (kcal/person/day)	*Share of DEC from protein (%)*	*Share of DEC from fat (%)*	*Share of DEC from total carbohydrates and alcohol (%)*
Total	2199	10.37	15.87	73.75
Quintiles of income				
Lowest quintile	1596	10.46	11.88	77.66
2	2043	10.53	14.08	75.39
3	2250	10.30	15.67	74.03
4	2604	10.13	17.40	72.47
Highest quintile	3051	10.45	20.50	69.02
Area				
Capital city	2064	10.36	22.72	66.89
Other urban areas	2179	10.24	20.19	69.53
Rural areas	2213	10.40	14.68	74.92
Household size				
One person	3667	10.42	18.70	70.78
Between 2 and 3 people	2667	10.36	17.34	72.28
Between 4 and 5 people	2253	10.45	16.16	73.39
Between 6 and 7 people	2097	10.15	15.04	74.81
More than 7	1951	10.50	15.14	74.36
Gender of the household head				
Male	2207	10.35	15.73	73.91
Female	2167	10.48	16.48	73.04
Age of the household head				
Less than 35	2310	10.42	16.36	73.20
Between 35 and 45	2155	10.30	15.93	73.76
Between 46 and 60	2206	10.37	15.84	73.79
More than 60	2089	10.46	14.85	74.69

nondeficiency diseases. So, according to the experts, a diet is determined to be balanced when

- The proportion of dietary energy provided by protein is in the range of 10–15 percent
- The proportion of dietary energy provided by fats is in the range of 15–30 percent
- The proportion of total dietary energy provided by the remaining macronutrients is in the range of 55–75 percent

Table 1.11: Nutrient Contribution to Dietary Energy Consumption at Income Quintile Levels This table indicates whether households classified by income quintile groups have access to a balanced diet. The main sources of dietary energy are protein, fats, and total carbohydrates. Households have

Table 1.11: Nutrient Contribution to Dietary Energy Consumption at Income Quintile Levels

	Average dietary energy consumption (kcal/person/ day)	*Share of DEC from protein (%)*	*Share of DEC from fat (%)*	*Share of DEC from total carbohydrates and alcohol (%)*	*Within range of population protein intake goal: 10%–15%*	*Within range of population fat intake goal: 15%–30%*	*Within range of population total carbohydrates and alcohol intake goal: 55%–75%*
Total							
Quantiles of income							
Lowest quintile	1596	10.46	11.88	77.66	OK	LOW	HIGH
2	2043	10.53	14.08	75.39	OK	LOW	HIGH
3	2250	10.30	15.67	74.03	OK	OK	OK
4	2604	10.13	17.40	72.47	OK	OK	OK
Highest quintile	3051	10.45	20.50	69.02	OK	OK	OK
Area							
Capital city							
Lowest quintile	938	9.86	15.51	74.63	LOW	OK	OK
2	1226	9.99	16.37	73.64	LOW	OK	OK
3	1434	10.02	20.45	69.52	OK	OK	OK
4	1922	10.32	22.47	67.21	OK	OK	OK
Highest quintile	2793	10.52	24.31	65.12	OK	OK	OK
Other urban areas							
Lowest quintile	1335	10.17	15.94	73.89	OK	OK	OK
2	1635	10.06	16.42	73.51	OK	OK	OK
3	1858	10.22	18.16	71.62	OK	OK	OK
4	2419	9.95	19.41	70.61	LOW	OK	OK
Highest quintile	2905	10.59	24.08	65.26	OK	OK	OK
Rural areas							
Lowest quintile	1621	10.48	11.63	77.89	OK	LOW	HIGH
2	2105	10.57	13.86	75.57	OK	LOW	HIGH
3	2382	10.32	15.10	74.57	OK	OK	OK
4	2749	10.16	16.39	73.45	OK	OK	OK
Highest quintile	3193	10.38	18.06	71.55	OK	OK	OK

access to a balanced diet if the proportion of dietary energy from each macronutrient is within the experts' recommendations.[7] When OK appears for the three calorie-yielding macronutrients, we can consider that households in that population have access to a balanced diet. However, little else is known about households' consuming a balanced diet because there is no information about how people combine the food they consume or about intrahousehold allocation of food.

Table 1.12: Nutrient Density per 1,000 Kcal This table shows the grams of protein, carbohydrates, and fats per 1,000 kcals (kilocalories) of households' consumption (macronutrient density).

Table 1.12: Nutrient Density per 1,000 Kcal

	Average dietary energy consumption (kcal/person/ day)	*Average protein consumption (g/1000 kcal)*	*Average carbohydrates consumption (g/1000 kcal)*	*Average fat consumption (g/1000 kcal)*
Total	2199	25.9	163.2	17.6
Quintiles of income				
Lowest quintile	1596	26.1	175.6	13.2
2	2043	26.3	166.8	15.6
3	2250	25.7	163.5	17.4
4	2604	25.3	158.3	19.3
Highest quintile	3051	26.1	151.7	22.8
Area				
Capital city	2064	25.9	158.9	25.2
Other urban areas	2179	25.6	159.3	22.4
Rural areas	2213	26.0	164.2	16.3
Household size				
One person	3667	26.1	139.5	20.8
Between 2 and 3 people	2667	25.9	155.2	19.3
Between 4 and 5 people	2253	26.1	161.6	18.0
Between 6 and 7 people	2097	25.4	166.7	16.7
More than 7	1951	26.3	169.2	16.8
Gender of the household head				
Male	2207	25.9	162.9	17.5
Female	2167	26.2	164.9	18.3
Age of the household head				
Less than 35	2310	26.1	162.5	18.2
Between 35 and 45	2155	25.7	163.4	17.7
Between 46 and 60	2206	25.9	163.2	17.6
More than 60	2089	26.2	164.4	16.5

Table 1.13: Share of Animal Protein in Total Protein Consumption This table shows the proportion of protein consumption coming from food of animal origin (animal proteins). The food commodities considered to be of animal origin are meat (red and white), fish, eggs, milk, and cheese. When households are classified by income quintiles, an increasing trend in the proportion of protein of animal origin consumed as one moves from the first to the last income quintile is expected. This is mainly because richer households can afford more expensive food products such as meat and fish. However, such a trend probably is not present in pastoral regions where poor

Table 1.13: Share of Animal Protein in Total Protein Consumption

	Share of animal protein in total protein consumption (%)
Total	21.2
Quintiles of income	
Lowest quintile	16.0
2	18.3
3	20.8
4	23.4
Highest quintile	28.0
Area	
Capital city	25.3
Other urban areas	24.2
Rural areas	20.5
Household size	
One person	26.1
Between 2 and 3 people	23.1
Between 4 and 5 people	21.4
Between 6 and 7 people	19.7
More than 7	20.8
Gender of the household head	
Male	21.3
Female	21.0
Age of the household head	
Less than 35	21.6
Between 35 and 45	21.9
Between 46 and 60	20.5
More than 60	20.7

communities/households derive a sizeable part of their consumption from livestock products (i.e., milk and cheese).

Table 1.14: Within-Region Differences in Nutrient Consumption, by Regional Income Quintiles This table shows the average macronutrients consumption by region and income quintile providing information on the intraregional differences. Such disaggregation can be used to explore income-based disparities on macronutrient consumption within each region and identify regions where disparities due to income are more pronounced. Note that the first row of the table for each region corresponds to table 1.9.

Table 1.14: Within-Region Differences in Nutrient Consumption, by Regional Income Quintiles

	Average protein consumption (g/person/day)	*Average fat consumption (g/person/day)*	*Average carbohydrates consumption (g/person/day)*
Region			
Region 1			
Total	69.11	43.87	372.61
Lowest quintile	58.41	28.75	340.23
2	65.74	39.17	354.78
3	66.16	41.44	345.70
4	78.96	55.96	414.48
Highest quintile	81.41	61.05	428.31
Region 2			
Total	52.38	39.78	316.28
Lowest quintile	40.42	21.95	267.46
2	45.01	29.18	273.65
3	49.31	34.52	289.86
4	62.47	52.19	380.20
Highest quintile	72.66	75.11	404.26
Region 3			
Total	46.13	41.07	310.79
Lowest quintile	31.13	19.66	232.15
2	36.71	31.13	263.27
3	45.50	38.97	313.15
4	53.25	50.73	349.16
Highest quintile	73.79	76.22	438.10
Region 4			
Total	50.35	37.53	380.04

Disaggregated by Food Commodity Group: Tables 2.1 to 2.9

The output tables showing statistics on consumption (grams/person/day) of protein, fats, and carbohydrates by food commodity group are useful in providing a picture of the consumption pattern. Note that this information also helps to identify which food commodity group contributes the most to the consumption of a given macronutrient. Note as well that the values of carbohydrates refer to available carbohydrates.[8]

Table 2.1: Food Consumption by Food Commodity Group This table shows the macronutrients (expressed in grams) and the food consumption (in dietary energy and monetary values) at the *national level* disaggregated by food commodity groups.

Table 2.1: Food Consumption by Food Commodity Groups

	Average food consumption in monetary value (LCU/ person/day)	*Average dietary energy consumption (kcal/person/ day)*	*Average protein consumption (g/person/day)*	*Average carbohydrates consumption (g/person/day)*	*Average fat consumption (g/person/day)*
Food group					
Cereals	67.4	1201	29.5	231.0	12.6
Roots and tubers	14.9	258	2.6	58.7	0.5
Sugars and syrups	12.7	89	0.0	22.3	0.0
Pulses	12.5	73	5.5	9.8	0.3
Tree nuts	0.1	3	0.1	0.0	0.2
Oil crops	4.9	79	2.6	1.1	6.8
Vegetables	18.6	30	1.7	4.0	0.3
Fruits	10.2	69	0.6	15.4	0.2
Stimulants	2.0	5	0.1	0.9	0.1
Spices	2.9	2	0.0	0.2	0.0
Alcoholic beverages	7.8	114	0.1	1.7	0.0
Meat	23.1	71	5.3	0.0	5.5
Eggs	0.8	1	0.1	0.0	0.1
Fish	15.1	27	5.4	0.0	0.6
Milk and cheese	7.2	24	1.3	1.8	1.3
Oils and fats (vegetable)	8.4	76	0.0	0.0	8.4
Oils and fats (animal)	0.3	2	0.0	0.0	0.2
Nonalcoholic beverages	2.8	2	0.0	0.5	0.0
Miscellaneous and prepared food	9.4	74	2.0	11.8	1.5

Table 2.2: Contribution of Food Commodity Groups to Total Nutrient Consumption This table shows the contribution (in percentage) of each food commodity group to the total dietary energy, protein, available carbohydrates, and fats consumed by the households at the *national level*. The disaggregation of these statistics by food commodity groups helps identify which food commodity groups are the main sources of calories, protein, total carbohydrates, and fats.

Table 2.3: Food Consumption by Food Commodity Group and Income Quintile This table has the same indicators as table 2.1 except that the statistics are disaggregated by income quintile. This table helps to identify the food item groups that contribute the most in terms of calories and macronutrients to the diet of population groups segmented according to their income level.

Table 2.4: Food Consumption by Food Commodity Group and Area This table shows the *urban/rural* food consumption statistics in terms of

Table 2.2: Contribution of Food Commodity Groups to Total Nutrient Consumption

	Share of total dietary energy consumption (%)	*Share of total protein consumption (%)*	*Share of total carbohydrates consumption (%)*	*Share of total fat consumption (%)*
Food group				
Cereals	54.6	51.6	64.3	32.5
Roots and tubers	11.7	4.6	16.4	1.3
Sugars and syrups	4.1	0.0	6.2	0.0
Pulses	3.3	9.7	2.7	0.8
Tree nuts	0.1	0.2	0.0	0.6
Oil crops	3.6	4.6	0.3	17.5
Vegetables	1.3	3.0	1.1	0.8
Fruits	3.1	1.1	4.3	0.5
Stimulants	0.2	0.2	0.2	0.2
Spices	0.1	0.1	0.1	0.1
Alcoholic beverages	5.2	0.3	0.5	0.0
Meat	3.2	9.4	0.0	14.3
Eggs	0.1	0.2	0.0	0.2
Fish	1.2	9.5	0.0	1.7
Milk and cheese	1.1	2.3	0.5	3.5
Oils and fats (vegetable)	3.5	0.0	0.0	21.8
Oils and fats (animal)	0.1	0.0	0.0	0.5
Nonalcoholic beverages	0.1	0.0	0.1	0.0
Miscellaneous and prepared food	3.4	3.5	3.3	3.8

Table 2.3: Food Consumption by Food Commodity Group and Income Quintile

	Average food consumption in monetary value (LCU/ person/day)	*Average dietary energy consumption (kcal/person/ day)*	*Average protein consumption (g/person/day)*	*Average carbohydrates consumption (g/person/day)*	*Average fat consumption (g/person/day)*
Quintiles of income					
Lowest quintile					
Cereals	35.75	987	24.6	188.5	10.5
Roots and tubers	11.62	267	2.6	60.9	0.5
Sugars and syrups	3.50	25	0.0	6.2	0.0
Pulses	6.76	53	4.0	7.1	0.2
Tree nuts	0.08	2	0.1	0.0	0.2
Oil crops	1.76	34	1.3	0.5	2.9
Vegetables	10.25	23	1.3	3.0	0.2
Fruits	3.34	30	0.3	6.8	0.1
Stimulants	0.47	2	0.0	0.4	0.0
Spices	1.61	0	0.0	0.0	0.0
Alcoholic beverages	2.25	59	0.1	0.8	0.0
Meat	8.06	29	2.4	0.0	2.2
Eggs	0.12	0	0.0	0.0	0.0
Fish	8.21	18	3.5	0.0	0.4
Milk and cheese	3.49	15	0.8	1.1	0.8

(continued)

Table 2.3: Food Consumption by Food Commodity Group and Income Quintile (continued)

	Average food consumption in monetary value (LCU/ person/day)	*Average dietary energy consumption (kcal/person/ day)*	*Average protein consumption (g/person/day)*	*Average carbohydrates consumption (g/person/day)*	*Average fat consumption (g/person/day)*
Oils and fats (vegetable)	2.20	21	0.0	0.0	2.4
Oils and fats (animal)	0.08	1	0.0	0.0	0.1
Nonalcoholic beverages	0.22	0	0.0	0.0	0.0
Miscellaneous and prepared food	1.87	29	0.8	4.9	0.5
Quintile 2					
Cereals	57.42	1192	29.4	228.9	12.4
Roots and tubers	14.53	265	2.6	60.6	0.5
Sugars and syrups	8.23	57	0.0	14.3	0.0
Pulses	10.63	65	4.9	8.7	0.3

Table 2.4: Food Consumption by Food Commodity Group and Area

	Average food consumption in monetary value (LCU/ person/day)	*Average dietary energy consumption (kcal/person/ day)*	*Average protein consumption (g/person/day)*	*Average carbohydrates consumption (g/person/day)*	*Average fat consumption (g/person/day)*
Area					
Capital city					
Cereals	110.12	1029	24.7	200.2	10.6
Roots and tubers	8.95	50	0.7	11.2	0.1
Sugars and syrups	22.55	184	0.0	46.1	0.0
Pulses	15.27	51	3.8	6.8	0.2
Tree nuts	0.10	1	0.0	0.0	0.1
Oil crops	10.86	105	1.5	1.8	9.6
Vegetables	33.88	24	1.2	3.5	0.2
Fruits	19.88	44	0.4	9.8	0.1
Stimulants	3.63	5	0.1	1.0	0.1
Spices	2.94	4	0.1	0.5	0.1
Alcoholic beverages	12.99	8	0.0	0.4	0.0
Meat	40.76	91	6.2	0.0	7.3
Eggs	2.99	3	0.2	0.0	0.2
Fish	23.49	34	6.5	0.0	0.9
Milk and cheese	7.52	11	0.6	0.8	0.6
Oils and fats (vegetable)	14.71	132	0.0	0.0	14.6
Oils and fats (animal)	1.32	6	0.0	0.0	0.6
Nonalcoholic beverages	11.32	9	0.0	2.3	0.0
Miscellaneous and prepared food	52.44	274	7.3	43.6	6.6
Other urban areas					
Cereals	87.62	1217	29.4	234.3	13.1
Roots and tubers	11.06	134	1.6	30.2	0.3
Sugars and syrups	20.73	148	0.0	36.9	0.0
Pulses	12.87	57	4.2	7.6	0.3

macronutrients,[9] dietary energy, and monetary values disaggregated by food commodity groups. This table allows the analyst to explore the macronutrient consumption patterns in urban and rural areas and detect differences, if any.

Table 2.5: Contribution of Food Commodity Groups to Total Nutrient Consumption by Area This table shows the contribution (in percentage) of each food commodity group to the total dietary energy, protein, available carbohydrates,[10] and fats consumed by the households in *rural and urban areas*. The disaggregation of these statistics by food commodity groups helps to identify which food commodity group(s) are the main sources of calories, protein, available carbohydrates, and fats within urban and rural areas and highlights urban/rural-based differences.

Table 2.6: Food Consumption by Food Commodity Group and Region This table shows, for the *first income quintile of each region*, the food consumption statistics in terms of macronutrients, dietary energy, and monetary values disaggregated by food commodity groups. Because the food consumption pattern varies among regions, the analysis of the first income quintile group by region is important to identify the main sources of each macronutrient for the poorest population.

Table 2.7: Food Consumption by Food Commodity Group and Region in the First Quintile This table shows, for the *first income quintile of each region*, the food consumption statistics in terms of macronutrients,[11] dietary energy, and monetary values disaggregated by food commodity groups. Because the first income quintile refers to the poorest population and the food consumption pattern varies among regions, the analysis is important in helping to identify regional differences within the poorest part of the population.

Table 2.8: Nutrient Costs by Food Commodity Group This table shows the unit value of dietary energy, protein, available carbohydrates,[12] and fats disaggregated by *food commodity groups*. The objective of this table is to identify the food commodity groups that are low-cost sources of dietary energy, protein, carbohydrates, or fats.

The unit values are estimated using expenditures of each food commodity group as well as their contribution to total calories or nutrient content

Table 2.5: Contribution of Food Commodity Groups to Total Nutrient Consumption by Area

	Share of total dietary energy consumption (%)	*Share of total protein consumption (%)*	*Share of total carbohydrates consumption (%)*	*Share of total fat consumption (%)*
Area				
Capital city				
Cereals	49.8	46.3	61.0	20.4
Roots and tubers	2.4	1.3	3.4	0.2
Sugars and syrups	8.9	0.0	14.0	0.0
Pulses	2.5	7.1	2.1	0.4
Tree nuts	0.1	0.1	0.0	0.2
Oil crops	5.1	2.8	0.5	18.5
Vegetables	1.1	2.3	1.1	0.4
Fruits	2.2	0.8	3.0	0.2
Stimulants	0.2	0.2	0.3	0.2
Spices	0.2	0.2	0.2	0.2
Alcoholic beverages	0.4	0.1	0.1	0.0
Meat	4.4	11.6	0.0	14.0
Eggs	0.1	0.4	0.0	0.4
Fish	1.7	12.1	0.0	1.7
Milk and cheese	0.5	1.0	0.2	1.1
Oils and fats (vegetable)	6.4	0.0	0.0	28.1
Oils and fats (animal)	0.3	0.0	0.0	1.2
Nonalcoholic beverages	0.5	0.0	0.7	0.0
Miscellaneous and prepared food	13.3	13.6	13.3	12.6
Other urban areas				
Cereals	55.8	52.7	67.5	26.9
Roots and tubers	6.2	2.9	8.7	0.5
Sugars and syrups	6.8	0.0	10.6	0.0
Pulses	2.6	7.6	2.2	0.5

(in kcal or grams, respectively). Then the dietary energy unit value is expressed in local currency per 1,000 kcal, while the cost of each macronutrient is expressed in local currency per 100 grams of the respective nutrient. Each time N/A replaces a unit value, it means that the dietary energy or nutrient content provided by the food commodity group is very low or null, or there was no acquisition of that food group.

Table 2.9: Food Consumption by Food Commodity Group and Food Sources (in Dietary Energy) This table shows the contribution of each food source to the dietary energy consumption for each food commodity group.

In table 2.9, own production contributes 47 percent to the total dietary energy consumption coming from cereals while most of the dietary energy consumption provided by vegetable oils and fats are coming from purchases (93.4 percent). This table makes it possible to identify the main sources of acquisition of the food group commodity being consumed.

Table 2.6: Food Consumption by Food Commodity Group and Region

	Average food consumption in monetary value (LCU/person/day)	Average dietary energy consumption (kcal/person/day)	Average protein consumption (g/person/day)	Average carbohydrates consumption (g/person/day)	Average fat consumption (g/person/day)
Region					
Region 1					
Cereals	69.53	1614	41.6	307.5	16.5
Roots and tubers	5.67	41	0.7	9.1	0.1
Sugars and syrups	9.62	67	0.0	16.7	0.0
Pulses	12.22	93	7.1	12.3	0.4
Tree nuts	0.05	1	0.0	0.0	0.1
Oil crops	5.36	124	5.2	1.6	10.3
Vegetables	22.22	47	2.7	5.6	0.5
Fruits	4.56	18	0.3	3.4	0.1
Stimulants	1.35	4	0.1	0.8	0.1
Spices	2.72	1	0.0	0.2	0.0
Alcoholic beverages	7.13	145	0.2	2.1	0.0
Meat	19.85	66	4.9	0.0	5.2
Eggs	0.65	1	0.1	0.0	0.1
Fish	6.53	12	2.4	0.0	0.3
Milk and cheese	7.53	33	1.8	2.4	1.8
Oils and fats (vegetable)	7.59	63	0.0	0.0	6.9
Oils and fats (animal)	0.22	1	0.0	0.0	0.1
Nonalcoholic beverages	1.24	1	0.0	0.2	0.0
Miscellaneous and prepared food	6.13	69	2.0	10.8	1.3
Region 2					
Cereals	73.76	1275	31.9	241.3	14.7
Roots and tubers	6.45	40	0.8	8.4	0.1
Sugars and syrups	16.91	113	0.0	28.3	0.0
Pulses	11.50	54	4.0	7.3	0.3

Table 2.7: Food Consumption by Food Commodity Group and Region in the First Quintile

	Average food consumption in monetary value (LCU/person/day)	Average dietary energy consumption (kcal/person/day)	Average protein consumption (g/person/day)	Average carbohydrates consumption (g/person/day)	Average fat consumption (g/person/day)
Region					
Region 1					
Cereals	44.76	1584	42.0	301.2	15.7
Roots and tubers	2.93	31	0.4	6.9	0.1
Sugars and syrups	2.66	19	0.0	4.7	0.0
Pulses	5.83	56	4.3	7.4	0.2
Tree nuts	0.00	0	0.0	0.0	0.0
Oil crops	3.67	74	3.2	1.0	6.2
Vegetables	18.62	47	2.7	5.6	0.5
Fruits	2.14	10	0.2	1.7	0.1

(continued)

Table 2.7: Food Consumption by Food Commodity Group and Region in the First Quintile (continued)

	Average food consumption in monetary value (LCU/person/day)	*Average dietary energy consumption (kcal/person/day)*	*Average protein consumption (g/person/day)*	*Average carbohydrates consumption (g/person/day)*	*Average fat consumption (g/person/day)*
Stimulants	0.65	9	0.1	1.9	0.1
Spices	1.60	0	0.0	0.1	0.0
Alcoholic beverages	3.66	98	0.1	1.4	0.0
Meat	5.31	20	1.5	0.0	1.6
Eggs	0.37	0	0.0	0.0	0.0
Fish	2.89	7	1.3	0.0	0.2
Milk and cheese	5.44	25	1.3	1.8	1.4
Oils and fats (vegetable)	2.52	20	0.0	0.0	2.2
Oils and fats (animal)	0.01	0	0.0	0.0	0.0
Nonalcoholic beverages	0.06	0	0.0	0.0	0.0
Miscellaneous and prepared food	1.96	41	1.2	6.6	0.6
Region 2					
Cereals	46.80	1264	31.9	236.4	15.5
Roots and tubers	3.08	21	0.4	4.7	0.0
Sugars and syrups	4.09	30	0.0	7.6	0.0
Pulses	5.90	35	2.6	4.7	0.2

Table 2.8: Nutrient Costs by Food Commodity Group

	Average dietary energy unit value (LCU/1000 kcal)	*Average protein unit value (LCU/100 g)*	*Average carbohydrates unit value (LCU/100 g)*	*Average fat unit value (LCU/100 g)*
Food group				
Cereals	56.11	228.83	29.19	534.81
Roots and tubers	57.88	567.85	25.40	N/A
Sugars and syrups	142.73	N/A	57.10	
Pulses	169.82	225.32	127.47	N/A
Tree nuts	49.99	146.84	343.23	58.57
Oil crops	61.88	186.93	431.55	71.81
Vegetables	626.42	N/A	462.75	N/A
Fruits	147.45	N/A	66.18	N/A
Stimulants	432.40	N/A	225.33	N/A
Spices	N/A	N/A	N/A	N/A
Alcoholic beverages	68.81	N/A	462.77	
Meat	324.50	433.27	N/A	417.05
Eggs	683.22	814.44	N/A	N/A
Fish	551.73	279.98		N/A
Milk and cheese	294.66	554.64	404.89	534.26
Oils and fats (vegetable)	110.11			99.10
Oils and fats (animal)	169.17	N/A		152.70
Nonalcoholic beverages	N/A	N/A	574.81	
Miscellaneous and prepared food	126.07	469.16	79.66	635.46

Note: N/A: very low or no nutrient content or no consumption.

Table 2.9: Food Consumption by Food Commodity Group and Food Sources (in Dietary Energy)

	Purchases		*Own Production*		*Food consumed away from home*		*Other sources*	
	Average dietary energy consumption (kcal/person/day)	*Share in food commodity group's total consumption (%)*	*Average dietary energy consumption (kcal/person/day)*	*Share in food commodity group's total consumption (%)*	*Average dietary energy consumption (kcal/person/day)*	*Share in food commodity group's total consumption (%)*	*Average dietary energy consumption (kcal/person/day)*	*Share in food commodity group's total consumption (%)*
Food group								
Cereals	593.8	49.4	565.5	47.1		0.0	42.0	3.5
Roots and tubers	90.8	35.2	157.7	61.2		0.0	9.2	3.6
Sugars and syrups	85.5	95.9	0.8	0.9		0.0	2.8	3.2
Pulses	35.2	48.0	35.0	47.7		0.0	3.2	4.3
Tree nuts	1.2	44.5	1.1	40.4		0.0	0.4	15.1
Oil crops	43.0	54.7	30.9	39.3		0.0	4.7	6.0
Vegetables	11.8	39.8	16.7	56.4		0.0	1.1	3.8
Fruits	27.8	40.2	36.4	52.7		0.0	4.9	7.1
Stimulants	3.0	65.7	0.8	18.1		0.0	0.7	16.2
Spices	1.4	88.5	0.1	8.4		0.0	0.0	3.0
Alcoholic beverages	93.9	82.7	6.6	5.8		0.0	13.1	11.5
Meat	57.3	80.4	10.2	14.4		0.0	3.7	5.2
Eggs	0.7	64.1	0.3	30.7		0.0	0.1	5.2
Fish	25.8	94.3	0.7	2.6		0.0	0.9	3.1
Milk and cheese	10.3	42.3	13.1	53.6		0.0	1.0	4.1
Oils and fats (vegetable)	70.9	93.4	3.8	5.1		0.0	1.2	1.6
Oils and fats (animal)	1.6	86.4	0.2	10.9		0.0	0.1	2.7
Nonalcoholic beverages	1.8	89.7	0.0	0.2		0.0	0.2	10.1
Miscellaneous and prepared food		0.0		0.0	74.8	100.0		0.0

Disaggregated by Food Commodity: Tables 3.1 to 3.9

The food commodities analyzed are those collected in the survey excluding those consumed away from home. Therefore, the total protein consumed from the listed commodities is underestimated. The food commodity quantities refer to edible portions, which mean that they exclude the nonedible parts (peels, bones, etc.).

Table 3.1: Consumption Statistics for Each Food Item at National Level This table shows *national* food consumption statistics by food commodities. The statistics comprise food commodity edible quantities and their respective monetary value, the calories they provide, and the calorie costs.

This table is useful to identify the food commodities providing more calories to the households' consumption at the national level and how much an individual living in the country has to pay to acquire those calories. Moreover, it enables one to do a comparison between food availability and food consumption. For instance, one could look for differences between daily calories[13] consumption per person (third column) obtained from NHS and those available obtained from the FBS.

Table 3.2: Food Item Protein Consumption at National Level This table shows *national* food consumption statistics by food commodities. The statistics comprise food commodity edible quantities and their respective monetary value, the amount of protein they provide, and the protein costs.

This table is useful to identify the food commodities providing more protein to the households' consumption at the national level and how much an individual living in the country has to pay to acquire that amount of protein. Moreover, it enables one to do a comparison between food availability as compiled in the FBS and food consumption as collected in NHS. For instance, one could look for differences between daily protein consumption per person (third column) obtained from NHS and those available obtained from the FBS.

Table 3.3: Consumption Statistics for Each Food Item by Area This table shows *urban/rural* food consumption statistics by food commodity. The statistics comprise food commodity edible quantities and their respective monetary value, the calories they provide, and the calorie costs. This table is useful to identify differences between urban and rural areas with respect to calorie consumption patterns and unit values.

Table 3.1: Consumption Statistics for Each Food Item at National Level

	Average edible quantity consumed (g/person/day)	*Average food consumption in monetary value (LCU/person/day)*	*Average dietary energy consumption (kcal/person/day)*	*Average dietary energy unit value (LCU/1000 kcals)*
Food item				
Rice paddy or rough	6.12	0.98	21.37	45.78
Rice husked	48.14	18.65	169.44	110.07
Maize cob fresh	9.29	1.71	6.10	279.42
Maize grain	72.72	7.01	263.81	26.57
Maize flour	163.94	27.63	586.57	47.10
Millet whole grain dried	1.68	0.33	5.34	62.75
Millet foxtail Italian whole grain	1.17	0.43	3.70	116.10
Sorghum whole grain brown	7.91	0.88	28.09	31.35
Sorghum average of all variety	20.38	2.68	72.43	36.97
Wheat durum whole grain	0.46	0.11	1.66	66.89
Wheat meal or flour unspecified wheat	4.41	1.54	14.90	103.16
Wheat	1.30	0.20	4.68	42.37
Bread	3.11	1.65	8.22	201.08
Baby cereals	0.09	0.04	0.33	128.53
Biscuits wheat from Europe	0.15	0.27	0.65	409.68
Buns cakes	3.25	3.10	10.50	295.24
Macaroni spaghetti	0.27	0.16	0.95	171.61
Oats	2.31	1.63	8.61	188.93
Cassava sweet roots raw	28.98	2.96	45.53	64.95
Cassava sweet roots dried	14.31	1.30	45.20	28.84
Cassava flour	35.67	4.18	112.69	37.12
Sweet potato	50.00	3.93	35.65	110.37
Coco yam tuber	5.45	0.65	5.76	112.69
Potatoes tubers raw	9.00	1.58	5.03	314.51
Banana cooking	42.28	5.41	52.93	102.12
Starch	2.17	0.31	7.93	39.20
Sugar refined white	21.67	12.21	86.60	141.02
Honey local product	0.58	0.17	1.92	88.47
Lemon sweet	0.22	0.34	0.62	551.97
Peas dry	4.15	1.13	13.34	84.99
Beans dry	36.36	9.40	38.36	244.94
Lentil seed dried whole	7.39	1.75	20.67	84.75
Pulse product	1.04	0.18	1.02	177.34

Table 3.4: Food Item Protein Consumption by Area This table shows *urban/rural* food consumption statistics by food commodity. The statistics comprise food commodity edible quantities and their respective monetary value, the amount of protein they provide, and the protein costs. This table is useful to identify differences between rural and urban areas with respect to protein consumption and protein unit values.

Table 3.5: Consumption Statistics for Each Food Item by Region This table shows *regional* food consumption statistics by food commodity. The statistics comprise food commodity edible quantities and their respective monetary

Table 3.2: Food Item Protein Consumption at National Level

	Average edible quantity consumed (g/person/day)	*Average food consumption in monetary value (LCU/person/day)*	*Average protein consumption (g/person/day)*	*Average protein unit value (LCU/100 g)*
Food item				
Rice paddy or rough	6.12	0.98	0.40	241.71
Rice husked	48.14	18.65	3.61	516.60
Maize cob fresh	9.29	1.71	0.17	1019.90
Maize grain	72.72	7.01	6.85	102.33
Maize flour	163.94	27.63	13.28	208.07
Millet whole grain dried	1.68	0.33	0.20	171.37
Millet foxtail Italian whole grain	1.17	0.43	0.08	556.05
Sorghum whole grain brown	7.91	0.88	0.89	98.58
Sorghum average of all varieties	20.38	2.68	2.30	116.25
Wheat durum whole grain	0.46	0.11	0.06	176.74
Wheat meal or flour unspecified wheat	4.41	1.54	0.60	254.43
Wheat	1.30	0.20	0.30	66.05
Bread	3.11	1.65	0.27	603.70
Baby cereals	0.09	0.04	0.01	345.18
Biscuits wheat from Europe	0.15	0.27	0.01	1886.31
Buns cakes	3.25	3.10	0.15	2027.08
Macaroni spaghetti	0.27	0.16	0.03	582.82
Oats	2.31	1.63	0.39	417.84
Cassava sweet roots raw	28.98	2.96	0.41	728.87
Cassava sweet roots dried	14.31	1.30	0.37	350.35
Cassava flour	35.67	4.18	0.93	451.02
Sweet potato	50.00	3.93	0.60	655.78
Coco yam tuber	5.45	0.65	0.08	793.35
Potatoes tubers raw	9.00	1.58	0.23	676.19
Banana cooking	42.28	5.41	0.34	1598.24
Starch	2.17	0.31	0.01	4775.90
Sugar refined white	21.67	12.21	0.00	

Table 3.3: Consumption Statistics for Each Food Item by Area

	Average edible quantity consumed (g/person/day)	*Average food consumption in monetary value (LCU/person/day)*	*Average dietary energy consumption (kcal/person/day)*	*Average dietary energy unit value (LCU/1000 kcals)*
Area				
Capital city				
Rice paddy or rough	0.60	0.12	2.11	58.75
Rice husked	108.34	43.98	381.34	115.32
Maize cob fresh	0.68	0.27	0.45	594.81
Maize grain	7.68	1.35	27.86	48.48
Maize flour	125.27	28.35	448.23	63.24
Millet whole grain dried	0.42	0.19	1.32	145.65
Millet foxtail Italian whole grain	0.71	0.49	2.23	220.35
Sorghum whole grain brown	0.13	0.05	0.46	98.48
Sorghum average of all varieties	0.04	0.01	0.15	58.51
Wheat durum whole grain	0.24	0.08	0.87	92.83

(continued)

Table 3.3: Consumption Statistics for Each Food Item by Area (continued)

	Average edible quantity consumed (g/person/day)	Average food consumption in monetary value (LCU/person/day)	Average dietary energy consumption (kcal/person/day)	Average dietary energy unit value (LCU/1000 kcals)
Wheat meal or flour unspecified wheat	10.40	3.22	35.14	91.71
Wheat	0.21	0.09	0.77	115.73
Bread	16.83	9.18	44.46	206.59
Baby cereals	0.04	0.06	0.14	427.80
Biscuits wheat from Europe	0.50	0.78	2.13	365.65
Buns cakes	10.44	9.75	33.69	289.34
Macaroni spaghetti	2.18	1.30	7.79	167.22
Oats	10.69	11.05	39.93	276.67
Cassava sweet roots raw	13.06	2.24	20.52	109.14
Cassava sweet roots dried	1.09	0.13	3.43	38.41
Cassava flour	0.83	0.15	2.64	55.26
Sweet potato	13.23	2.20	9.43	233.14
Coco yam tuber	2.45	0.54	2.59	210.14
Potatoes tubers raw	10.96	3.26	6.12	531.81
Banana cooking	13.80	5.55	17.27	321.50

Table 3.4: Food Item Protein Consumption by Area

	Average edible quantity consumed (g/person/day)	Average food consumption in monetary value (LCU/person/day)	Average protein consumption (g/person/day)	Average protein unit value (LCU/100 g)
Area				
Capital city				
Rice paddy or rough	0.60	0.12	0.04	310.22
Rice husked	108.34	43.98	8.13	541.26
Maize cob fresh	0.68	0.27	0.01	2171.06
Maize grain	7.68	1.35	0.72	186.70
Maize flour	125.27	28.35	10.15	279.37
Millet whole grain dried	0.42	0.19	0.05	397.77
Millet foxtail Italian whole grain	0.71	0.49	0.05	1055.33
Sorghum whole grain brown	0.13	0.05	0.01	309.64
Sorghum average of all varieties	0.04	0.01	0.00	183.97
Wheat durum whole grain	0.24	0.08	0.03	245.30
Wheat meal or flour unspecified wheat	10.40	3.22	1.42	226.19
Wheat	0.21	0.09	0.05	180.40
Bread	16.83	9.18	1.48	620.23
Baby cereals	0.04	0.06	0.01	1148.95
Biscuits wheat from Europe	0.50	0.78	0.05	1683.56
Buns cakes	10.44	9.75	0.49	1986.58
Macaroni spaghetti	2.18	1.30	0.23	567.92
Oats	10.69	11.05	1.81	611.88
Cassava sweet roots raw	13.06	2.24	0.18	1224.71
Cassava sweet roots dried	1.09	0.13	0.03	466.67
Cassava flour	0.83	0.15	0.02	671.35
Sweet potato	13.23	2.20	0.16	1385.23
Coco yam tuber	2.45	0.54	0.04	1479.39
Potatoes tubers raw	10.96	3.26	0.28	1143.39
Banana cooking	13.80	5.55	0.11	5031.45

Table 3.5: Consumption Statistics for Each Food Item by Region

	Average edible quantity consumed (g/person/day)	*Average food consumption in monetary value (LCU/person/day)*	*Average dietary energy consumption (kcal/person/day)*	*Average dietary energy unit value (LCU/1000 kcals)*
Region				
Region 1				
Rice paddy or rough	0.29	0.06	1.02	60.51
Rice husked	26.25	10.63	92.39	115.08
Maize cob fresh	11.73	2.57	7.70	333.78
Maize grain	50.15	4.09	181.92	22.47
Maize flour	240.10	33.01	859.08	38.43
Millet whole grain dried	1.68	0.26	5.32	48.86
Millet foxtail Italian whole grain	1.67	1.11	5.28	210.73
Sorghum whole grain brown	20.07	1.84	71.29	25.80
Sorghum average of all varieties	102.41	12.41	363.88	34.12
Wheat durum whole grain	0.05	0.02	0.19	110.07
Wheat meal or flour unspecified wheat	3.35	1.17	11.30	103.09
Wheat	1.40	0.24	5.05	48.00
Bread	1.07	0.70	2.84	247.73
Baby cereals	0.04	0.01	0.13	52.66
Biscuits wheat from Europe	0.09	0.17	0.40	428.68
Buns cakes	2.69	2.38	8.67	274.66
Macaroni spaghetti	0.12	0.10	0.43	225.61
Oats	1.38	1.28	5.15	249.02
Cassava sweet roots raw	11.73	1.63	18.42	88.51
Cassava sweet roots dried	0.16	0.03	0.50	63.99
Cassava flour	0.29	0.06	0.91	64.02
Sweet potato	22.52	2.32	16.06	144.43
Coco yam tuber	0.24	0.05	0.25	197.18
Potatoes tubers raw	7.74	1.52	4.33	351.12
Banana cooking	2.33	0.52	2.92	179.39

value, the calories they provide, and the calorie costs. This table is useful to identify differences across regions with respect to calorie consumption and unit values.

Table 3.6: Food Item Protein Consumption by Region This table shows *regional* food consumption statistics by food commodity. The statistics comprise food commodity edible quantities and their respective monetary value, the amount of protein they provide, and the protein costs. This table is useful to identify differences across regions with respect to protein consumption and unit values.

Table 3.7: Food Item Quantities by Food Source For food fortification programs, the distinction of the food source plays an important role. Food that is home-produced is assumed not to have been fortified and

Table 3.6: Food Item Protein Consumption by Region

	Average edible quantity consumed (g/person/day)	Average food consumption in monetary value (LCU/person/day)	Average protein consumption (g/person/day)	Average protein unit value (LCU/100 g)
Region				
Region 1				
Rice paddy or rough	0.29	0.06	0	319.53
Rice husked	26.25	10.63	2	540.13
Maize cob fresh	11.73	2.57	0	1218.31
Maize grain	50.15	4.09	5	86.53
Maize flour	240.10	33.01	19	169.75
Millet whole grain dried	1.68	0.26	0	133.43
Millet foxtail Italian whole grain	1.67	1.11	0	1009.28
Sorghum whole grain brown	20.07	1.84	2	81.11
Sorghum average of all varieties	102.41	12.41	12	107.27
Wheat durum whole grain	0.05	0.02	0	290.83
Wheat meal or flour unspecified wheat	3.35	1.17	0	254.26
Wheat	1.40	0.24	0	74.82
Bread	1.07	0.70	0	743.74
Baby cereals	0.04	0.01	0	141.42
Biscuits wheat from Europe	0.09	0.17	0	1973.79
Buns cakes	2.69	2.38	0	1885.79
Macaroni spaghetti	0.12	0.10	0	766.22
Oats	1.38	1.28	0	550.72
Cassava sweet roots raw	11.73	1.63	0	993.23
Cassava sweet roots dried	0.16	0.03	0	777.43
Cassava flour	0.29	0.06	0	777.84
Sweet potato	22.52	2.32	0	858.17
Coco yam tuber	0.24	0.05	0	1388.15
Potatoes tubers raw	7.74	1.52	0	754.90
Banana cooking	2.33	0.52	0	2807.42

Table 3.7: Food Item Quantities by Food Source

	Purchases		Own production		Other sources	
	Quantity "as purchased," g/person/day	Proportion of households in total households (%)	Quantity "as produced," g/person/day	Proportion of households in total households (%)	Quantity "as received," g/person/day	Proportion of households in total households (%)
Food item						
Rice husked	56.1	67.4	63.7	14.4	19.2	8.9
Maize grain	134.6	30.6	82.1	25.9	73.6	5.0
Maize flour	113.3	52.7	189.3	52.2	29.8	11.3
Buns cakes	6.0	51.6	2.8	1.2	1.2	7.2
Oats	5.4	30.7	18.6	2.1	1.4	4.8
Sweet potato	48.7	28.4	124.2	24.0	25.4	6.5
Sugar refined white	28.4	72.5	5.2	1.5	8.6	7.2
Beans dry	30.2	66.8	40.5	34.3	14.6	9.0
Groundnuts shelled	7.8	31.0	15.7	15.5	4.1	5.1
Onion garden	8.8	70.8	5.1	5.8	2.5	5.0

(continued)

Table 3.7: Food Item Quantities by Food Source (continued)

	Purchases		Own production		Other sources	
	Quantity "as purchased," g/person/ day	Proportion of households in total households (%)	Quantity "as produced," g/person/ day	Proportion of households in total households (%)	Quantity "as received," g/person/ day	Proportion of households in total households (%)
Food item						
Spinach raw	12.9	43.4	10.9	24.9	5.6	4.0
Tomato raw ripe whole	23.7	73.8	12.2	11.7	7.5	7.6
Cattle meat	25.5	63.9	18.2	2.4	9.1	7.7
Fish average of all kinds raw	32.3	34.7	24.4	3.0	13.6	4.9
Sardine salted dried	17.0	77.7	4.6	3.8	4.9	8.4
Milk cow fluid whole	39.1	30.8	90.3	9.0	16.2	5.0
Cooking oil other	9.9	43.8	6.7	2.0	2.8	2.4
Salt	12.3	85.4	6.5	2.3	4.2	4.4
Tea common dried black	1.0	47.0	0.9	0.7	0.3	1.9
Soft drinks	15.7	28.7	3.6	0.3	6.4	7.4

not to be fortifiable. In assessing the potential coverage of a fortified or fortifiable food, only food that is purchased should be included in the analysis. Information on the proportion of households purchasing a food commodity is also relevant for food fortification programs. For instance, in some cases processed staple foods (e.g., flour) are purchased by households in larger quantities than the respective unprocessed food commodity (e.g., grain). So, from a fortification policy perspective, the processed food can be as important as, or more important, than the staple (Fiedler 2009).

Table 3.7 shows information for the 20 food commodities most purchased by households at the *national level*. The information comprises (1) the food quantities acquired from purchases (i.e., at the market, from street vendors, at shops, etc.) and the percentage of households that purchased these food quantities; (2) the food quantities from own production and the percentage of households that reported own production; and (3) the food quantities from other sources and the percentage of households that reported other sources.[14]

Note that the sum of the proportion of households that acquired the product through purchase, or received in kind, or from own consumption does not necessarily equal 100 percent because not all households might have consumed the food.

Table 3.8: Food Item Quantities by Food Source and Area This table shows information for the 20 food commodities most purchased by households by *rural and urban* areas separately. The information comprises (1) the food quantities acquired from purchases (i.e., at the market, from street vendors, at shops, etc.) and the percentage of households that purchased these food quantities; (2) the food quantities from own production and the percentage of households that reported own production; and (3) the food quantities from other sources and the percentage of households that reported other sources.[15]

Note that the sum of the proportion of households that acquired the product through purchase, or received in kind, or from own consumption does not necessarily equal 100 percent because not all households might have consumed the food.

Table 3.9: Food Item Quantities by Food Source and Region This table shows information for the 20 food commodities most purchased by households at the *regional* level. The information comprises (1) the food quantities acquired from purchases (i.e., at the market, from street vendors, at shops, etc.) and the percentage of households that purchased these food quantities; (2) the food quantities from own production and the percentage of households that reported own production; and (3) the food quantities from other sources and the percentage of households that reported other sources.[16]

Note that the sum of the proportion of households that acquired the product through purchase, or received in kind, or from own consumption does not necessarily equal 100 percent because not all households might have consumed the food.

Inequality

The dispersion ratios measure inequality between the two extreme income quintile groups. They are calculated using as reference the average values corresponding to the first quintile. The following tables show dispersion ratios related to food and nonfood consumption, as well as between *each income quintile* and the *first quintile*. Since consumption is positively correlated with income, when the *first* income quintile is used as reference, all the ratios are expected to be greater than 1. In this case, a higher ratio value implies higher inequality between the poorest and the richest groups. For instance, a dietary energy dispersion ratio of

Table 3.8: Food Item Quantities by Food Source and Area

	Purchases		*Own production*		*Other sources*	
	Quantity "as purchased," g/person/day	*Proportion of households in total households (%)*	*Quantity "as produced," g/person/day*	*Proportion of households in total households (%)*	*Quantity "as received," g/person/day*	*Proportion of households in total households (%)*
Area						
Capital city						
Rice husked	110.6	90.6	44.9	0.5	43.2	6.9
Maize flour	128.8	91.1	18.4	1.3	35.6	6.9
Bread	20.9	74.0	5.7	0.5	4.1	1.6
Buns cakes	11.7	86.2	1.0	0.8	1.7	5.1
Oats	13.4	78.9	1.8	1.3	1.8	5.2
Potatoes tubers raw	22.8	56.9	3.4	0.1	17.7	1.2
Sugar refined white	44.8	93.2	3.2	1.0	22.0	4.0
Beans dry	32.1	86.6	4.3	0.2	8.4	3.8
Coconut mature kernel	52.7	77.7	44.8	0.6	16.2	3.0
Onion garden common	12.5	88.2	0.5	0.7	3.8	2.8
Spinach raw	18.4	78.8	4.7	1.7	5.6	3.4
Tomato raw ripe whole	45.5	92.2	5.0	1.0	10.9	3.3
Cattle meat	34.2	83.1	16.0	0.1	9.4	2.4
Fish dried	10.7	56.9	5.1	0.1	1.6	0.8
Sardine salted dried	9.3	67.1	12.6	0.3	4.3	2.0
Cooking oil other	15.2	73.0	0.7	0.9	2.4	1.5
Salt	9.9	77.2	3.3	0.3	2.1	1.0
Tea common dried black	1.1	78.2	0.0	0.5	0.1	0.4
Soft drinks	33.4	66.0	3.6	0.3	9.1	12.8
Orange sweet juice fresh	17.8	56.5	0.6	0.1	6.6	6.6
Other urban areas						
Rice husked	81.6	90.2	64.6	6.8	20.6	10.6
Maize flour	116.4	70.6	159.4	25.1	27.5	7.0

Table 3.9: Food Item Quantities by Food Source and Region

	Purchases		*Own production*		*Other sources*	
	Quantity "as purchased," g/person/day	*Proportion of households in total households (%)*	*Quantity "as produced," g/person/day*	*Proportion of households in total households (%)*	*Quantity "as received," g/person/day*	*Proportion of households in total households (%)*
Region						
Region 1						
Rice husked	46.1	54.1	16.8	6.0	9.9	5.2
Maize flour	167.9	35.7	244.5	71.2	29.6	13.7
Buns cakes	5.6	44.3	2.2	1.0	0.6	6.9
Sweet potato	34.7	32.4	49.4	15.1	13.6	6.1
Potatoes tubers raw	26.2	25.4	38.7	4.1	14.6	5.4
Sugar refined white	23.8	59.1	4.8	2.6	7.4	6.5
Beans dry	25.5	63.6	26.1	22.7	8.0	8.2
Groundnuts shelled	12.2	31.5	18.8	34.7	4.9	6.8
Onion garden common	7.2	65.9	9.3	7.1	2.6	8.7
Spinach raw	9.3	32.0	9.2	23.4	4.5	3.5
Tomato raw ripe whole	17.0	59.8	10.6	15.0	2.8	8.0
Other fruit	38.0	27.4	11.0	3.5	21.3	9.5
Cattle meat	22.6	68.4	8.4	4.6	9.0	15.2
Sardine salted dried	8.9	71.4	4.0	2.1	1.8	7.8
Milk cow fluid whole	35.0	25.3	56.0	10.3	21.4	11.3
Yogurt made from whole milk	28.4	34.6	85.5	16.0	28.7	22.7
Oil sunflower seed	6.2	33.9	2.2	0.8	2.0	2.8
Cooking oil other	7.9	46.8	0.5	3.9	2.6	1.7
Salt	12.6	77.9	3.8	2.3	4.5	6.3
Tea common dried black	0.9	30.0	0.3	1.8	0.2	1.4
Region 2						
Rice husked	53.8	70.3	15.3	3.1	14.0	3.8
Maize grain	160.0	67.0	155.5	35.7	105.7	13.9

2 (between the fifth and first income groups) indicates that households belonging to the highest income quintile consume twice as many calories as those in the lowest quintile. As the first income quintile values are used as reference, the dispersion ratio of the first quintile using itself as reference is 1.

Another way to measure inequality in food consumption is with elasticities. An income elasticity of demand is used to measure how sensitive the demand for food consumed is with respect to a change in income. The income elasticity of the demand of food could be measured through the responsiveness of dietary energy, food expenditure, or Engel ratio to a variation in income.

Disaggregated by Population Group: Tables 4.1 to 4.5

Table 4.1: Dispersion Ratio of Food Consumption by Income Quintile within Population Groups This table shows dispersion ratios related to food and nonfood consumption. These dispersion ratios measure the inequality between *each* income quintile and the *first* quintile.

While the amount of calories consumed has a limit due to biological factors, expenditures and income do not. Thus, the dispersion ratios of dietary energy consumption are expected to be smaller than those related to monetary values.

Table 4.2: Dispersion Ratios of Share of Food Consumption (in Dietary Energy) by Food Source, Income Quintile, and Population Groups This table shows the dispersion ratios of the percentage of total dietary energy provided by each of the four sources of food acquisition. These dispersion ratios measure the inequality between *each* income quintile and the *first* quintile.

Table 4.3: Dispersion Ratios of Share of Food Consumption (in Monetary Values) by Food Source and Income Quintile within Population Groups This table shows the dispersion ratios of the percentage of total food expenditure that each of the four sources of food acquisition represents. These dispersion ratios measure the inequality between *each* income quintile and the *first* quintile.

It is expected that the dispersion ratio of the food consumed away from home increases with income in general, because rich people spend more on food outside the home than do poor ones.

Table 4.1: Dispersion Ratio of Food Consumption by Income Quintile within Population Groups

		Average dietary energy consumption (kcal/person/day)		*Average food consumption in monetary value (LCU/person/day)*		*Average total consumption in monetary value (LCU/person/day)*	
	Average household size	*Average*	*Ratio to the first reference group*	*Average*	*Ratio to the first reference group*	*Average*	*Ratio to the first reference group*
Total							
Quintiles of income							
Lowest quintile	6.6	1596.05	1.00	101.64	1.00	142.49	1.00
2	5.6	2043.27	1.28	166.18	1.64	241.69	1.70
3	5.0	2249.74	1.41	218.23	2.15	333.62	2.34
4	4.2	2604.28	1.63	299.82	2.95	480.80	3.37
Highest quintile	3.4	3051.03	1.91	448.89	4.42	812.53	5.70
Area							
Capital city							
Quintiles of income							
Lowest quintile	5.6	938.17	1.00	119.22	1.00	164.03	1.00
2	6.5	1226.18	1.31	176.55	1.48	262.98	1.60
3	5.8	1433.59	1.53	226.33	1.90	378.08	2.30
4	4.5	1921.70	2.05	342.83	2.88	592.30	3.61
Highest quintile	3.5	2793.36	2.98	597.27	5.01	1160.41	7.07
Other urban areas							
Quintiles of income							
Lowest quintile	6.5	1334.91	1.00	110.75	1.00	152.28	1.00
2	5.7	1634.69	1.22	164.35	1.48	244.85	1.61
3	5.2	1858.39	1.39	221.00	2.00	350.09	2.30
4	4.5	2418.62	1.81	308.50	2.79	517.64	3.40
Highest quintile	3.2	2904.60	2.18	491.51	4.44	894.70	5.88

Table 4.4: Dispersion Ratios of Food Dietary Energy Unit Values, Total Income, and Engel Ratio by Income Quintile within Population Groups This table shows dispersion ratios of dietary energy unit value and income. These dispersion ratios measure the inequality between *each* income quintile and the *first* quintile.

Table 4.5: Income Demand Elasticities by Income Decile within Population Groups This table shows values of the demand elasticity of food consumption with respect to income. The demand for food is analyzed in terms of dietary energy, monetary values, and Engel ratio.

The elasticity values can be 0, negative, or positive. A value of 0 means the demand for food consumption is not sensitive to an income change

Table 4.2: Dispersion Ratios of Share of Food Consumption (in Dietary Energy) by Food Source, Income Quintile, and Population Groups

		Share of purchased food in total food consumption (%)		*Share of own produced food in total food consumption (%)*		*Share of food consumed away from home in total food consumption (%)*		*Share of food from other sources in total food consumption (%)*	
	Average household size	*Shares*	*Ratio to the first reference group*	*Shares*	*Ratio to the first reference group*	*Shares*	*Ratio to the first reference group*	*Shares*	*Ratio to the first reference group*
Total									
Quintiles of income									
Lowest quintile	6.6	40.31	1.00	53.34	1.00	1.82	1.00	4.54	1.00
2	5.6	42.01	1.04	51.83	0.97	2.11	1.16	4.06	0.89
3	5.0	51.34	1.27	41.80	0.78	2.62	1.44	4.25	0.94
4	4.2	62.10	1.54	30.64	0.57	3.37	1.86	3.88	0.86
Highest quintile	3.4	67.84	1.68	21.44	0.40	7.19	3.96	3.54	0.78
Area									
Capital city									
Quintiles of income									
Lowest quintile	5.6	87.24	1.00	3.89	1.00	3.53	1.00	5.34	1.00
2	6.5	87.92	1.01	0.98	0.25	7.01	1.98	4.10	0.77
3	5.8	92.12	1.06	0.33	0.08	6.03	1.71	1.51	0.28
4	4.5	87.59	1.00	0.06	0.02	10.61	3.00	1.73	0.32
Highest quintile	3.5	80.00	0.92	0.48	0.12	17.36	4.91	2.17	0.41
Other urban areas									
Quintiles of income									
Lowest quintile	6.5	76.84	1.00	17.09	1.00	1.97	1.00	4.10	1.00
2	5.7	73.27	0.95	19.53	1.14	2.57	1.30	4.63	1.13
3	5.2	78.68	1.02	15.65	0.92	2.32	1.18	3.34	0.82
4	4.5	81.11	1.06	12.29	0.72	4.16	2.11	2.43	0.59
Highest quintile	3.2	81.68	1.06	7.97	0.47	7.68	3.90	2.66	0.65

(i.e., that the demand for food consumption is income inelastic). When the value is negative, it means that the demand for the current food consumed decreases with an increase of income. A positive value could be classified into less than 1 (necessary foods) or more than 1 (luxurious foods) and means that an increase in income would increase the demand for food consumption.

As far as dietary energy is concerned, small values of calorie-income elasticity suggest that an increase in income would not affect much of the calorie intake, but it may improve the quality of the diet consumed by moving from cheap to more expensive food. On the other hand, Engel's law states that given a set of tastes and preferences, an increase in income will

Table 4.3: Dispersion Ratios of Share of Food Consumption (in Monetary Values) by Food Source and Income Quintile within Population Groups

		Share of purchased food in total food consumption (%)		*Share of own produced food in total food consumption (%)*		*Share of food consumed away from home in total food consumption (%)*		*Share of food from other sources in total food consumption (%)*	
	Average household size	*Shares*	*Ratio to the first reference group*	*Shares*	*Ratio to the first reference group*	*Shares*	*Ratio to the first reference group*	*Shares*	*Ratio to the first reference group*
Total									
Quintiles of income									
Lowest quintile	6.6	50.15	1.00	42.89	1.00	1.84	1.00	5.12	1.00
2	5.6	53.50	1.07	40.11	0.94	2.19	1.19	4.20	0.82
3	5.0	62.04	1.24	30.88	0.72	2.76	1.50	4.33	0.85
4	4.2	71.56	1.43	20.92	0.49	3.79	2.06	3.73	0.73
Highest quintile	3.4	76.58	1.53	12.06	0.28	7.96	4.32	3.40	0.66
Area									
Capital city									
Quintiles of income									
Lowest quintile	5.6	88.97	1.00	3.10	1.00	3.46	1.00	4.47	1.00
2	6.5	88.32	0.99	1.49	0.48	7.32	2.12	2.88	0.64
3	5.8	91.80	1.03	0.32	0.10	6.09	1.76	1.79	0.40
4	4.5	87.64	0.99	0.12	0.04	10.69	3.09	1.55	0.35
Highest quintile	3.5	80.22	0.90	0.47	0.15	16.22	4.69	3.08	0.69

correspond to an increase in food expenditure, but at a slower rate than that of income. Regarding the Engel ratio, the proportion of income dedicated to acquiring food decreases with an increase in income.

On the whole, the elasticity of food consumption with respect to income is higher for lower income groups of the population than for higher ones. However, the elasticity of food in dietary energy terms with respect to income is lower than its elasticity in monetary terms. In other words, for higher income groups the variation of dietary energy consumption due to a variation in income is lower than the variation of food expenditure with respect to the same income variation.

Availability of Micronutrients

The micronutrients analyzed in the ADePT-Food Security Module are vitamin A, ascorbic acid, thiamine (B1), riboflavin (B2), B6, and cobalamin (B12), as well as the minerals calcium and iron. It is important to remember

Table 4.4: Dispersion Ratios of Food Dietary Energy Unit Values, Total Income, and Engel Ratio by Income Quintile within Population Groups

	Average income (LCU/person/day)		Average dietary energy unit value (LCU/1000 kcals)		Share of food consumption in total income (%) (Engel ratio)
	Mean	*Ratio to the first reference group*	*Shares*	*Ratio to the first reference group*	
Total					
Quintiles of income					
Lowest quintile	152.11	1.00	63.68	1.00	66.8
2	265.79	1.75	81.33	1.28	62.5
3	395.28	2.60	97.00	1.52	55.2
4	614.73	4.04	115.12	1.81	48.8
Highest quintile	1898.29	12.48	147.13	2.31	23.6
Area					
Capital city					
Quintiles of income					
Lowest quintile	172.46	1.00	127.08	1.00	69.1
2	271.33	1.57	143.99	1.13	65.1
3	396.97	2.30	157.87	1.24	57.0
4	645.43	3.74	178.40	1.40	53.1
Highest quintile	2034.94	11.80	213.82	1.68	29.4
Other urban areas					
Quintiles of income					
Lowest quintile	160.34	1.00	82.97	1.00	69.1
2	268.20	1.67	100.54	1.21	61.3
3	404.36	2.52	118.92	1.43	54.7
4	633.34	3.95	127.55	1.54	48.7
Highest quintile	2208.29	13.77	169.22	2.04	22.3

that the statistics on micronutrients shown in the tables exclude food consumed away from home. Therefore, the statistics of total available vitamins and minerals are underestimated.

In the tables, the available amount of micronutrients (derived from national household survey data) is compared to the estimated average requirement (EAR) and recommended nutrient intake (RNI)[17] through ratios. The available amount is the numerator, and the EAR or RNI are the denominators of the ratios. For instance, if the ratio of vitamin A available to vitamin A required is 100 percent, we could expect (under equal distribution of vitamin A within the population) that half of the healthy individuals in the population meet its required level of vitamin A. As for the ratio of availability to recommend safe intake, if it is greater than 100 percent (under equal distribution of vitamin A within the population) we could expect that almost all apparently healthy individuals meet their requirement.

Table 4.5: Income Demand Elasticities by Income Decile within Population Groups

	Average income (LCU/person/day)	*Demand elasticity of food in dietary energy consumption*	*Demand elasticity of food consumption in monetary value*	*Demand elasticity of the share of food consumption in monetary value*
Total				
Deciles of income				
1	122.00	0.36	2.28	0.91
2	188.26	0.31	1.15	0.91
3	240.31	0.29	0.90	0.90
4	294.60	0.27	0.76	0.90
5	355.73	0.26	0.66	0.90
6	436.89	0.25	0.58	0.90
7	539.38	0.24	0.52	0.90
8	698.55	0.22	0.46	0.89
9	1001.65	0.21	0.39	0.89
10	2927.19	0.17	0.28	0.87
Area				
Capital city				
Deciles of income				
1	127.09	1.20	5.35	0.86
2	185.05	0.83	1.78	0.85
3	243.00	0.68	1.20	0.85
4	300.12	0.59	0.96	0.84
5	357.19	0.54	0.82	0.84
6	439.84	0.48	0.70	0.83
7	540.86	0.44	0.61	0.82
8	720.97	0.39	0.52	0.81
9	1030.48	0.34	0.44	0.80
10	2999.53	0.25	0.30	0.75

If the mean micronutrient intake is equal or exceeds mean micronutrient requirements, it cannot be concluded that group diets (group mean intakes, not individual diets) were adequate and conformed to recognized nutritional standards. The reason is that the prevalence of inadequacy depends on the shape and variation of the usual intake distribution, not on mean intake. If the mean intake equals the EAR, it is likely that a very high proportion of the population will have inadequate usual intake. In fact, roughly half of the population is expected to have intakes less than its requirement (except for energy). (NAS 2000)

Disaggregated by Population Group: Tables 5.1 to 5.7

Table 5.1: Availability of Vitamin A This table shows the daily per person retinol, beta-carotene, and vitamin A available for human consumption. It also shows the vitamin A estimated average requirement and recommended nutrient intake for a representative individual of the population of analysis.

Table 5.1: Availability of Vitamin A

	Average vitamin A availability (mcg RAE/ person/day)	*Vitamin A mean requirement (mcg RAE/ person/day)*	*Ratio of vitamin A available to required (%)*	*Vitamin A recommended safe intake (mcg RAE/ person/day)*	*Ratio of vitamin A available to recommended (%)*	*Average retinol availability (mcg/person/ day)*	*Ratio of retinol available to vitamin A available (%)*	*Average beta-carotene availability (mcg/person/ day)*
Total	717	279	257	527	136	22	3.0	8320
Quintiles of income								
Lowest quintile	682	276	247	522	131	12	1.8	8011
2	705	279	252	526	134	18	2.6	8211
3	714	280	255	528	135	21	2.9	8285
4	785	281	279	531	148	28	3.6	9048
Highest quintile	729	282	259	535	136	39	5.3	8247
Area								
Capital city	313	285	110	537	58	22	6.9	3487
Other urban areas	581	283	205	531	109	23	3.9	6660
Rural areas	770	278	277	526	146	22	2.8	8953
Household size								
One person	797	291	274	569	140	36	4.5	9107
Between 2 and 3 people	766	281	272	536	143	27	3.5	8843
Between 4 and 5 people	677	273	248	520	130	22	3.2	7837
Between 6 and 7 people	711	279	255	525	135	19	2.6	8280
More than 7	731	283	258	529	138	21	2.9	8482
Gender of the household head								
Male	694	279	249	528	131	22	3.1	8035
Female	820	282	291	524	156	22	2.7	9556

Vitamin A is an essential nutrient needed in small amounts by humans for the normal functioning of vision, growth and development, maintenance of epithelial cellular integrity, immune system functioning, and reproduction (FAO/WHO 2004). It can be found in food of animal origin or under the form of a precursor of vitamin A in vegetal origin food. Low intake of vitamin A (as carotenoids) tends to reflect low intake of fruits and vegetables (USHHS/USDA 2005). The main consequence of vitamin A deficiency is night blindness, which can develop into irreversible blindness.

Table 5.2: Availability of B Vitamins This table shows daily per person quantities of the vitamins B1 (thiamine), B2 (riboflavin), B6, and B12 (cobalamin) available for human consumption. It also shows their estimated average requirement and the cobalamin recommended nutrient intake for a representative individual of the population of analysis.

Thiamine deficiency occurs when the diet consists mainly of milled white cereals, including polished rice and wheat flour, all of which are very poor sources of thiamine (WHO/UNHCR 1999). A deficiency of thiamine results in the disease beriberi, which provokes damage to the nervous system, heart failure, and gastrointestinal illnesses. A deficiency of riboflavin, which is present in a wide variety of food, results in the condition of hypo- or ariboflavinosis. The major cause of the former is inadequate dietary intake, which is sometimes exacerbated by poor food storage or processing (FAO/WHO 2004).

Vitamin B6 can be found in a wide variety of food, and its deficiency results in an impairment of the immune system. “A deficiency of vitamin B6 alone is uncommon because it usually occurs in association with a deficit in other B-complex vitamins” (FAO/WHO 2004).

A deficiency in cobalamin could cause the autoimmune disease pernicious anemia. “Products from herbivorous animals, such as milk, meat, and eggs, constitute important dietary sources of cobalamin, unless the animal is subsisting in one of the many regions known to be geochemically deficient in cobalt” (FAO/WHO 2004).

Table 5.3: Availability of Vitamin C and Calcium This table shows the daily per person amount of vitamin C (ascorbic acid) and calcium available for human consumption, their recommended intake for a representative individual of the population of analysis, and the ratios between available and recommended quantities.

Table 5.2: Availability of B Vitamins

	Average vitamin B1 availability (mg/person/day)	*Vitamin B1 recommended intake (mg/person/day)*	*Ratio vitamin B1 available to recommended (%)*	*Average vitamin B2 availability (mg/person/day)*	*Vitamin B2 recommended intake (mg/person/day)*	*Ratio vitamin B2 available to recommended (%)*	*Average vitamin B6 availability (mg/person/day)*	*Vitamin B6 recommended intake (mg/person/day)*	*Ratio vitamin B6 available to recommended (%)*	*Average vitamin B12 availability (mcg/person/day)*	*Vitamin B12 average requirement (mcg/person/day)*	*Ratio vitamin B12 available to required (%)*	*Vitamin B12 recommended intake (mcg/person/day)*	*Ratio vitamin B12 available to recommended (%)*
Total	2.13	0.98	217	1.75	1.01	174	2.31	1.11	207	1.63	1.68	97	2.03	80
Quintiles of income														
Lowest quintile	1.86	0.95	195	1.32	0.97	136	1.90	1.08	177	0.97	1.63	60	1.97	49
2	2.20	0.98	225	2.25	1.00	225	2.22	1.11	200	1.33	1.67	80	2.01	66
3	2.16	0.98	220	1.70	1.01	169	2.37	1.11	213	1.69	1.69	100	2.03	83
4	2.28	1.00	229	1.73	1.02	169	2.59	1.14	228	2.04	1.71	119	2.06	99
Highest quintile	2.29	1.02	224	1.85	1.05	176	2.78	1.16	240	2.77	1.75	158	2.11	132
Area														
Capital city	1.41	1.03	137	1.19	1.05	113	1.49	1.16	128	1.79	1.77	102	2.12	84
Other urban areas	1.80	1.00	180	1.43	1.02	139	2.15	1.13	189	1.76	1.72	102	2.07	85
Rural areas	2.23	0.97	229	1.85	1.00	185	2.39	1.11	216	1.59	1.67	95	2.01	79
Household size														
One person	2.87	1.16	247	2.37	1.22	195	2.77	1.40	198	3.58	2.00	179	2.39	149
Between 2 and 3 people	2.40	1.04	231	1.99	1.07	186	2.62	1.21	215	2.20	1.80	122	2.16	102
Between 4 and 5 people	2.19	0.96	228	2.10	0.99	214	2.30	1.08	212	1.72	1.64	105	1.98	87
Between 6 and 7 people	2.17	0.96	225	1.50	0.98	152	2.31	1.08	213	1.40	1.65	85	1.99	71
More than 7	1.88	0.98	193	1.52	1.00	152	2.15	1.10	195	1.36	1.67	82	2.01	68
Gender of the household head														
Male	2.13	0.98	217	1.74	1.01	172	2.31	1.11	208	1.62	1.68	97	2.02	80
Female	2.11	0.99	215	1.81	1.00	182	2.29	1.13	203	1.66	1.71	97	2.06	80

Table 5.3: Availability of Vitamin C and Calcium

	Average vitamin C availability (mg/person/day)	*Vitamin C recommended intake (mg/person/day)*	*Ratio vitamin C available to recommended (%)*	*Average calcium availability (mg/person/day)*	*Calcium recommended intake (mg/person/day)*	*Ratio calcium available to recommended (%)*
Total	92.42	39.57	233.58	295.91	747.00	39.61
Quintiles of income						
Lowest quintile	77.54	38.83	199.69	215.85	739.24	29.20
2	92.20	39.34	234.40	286.70	748.71	38.29
3	94.11	39.60	237.63	301.06	749.28	40.18
4	100.28	40.02	250.58	340.82	752.37	45.30
Highest quintile	109.37	40.75	268.39	403.01	749.18	53.79
Area						
Capital city	50.46	40.70	124.00	279.42	760.71	36.73
Other urban areas	66.81	40.06	166.76	251.53	756.31	33.26
Rural areas	99.83	39.40	253.39	304.71	744.41	40.93
Household size						
One person	109.56	44.92	243.92	514.17	762.83	67.40
Between 2 and 3 people	103.95	41.72	249.18	373.42	749.59	49.82
Between 4 and 5 people	92.29	39.30	234.84	303.94	727.36	41.79
Between 6 and 7 people	93.23	38.92	239.54	270.75	748.61	36.17
More than 7	85.88	39.07	219.81	263.52	760.00	34.67
Gender of the household head						
Male	92.44	39.52	233.89	291.44	742.78	39.24
Female	92.31	39.75	232.23	315.23	765.26	41.19
Age of the household head						
Less than 35	92.49	38.82	238.24	298.52	696.02	42.89
Between 35 and 45	89.91	39.07	230.14	278.70	748.63	37.23
Between 46 and 60	92.93	40.12	231.61	311.90	774.21	40.29
More than 60	96.70	40.87	236.58	297.31	780.73	38.08

Ascorbic acid is an antioxidant, and it is found in many fruits and vegetables. The vitamin C content of food is strongly influenced by many factors, including transport to market, storage, and cooking practices. A common feature of vitamin C deficiency is anemia, because ascorbic acid is a promoter of nonheme iron absorption (FAO/WHO 2004). It is not possible to relate servings of fruits and vegetables to an exact amount of vitamin C, but the WHO dietary goal of 400g of fruits and vegetables consumed per day (five portions of them) is aimed at providing sufficient vitamin C to meet the 1970 FAO/WHO guidelines (FAO/WHO 2004).

Low intake of calcium tends to reflect low intake of milk and milk products (USHHS/USDA 2005). There is a wide variation in calcium intake between countries, generally following the animal protein intake and depending largely on dairy product consumption. Calcium salts provide rigidity to the skeleton, and calcium ions play a role in many, if not most, metabolic processes (FAO/WHO 2004). The populations at risk of calcium deficiency comprise children during the first two years of life, puberty, and adolescence; pregnant, lactating, and postmenopausal women; and, possibly, elderly men (FAO/WHO 2004).

Table 5.4: Availability of Iron This table shows daily per person iron availability for human consumption according to its source (animal or nonanimal origin) and its status (heme or nonheme). The food commodities considered animal origin are meat (red and white), fish, eggs, milk, and cheese. It also shows the median and the 95th percentile[18] daily requirements of total iron intake for a representative individual of the population.

Table 5.4: Availability of Iron

	Average iron availability from animal sources (mg/person/day)	*Average iron availability from nonanimal sources (mg/person/day)*	*Average heme iron availability (mg/person/day)*	*Average nonheme iron availability (mg/person/day)*	*Median of the average absolute iron intake required (mg/person/day)*	*95th percentile of the average absolute iron intake required (mg/person/day)*
Total	0.29	16.14	0.11	16.33	1.09	1.59
Quintiles of income						
Lowest quintile	0.17	13.45	0.07	13.56	1.06	1.52
2	0.24	16.58	0.09	16.73	1.10	1.57
3	0.30	16.44	0.11	16.64	1.10	1.60
4	0.36	17.55	0.13	17.79	1.11	1.62
Highest quintile	0.50	18.47	0.18	18.80	1.12	1.69
Area						
Capital city	0.35	11.13	0.12	11.36	1.17	1.74
Other urban areas	0.30	13.79	0.10	13.99	1.13	1.68
Rural areas	0.29	16.91	0.11	17.09	1.08	1.56
Household size						
One person	0.65	19.81	0.24	20.22	1.11	1.66
Between 2 and 3 people	0.39	18.89	0.14	19.14	1.12	1.73
Between 4 and 5 people	0.32	16.53	0.11	16.74	1.06	1.57
Between 6 and 7 people	0.25	15.47	0.09	15.63	1.09	1.55
More than 7	0.25	14.97	0.09	15.12	1.11	1.58
Gender of the household head						
Male	0.29	16.06	0.11	16.24	1.08	1.57
Female	0.29	16.53	0.10	16.72	1.13	1.68
Age of the household head						
Less than 35	0.33	16.68	0.12	16.89	1.02	1.55
Between 35 and 45	0.29	15.54	0.10	15.73	1.11	1.61
Between 46 and 60	0.28	16.38	0.10	16.56	1.13	1.64
More than 60	0.26	16.05	0.10	16.21	1.10	1.54

Iron has several vital functions in the body, including the transportation of oxygen to the tissues from the lungs by red blood cell hemoglobin (WHO 2004). There are two kinds of iron compounds in the diet with respect to the mechanism of absorption: heme iron (derived from hemoglobin and myoglobin) and nonheme iron (derived mainly from cereals, fruits, and vegetables). Heme iron forms a relatively minor part of iron intake. Even in diets with high meat content it accounts for only 10–15 percent of the total iron intake. Diets in developing countries usually contain negligible amounts of heme iron. Nonheme iron is thus the main source of dietary iron (Hallberg 1981).

Table 5.5: Density of Calcium per 1,000 Kcal This table shows the nutrient density[19] of calcium (mg/1,000 kcal) present in the food consumed by the population, the recommended intake, and the ratio between available and recommended. The first is estimated based on calcium and calorie consumption (using the food consumption data from the survey), and the second is based

Table 5.5: Density of Calcium per 1,000 Kcal

	Average calcium availability (mg/1000 kcal)	*Calcium recommended intake (mg/1000 kcal)*	*Ratio calcium available to recommended (%)*	*Average dietary energy requirement (kcal/person/day)*
Total	139	355	39	2106
Quintiles of income				
Lowest quintile	138	366	38	2018
2	143	360	40	2080
3	137	356	39	2107
4	135	349	39	2155
Highest quintile	142	332	43	2254
Area				
Capital city	156	337	46	2256
Other urban areas	121	349	35	2166
Rural areas	141	357	40	2085
Household size				
One person	171	287	59	2654
Between 2 and 3 people	145	328	44	2285
Between 4 and 5 people	139	351	40	2071
Between 6 and 7 people	133	367	36	2041
More than 7	138	366	38	2078
Gender of the household head				
Male	137	351	39	2117
Female	149	372	40	2059
Age of the household head				
Less than 35	135	344	39	2024
Between 35 and 45	134	361	37	2072
Between 46 and 60	145	354	41	2189
More than 60	146	361	41	2162

on the recommended calcium intake and the average dietary energy requirement.[20] The ratio compares the available and the recommended amounts and can be used to understand if (and to what extent) the amounts available are above or below the requirements.

The notion of nutrient density is helpful for defining food-based dietary guidelines (FBDG) and evaluating the adequacy of diets. Unlike recommended intakes, FBDG can be used to educate the public through the mass media and provide a practical guide to selecting foods by defining dietary adequacy (WHO 2004).

Table 5.6: Density of Vitamin A and Vitamin C per 1,000 Kcal This table shows the nutrient density[21] of vitamins A and C (mcg retinol activity equivalent [RAE] or mg/1,000 kcal) present in the food consumed by the population, the respective required densities, and the available to recommended ratios.

The nutrient density in the food consumed is estimated based on the nutrient and calorie consumption. The required nutrient density considers the estimated nutrient average requirement,[22] while the recommended one uses the recommended nutrient intake. Both required and recommended nutrient densities are based on the average dietary energy requirement.

The notion of nutrient density is helpful for defining food-based dietary guidelines and evaluating the adequacy of diets. Unlike recommended intakes, FBDG can be used to educate the public through the mass media and provide a practical guide to selecting foods by defining dietary adequacy (WHO 2004).

Table 5.7: Density of B Vitamins per 1,000 Kcal This table shows the nutrient density[23] of vitamins B1, B2, and B6 (in mg/1,000 kcal), and B12 (in mcg/1,000 kcal) present in the food consumed by the population. It also shows the required densities of the vitamins, the recommended one for vitamin B12, and the respective available to recommended ratios.

The nutrient density (grams of nutrient per 1,000 kcal) of the food consumed is estimated based on the nutrient and calorie consumption. Similarly, the required and recommended nutrient densities are calculated using the estimated nutrient average requirement and recommended nutrient intake, respectively. In both cases the 1,000 required calories are based on the average dietary energy requirements for a representative individual of the population.[24] For instance, the protein density of the food consumed refers to how many grams of protein are consumed per 1,000 calories consumed.

Table 5.6: Density of Vitamin A and Vitamin C per 1,000 Kcal

	Average vitamin A availability (mcg RAE/1000 kcal)	*Vitamin A mean requirement, mcg RAE/1000 kcal*	*Ratio of vitamin A available to required (%)*	*Vitamin A recommended safe intake, mcg RAE/1000 kcal*	*Ratio of vitamin A available to recommended (%)*	*Average vitamin C availability (mg/1000 kcal)*	*Vitamin C recommended safe intake, mg/1000 kcal*	*Ratio vitamin C available to recommended (%)*
Total	338	133	255	250	135	43	19	231
Quintiles of income								
Lowest quintile	435	137	318	259	168	49	19	257
2	352	134	262	253	139	46	19	244
3	326	133	245	250	130	43	19	229
4	312	131	239	246	127	40	19	215
Highest quintile	257	125	206	237	108	39	18	214
Area								
Capital city	175	126	138	238	74	28	18	156
Other urban areas	280	131	214	245	114	32	18	174
Rural areas	357	133	267	252	141	46	19	245
Household size								
One person	264	110	241	214	123	36	17	215
Between 2 and 3 people	298	123	242	234	127	40	18	221
Between 4 and 5 people	310	132	236	251	124	42	19	223
Between 6 and 7 people	349	137	255	257	136	46	19	240
More than 7	384	136	281	255	151	45	19	240
Gender of the household head								
Male	326	132	248	249	131	43	19	233
Female	388	137	284	255	153	44	19	226
Age of the household head								
Less than 35	342	130	263	252	136	42	19	219
Between 35 and 45	326	135	241	254	128	43	19	229
Between 46 and 60	316	132	240	245	129	43	18	236
More than 60	398	134	298	252	158	48	19	252

Table 5.7: Density of B Vitamins per 1,000 Kcal

	Average vitamin B1 availability (mg/1000 kcal)	*Vitamin B1 recommended safe intake, mg/1000 kcal*	*Ratio vitamin B1 available to recommended (%)*	*Average vitamin B2 availability (mg/1000 kcal)*	*Vitamin B2 recommended safe intake, mg/1000 kcal*	*Ratio vitamin B2 available to recommended (%)*	*Average vitamin B6 availability (mg/1000 kcal)*	*Vitamin B6 recommended safe intake, mg/1000 kcal*	*Ratio vitamin B6 available to recommended (%)*	*Average vitamin B12 availability (mcg/1000 kcal)*	*Vitamin B12 average requirement, mcg/1000 kcal*	*Ratio vitamin B12 available to required (%)*	*Vitamin B12 recommended safe intake (mcg/1000 kcal)*	*Ratio vitamin B12 available to recommended (%)*
Total	1.00	0.47	215	0.82	0.48	172	1.08	0.53	205	0.76	0.80	96	0.96	80
Quintiles of income														
Lowest quintile	1.18	0.47	251	0.84	0.48	175	1.21	0.53	228	0.62	0.81	77	0.97	64
2	1.10	0.47	234	1.13	0.48	234	1.11	0.53	208	0.67	0.80	83	0.97	69
3	0.99	0.47	212	0.78	0.48	163	1.08	0.53	205	0.77	0.80	96	0.96	80
4	0.91	0.46	196	0.69	0.48	145	1.03	0.53	195	0.81	0.80	102	0.96	85
Highest quintile	0.81	0.45	179	0.65	0.47	140	0.98	0.51	191	0.98	0.78	126	0.94	105
Area														
Capital city	0.79	0.45	173	0.67	0.47	142	0.83	0.51	161	1.00	0.78	128	0.94	106
Other urban areas	0.87	0.46	188	0.69	0.47	145	1.03	0.52	198	0.85	0.80	107	0.96	89
Rural areas	1.04	0.47	222	0.86	0.48	179	1.11	0.53	209	0.74	0.80	92	0.96	76
Household size														
One person	0.95	0.44	218	0.79	0.46	171	0.92	0.53	175	1.19	0.75	158	0.90	132
Between 2 and 3 people	0.93	0.46	205	0.77	0.47	165	1.02	0.53	191	0.85	0.79	108	0.95	90
Between 4 and 5 people	1.00	0.46	216	0.96	0.48	203	1.05	0.52	202	0.79	0.79	100	0.96	83
Between 6 and 7 people	1.06	0.47	226	0.74	0.48	153	1.13	0.53	214	0.69	0.81	85	0.97	71
More than 7	0.99	0.47	210	0.80	0.48	166	1.13	0.53	213	0.72	0.81	89	0.97	74
Gender of the household head														
Male	1.00	0.46	216	0.82	0.48	171	1.09	0.52	207	0.76	0.79	96	0.95	80
Female	1.00	0.48	209	0.86	0.48	178	1.08	0.55	198	0.79	0.83	94	1.00	78

Similarly, the nutrient density required/recommended of protein are the grams of protein required/recommended per 1,000 calories required.

The notion of nutrient density is helpful for defining food-based dietary guidelines and evaluating the adequacy of diets. Unlike recommended intakes, FBDG can be used to educate the public through the mass media and provide a practical guide to selecting foods by defining dietary adequacy (WHO 2004).

Disaggregated by Food Commodity Group: Tables 6.1 to 6.6

The micronutrients analyzed in the ADePT-Food Security Module are vitamin A, ascorbic acid, thiamine, riboflavin, B6, cobalamin, and the minerals calcium and iron. It is important to remember that the statistics shown in the tables exclude the food consumed away from home. Therefore, the values of total available vitamins and minerals are underestimated.

Table 6.1: Micronutrient Availability by Food Group This table shows how much each food commodity group contributes, in quantitative terms, to the total micronutrient availability at the *national level*. Each time N/A replaces a nutrient quantity, it means that the amount of nutrient available from the food commodity group is very low or null, or there was no acquisition of that food group.

Table 6.2: Micronutrient Availability by Food Group and Income Quintile This table shows how much each food commodity group contributes, in quantitative terms, to the total micronutrient availability in *each income quintile group*. Each time N/A replaces a nutrient quantity, it means that the amount of nutrient available from the food commodity group is very low or null, or there was no acquisition of that food group.

Table 6.3: Micronutrient Availability by Food Group and Area This table shows how much each food commodity group contributes, in quantitative terms, to the total micronutrient availability in *urban and rural areas*. Each time N/A replaces a nutrient quantity, it means that the amount of nutrient available from the food commodity group is very low or null, or there was no acquisition of that food group.

Table 6.4: Micronutrient Availability by Food Group and Region This table shows how much each food commodity group contributes, in quantitative terms, to the total micronutrient availability in *each region*. Each time N/A

Table 6.1: Micronutrient Availability by Food Group

	Average micronutrient availability												
	RAE of vitamin A (mcg/ person/ day)	*Retinol (mcg/ person/ day)*	*Beta-carotene (mcg/ person/ day)*	*Vitamin B1 (mg/ person/ day)*	*Vitamin B2 (mg/ person/ day)*	*Vitamin B6 (mg/ person/ day)*	*Vitamin B12 (mcg/ person/ day)*	*Vitamin C (mg/ person/ day)*	*Calcium (mg/ person/ day)*	*Animal iron (mg/ person/ day)*	*Nonanimal iron (mg/ person/ day)*	*Heme iron (mg/ person/ day)*	*Nonheme iron (mg/ person/ day)*
Food group													
Cereals	29.23	1.11	312.05	1.26	0.55	1.28	0.01	0.00	44.16	N/A	10.25	N/A	10.25
Roots and tubers	535.47	N/A	6425.59	0.23	0.14	0.51	N/A	52.23	42.72	N/A	1.54	N/A	1.54
Sugars and syrups	0.01	N/A	0.08	N/A	0.14	N/A	N/A	0.02	0.30	N/A	0.03	N/A	0.03
Pulses	1.04	N/A	12.56	0.14	0.06	0.09	N/A	0.37	17.17	N/A	1.75	N/A	1.75
Tree nuts	N/A	N/A	N/A	0.00	0.00	0.00	N/A	0.00	1.20	N/A	0.02	N/A	0.02
Oil crops	0.02	N/A	0.21	0.03	0.01	0.03	N/A	0.22	12.99	N/A	0.62	N/A	0.62
Vegetables	84.39	N/A	1011.89	0.37	0.67	0.11	N/A	14.65	58.88	N/A	1.23	N/A	1.23
Fruits	46.11	N/A	551.57	0.06	0.05	0.12	N/A	24.70	5.62	N/A	0.31	N/A	0.31
Stimulants	0.30	0.30	N/A	0.00	0.00	0.00	0.00	0.01	0.89	N/A	0.04	N/A	0.04
Spices	0.53	N/A	6.01	0.00	0.00	0.00	N/A	0.13	7.51	N/A	0.08	N/A	0.08
Alcoholic beverages	N/A	N/A	N/A	0.00	0.00	0.05	N/A	N/A	2.41	N/A	0.00	N/A	0.00
Meat	1.20	1.20	N/A	0.03	0.04	0.07	0.35	0.00	1.33	0.08	0.27	0.04	0.31
Eggs	1.27	1.27	N/A	0.00	0.00	0.00	0.01	N/A	0.38	0.01	N/A	N/A	0.01
Fish	5.49	5.49	N/A	0.01	0.02	0.04	1.10	0.02	54.76	0.18	N/A	0.06	0.12
Milk and cheese	11.38	11.38	N/A	N/A	0.06	0.00	0.16	0.07	45.41	0.03	N/A	N/A	0.03
Oils and fats (vegetable)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.00	N/A	N/A	0.00	N/A	0.00
Oils and fats (animal)	1.03	1.03	N/A	N/A	0.00	N/A	0.00	N/A	0.03	N/A	N/A	N/A	N/A
Nonalcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.15	N/A	N/A	N/A	N/A
Miscellaneous and prepared food													

Table 6.2: Micronutrient Availability by Food Group and Income Quintile

	Average micronutrient availability												
	RAE of vitamin A (mcg/ person/ day)	Retinol (mcg/ person/ day)	Beta-carotene (mcg/ person/ day)	Vitamin B1 (mg/ person/ day)	Vitamin B2 (mg/ person/ day)	Vitamin B6 (mg/ person/ day)	Vitamin B12 (mcg/ person/ day)	Vitamin C (mg/ person/ day)	Calcium (mg/ person/ day)	Animal iron (mg/ person/ day)	Nonanimal iron (mg/ person/ day)	Heme iron (mg/ person/ day)	Nonheme iron (mg/ person/ day)
Quintiles of income													
Lowest quintile													
Cereals	26.23	0.27	288.14	1.04	0.49	1.05	0.00	0.00	28.30	N/A	8.99	N/A	8.99
Roots and tubers	566.08	N/A	6792.96	0.25	0.13	0.56	N/A	56.86	43.26	N/A	1.52	N/A	1.52
Sugars and syrups	0.00	N/A	0.02	N/A	0.04	N/A	N/A	0.01	0.09	N/A	0.01	N/A	0.01
Pulses	0.73	N/A	8.80	0.10	0.04	0.06	N/A	0.27	12.51	N/A	1.27	N/A	1.27
Tree nuts	N/A	N/A	N/A	0.00	0.00	0.00	N/A	0.00	0.90	N/A	0.01	N/A	0.01
Oil crops	0.02	N/A	0.21	0.02	0.01	0.02	N/A	0.05	8.08	N/A	0.30	N/A	0.30
Vegetables	55.92	N/A	670.31	0.40	0.52	0.08	N/A	9.04	53.36	N/A	1.07	N/A	1.07
Fruits	20.82	N/A	249.41	0.03	0.02	0.05	N/A	11.20	2.17	N/A	0.14	N/A	0.14
Stimulants	0.10	0.10	N/A	0.00	0.00	0.00	0.00	0.00	0.21	N/A	0.01	N/A	0.01
Spices	0.08	N/A	0.95	0.00	0.00	0.00	N/A	0.03	3.83	N/A	0.03	N/A	0.03
Alcoholic beverages	N/A	N/A	N/A	0.00	0.00	0.02	N/A	N/A	1.21	N/A	0.00	N/A	0.00
Meat	0.68	0.68	N/A	0.01	0.02	0.03	0.15	0.00	0.59	0.04	0.11	0.02	0.13
Eggs	0.19	0.19	N/A	0.00	0.00	0.00	0.00	N/A	0.06	0.00	N/A	N/A	0.00
Fish	3.72	3.72	N/A	0.01	0.01	0.03	0.73	0.01	33.48	0.12	N/A	0.04	0.08
Milk and cheese	6.86	6.86	N/A	N/A	0.03	0.00	0.09	0.06	27.77	0.01	N/A	N/A	0.01
Oils and fats (vegetable)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.00	N/A	N/A	0.00	N/A	0.00
Oils and fats (animal)	0.56	0.56	N/A	N/A	0.00	N/A	0.00	N/A	0.01	N/A	N/A	N/A	N/A
Nonalcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.01	N/A	N/A	N/A	N/A
Quintile 2													
Cereals	30.37	0.53	334.37	1.23	0.56	1.23	0.00	0.00	37.44	N/A	10.71	N/A	10.71
Roots and tubers	529.41	N/A	6352.88	0.24	0.13	0.52	N/A	53.83	42.79	N/A	1.51	N/A	1.51
Sugars and syrups	0.00	N/A	0.04	N/A	0.08	N/A	N/A	0.01	0.19	N/A	0.02	N/A	0.02
Pulses	0.93	N/A	11.15	0.12	0.05	0.08	N/A	0.32	15.34	N/A	1.55	N/A	1.55

Table 6.3: Micronutrient Availability by Food Group and Area

	Average micronutrient availability												
	RAE of vitamin A (mcg/ person/day)	*Retinol (mcg/ person/ day)*	*Beta-carotene (mcg/ person/day)*	*Vitamin B1 (mg/ person/ day)*	*Vitamin B2 (mg/ person/ day)*	*Vitamin B6 (mg/ person/ day)*	*Vitamin B12 (mcg/ person/ day)*	*Vitamin C (mg/ person/ day)*	*Calcium (mg/ person/ day)*	*Animal iron (mg/ person/ day)*	*Nonanimal iron (mg/ person/ day)*	*Heme iron (mg/ person/ day)*	*Nonheme iron (mg/ person/ day)*
Area													
Capital city													
Cereals	18.30	3.55	174.36	1.10	0.36	1.00	0.02	0.00	60.20	N/A	7.42	N/A	7.42
Roots and tubers	140.16	N/A	1681.92	0.03	0.06	0.08	N/A	7.81	9.99	N/A	0.49	N/A	0.49
Sugars and syrups	0.01	N/A	0.13	N/A	0.22	N/A	N/A	0.04	0.65	N/A	0.06	N/A	0.06
Pulses	0.81	N/A	9.74	0.09	0.04	0.06	N/A	0.20	11.79	N/A	1.22	N/A	1.22
Tree nuts	N/A	N/A	N/A	0.00	0.00	0.00	N/A	0.00	0.49	N/A	0.01	N/A	0.01
Oil crops	0.00	N/A	0.02	0.01	0.00	0.03	N/A	0.82	6.42	N/A	0.72	N/A	0.72
Vegetables	111.82	N/A	1341.86	0.07	0.35	0.10	N/A	17.56	20.21	N/A	0.54	N/A	0.54
Fruits	22.22	N/A	256.61	0.05	0.03	0.08	N/A	23.59	11.06	N/A	0.17	N/A	0.17
Stimulants	0.03	0.03	N/A	0.00	0.01	0.00	0.00	0.00	0.68	N/A	0.03	N/A	0.03
Spices	1.96	N/A	22.11	0.00	0.00	0.01	N/A	0.39	10.38	N/A	0.15	N/A	0.15
Alcoholic beverages	N/A	N/A	N/A	0.00	0.00	0.01	N/A	N/A	0.63	N/A	0.00	N/A	0.00
Meat	0.90	0.90	N/A	0.04	0.04	0.07	0.44	0.01	1.51	0.06	0.34	0.03	0.36
Eggs	3.22	3.22	N/A	0.00	0.01	0.00	0.02	N/A	0.95	0.02	N/A	N/A	0.02
Fish	3.79	3.79	N/A	0.01	0.03	0.05	1.24	0.01	123.59	0.25	N/A	0.09	0.16
Milk and cheese	4.98	4.98	N/A	N/A	0.03	0.00	0.07	0.01	20.02	0.02	N/A	N/A	0.02
Oils and fats (vegetable)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.00	N/A	N/A	0.00	N/A	0.00
Oils and fats (animal)	5.04	5.04	N/A	N/A	0.00	N/A	0.00	N/A	0.13	N/A	N/A	N/A	N/A
Nonalcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.73	N/A	N/A	N/A	N/A
Other urban areas													
Cereals	27.61	2.46	268.48	1.29	0.52	1.43	0.02	0.00	53.25	N/A	9.27	N/A	9.27
Roots and tubers	400.48	N/A	4805.70	0.12	0.11	0.27	N/A	26.31	25.01	N/A	1.03	N/A	1.03
Sugars and syrups	0.01	N/A	0.13	N/A	0.23	N/A	N/A	0.03	0.48	N/A	0.04	N/A	0.04
Pulses	0.88	N/A	10.63	0.10	0.05	0.06	N/A	0.22	13.14	N/A	1.35	N/A	1.35

Table 6.4: Micronutrient Availability by Food Group and Region

	Average micronutrient availability												
	RAE of vitamin A (mcg/ person/day)	*Retinol (mcg/ person/ day)*	*Beta-carotene (mcg/ person/day)*	*Vitamin B1 (mg/ person/ day)*	*Vitamin B2 (mg/ person/ day)*	*Vitamin B6 (mg/ person/ day)*	*Vitamin B12 (mcg/ person/ day)*	*Vitamin C (mg/ person/ day)*	*Calcium (mg/ person/ day)*	*Animal iron (mg/ person/ day)*	*Nonanimal iron (mg/ person/ day)*	*Heme iron (mg/ person/ day)*	*Nonheme iron (mg/ person/ day)*
Region													
Region 1													
Cereals	40.61	0.91	458.79	1.56	0.73	1.44	0.01	0.00	53.45	N/A	15.65	N/A	15.65
Roots and tubers	238.05	N/A	2856.58	0.04	0.05	0.08	N/A	7.68	9.02	N/A	0.38	N/A	0.38
Sugars and syrups	0.00	N/A	0.05	N/A	0.09	N/A	N/A	0.02	0.24	N/A	0.02	N/A	0.02
Pulses	1.04	N/A	12.49	0.17	0.07	0.12	N/A	0.70	22.12	N/A	2.22	N/A	2.22
Tree nuts	N/A	N/A	N/A	0.00	0.00	0.00	N/A	0.00	0.47	N/A	0.01	N/A	0.01
Oil crops	0.00	N/A	0.03	0.06	0.02	0.06	N/A	0.04	19.44	N/A	0.96	N/A	0.96
Vegetables	95.08	N/A	1137.44	0.49	0.70	0.25	N/A	17.67	190.03	N/A	3.81	N/A	3.81
Fruits	8.78	N/A	104.99	0.02	0.01	0.03	N/A	23.13	4.33	N/A	0.07	N/A	0.07
Stimulants	0.17	0.17	N/A	0.00	0.00	0.00	0.00	0.01	0.50	N/A	0.03	N/A	0.03
Spices	0.74	N/A	8.33	0.00	0.00	0.00	N/A	0.12	6.45	N/A	0.04	N/A	0.04
Alcoholic beverages	N/A	N/A	N/A	0.00	0.00	0.06	N/A	N/A	2.98	N/A	0.00	N/A	0.00
Meat	0.89	0.89	N/A	0.03	0.04	0.06	0.33	0.01	1.16	0.06	0.26	0.03	0.29
Eggs	0.99	0.99	N/A	0.00	0.00	0.00	0.01	N/A	0.29	0.01	N/A	N/A	0.01
Fish	1.70	1.70	N/A	0.00	0.01	0.02	0.56	0.00	33.65	0.08	N/A	0.03	0.05
Milk and cheese	14.84	14.84	N/A	N/A	0.07	0.00	0.21	0.16	62.35	0.02	N/A	N/A	0.02
Oils and fats (vegetable)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.00	N/A	N/A	0.00	N/A	0.00
Oils and fats (animal)	0.45	0.45	N/A	N/A	0.00	N/A	0.00	N/A	0.01	N/A	N/A	N/A	N/A
Nonalcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.06	N/A	N/A	N/A	N/A

replaces a nutrient quantity, it means that the amount of nutrient available from the food commodity group is very low or null, or there was no acquisition of that food group.

Table 6.5: Contribution of Food Groups to Micronutrient Availability This table shows how much each food commodity group contributes, in percentage, to the total micronutrient availability at the *national level*. The total of each column is equal to 100 percent. The disaggregation of these statistics by food commodity groups helps identify the main food commodity group or groups as sources of each micronutrient.

Table 6.6: Contribution of Food Groups to Micronutrient Availability by Area This table shows how much each food commodity group contributes, in percentage, to the total micronutrient availability in *urban and rural areas*. The total of each column is equal to 100 percent. The disaggregation of these statistics by food commodity groups helps identify differences in urban and rural areas for the main food commodity group or groups as sources of each micronutrient.

Disaggregated by Food Commodity: Tables 6.7 to 6.9

The food commodities analyzed are those collected in the survey excluding those consumed away from home. The food commodity quantities refer to edible portions, which mean they exclude the nonedible parts (peels, bones, etc.).

Table 6.7: Micronutrient Availability by Food Item This table shows food commodity edible quantities and their contribution to the total micronutrient availability for human consumption at the *national level*. This table is useful to identify which food commodities are the main providers of micronutrients at the national level.

Table 6.8: Micronutrient Availability by Food Item and Area This table shows food commodity edible quantities and their contribution to the total amount of micronutrients available for human consumption in *urban and rural areas*. This table is useful to identify which food commodities are the main providers of micronutrients within rural and urban areas as well as differences between rural and urban patterns.

Table 6.9: Micronutrient Availability by Food Item and Region This table shows food commodity edible quantities and their contribution to the total

Table 6.5: Contribution of Food Groups to Micronutrient Availability

	Average micronutrient availability, % of total availability												
	RAE of vitamin A	*Retinol*	*Beta-carotene*	*Vitamin B1*	*Vitamin B2*	*Vitamin B6*	*Vitamin B12*	*Vitamin C*	*Calcium*	*Animal iron*	*Nonanimal iron*	*Heme iron*	*Nonheme iron*
Food group													
Cereals	4.07	5.09	3.75	59.10	31.32	55.69	0.46	0.00	14.92	0.00	63.46	0.00	62.73
Roots and tubers	74.63	0.00	77.23	11.02	7.77	22.26	0.00	56.51	14.44	0.00	9.54	0.00	9.43
Sugars and syrups	0.00	0.00	0.00	0.00	8.14	0.00	0.00	0.02	0.10	0.00	0.16	0.00	0.16
Pulses	0.15	0.00	0.15	6.38	3.42	3.76	0.00	0.40	5.80	0.00	10.87	0.00	10.74
Tree nuts	0.00	0.00	0.00	0.04	0.21	0.02	0.00	0.00	0.41	0.00	0.11	0.00	0.11
Oil crops	0.00	0.00	0.00	1.39	0.55	1.50	0.00	0.23	4.39	0.00	3.85	0.00	3.80
Vegetables	11.76	0.00	12.16	17.17	38.05	4.74	0.00	15.85	19.90	0.00	7.64	0.00	7.56
Fruits	6.43	0.00	6.63	2.80	2.90	5.11	0.00	26.73	1.90	0.00	1.91	0.00	1.89
Stimulants	0.04	1.37	0.00	0.02	0.26	0.05	0.16	0.01	0.30	0.00	0.25	0.00	0.24
Spices	0.07	0.00	0.07	0.03	0.06	0.12	0.00	0.14	2.54	0.00	0.51	0.00	0.51
Alcoholic beverages	0.00	0.00	0.00	0.00	0.00	2.04	0.00	0.00	0.82	0.00	0.00	0.00	0.00
Meat	0.17	5.51	0.00	1.58	2.30	2.89	21.73	0.00	0.45	26.40	1.68	40.32	1.88
Eggs	0.18	5.82	0.00	0.04	0.21	0.03	0.51	0.00	0.13	3.07	0.00	0.00	0.06
Fish	0.77	25.20	0.00	0.43	1.16	1.79	67.59	0.02	18.51	61.40	0.00	59.68	0.72
Milk and cheese	1.59	52.26	0.00	0.00	3.62	0.00	9.54	0.08	15.35	9.13	0.00	0.00	0.16
Oils and fats (vegetable)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oils and fats (animal)	0.14	4.74	0.00	0.00	0.02	0.00	0.02	0.00	0.01	0.00	0.00	0.00	0.00
Nonalcoholic beverages	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.00	0.00	0.00	0.00

Table 6.6: Contribution of Food Groups to Micronutrient Availability by Area

	Average micronutrient availability, % of total availability												
	RAE of vitamin A	*Retinol*	*Beta-carotene*	*Vitamin B1*	*Vitamin B2*	*Vitamin B6*	*Vitamin B12*	*Vitamin C*	*Calcium*	*Animal iron*	*Nonanimal iron*	*Heme iron*	*Nonheme iron*
Food group/area													
Capital city													
Cereals	5.84	16.50	5.00	77.84	30.13	66.99	1.19	0.00	21.54	0.00	66.63	0.00	65.28
Roots and tubers	44.74	0.00	48.24	2.45	4.95	5.53	0.00	15.49	3.58	0.00	4.40	0.00	4.31
Sugars and syrups	0.00	0.00	0.00	0.00	18.89	0.00	0.00	0.09	0.23	0.00	0.50	0.00	0.49
Pulses	0.26	0.00	0.28	6.65	3.59	3.87	0.00	0.41	4.22	0.00	10.92	0.00	10.70
Tree nuts	0.00	0.00	0.00	0.03	0.13	0.01	0.00	0.00	0.17	0.00	0.07	0.00	0.07
Oil crops	0.00	0.00	0.00	0.58	0.23	2.15	0.00	1.62	2.30	0.00	6.48	0.00	6.35
Vegetables	35.70	0.00	38.48	4.84	29.43	6.56	0.00	34.79	7.23	0.00	4.84	0.00	4.75
Fruits	7.09	0.00	7.36	3.53	2.28	5.24	0.00	46.76	3.96	0.00	1.51	0.00	1.48
Stimulants	0.01	0.15	0.00	0.03	0.45	0.13	0.02	0.00	0.24	0.00	0.26	0.00	0.25
Spices	0.63	0.00	0.63	0.11	0.24	0.39	0.00	0.78	3.72	0.00	1.36	0.00	1.34
Alcoholic beverages	0.00	0.00	0.00	0.00	0.00	0.61	0.00	0.00	0.23	0.00	0.00	0.00	0.00
Meat	0.29	4.18	0.00	2.89	3.48	4.99	24.30	0.03	0.54	17.18	3.03	27.37	3.21
Eggs	1.03	14.96	0.00	0.14	0.80	0.13	1.17	0.00	0.34	6.50	0.00	0.00	0.20
Fish	1.21	17.63	0.00	0.90	2.47	3.39	69.40	0.02	44.23	71.63	0.00	72.63	1.44
Milk and cheese	1.59	23.15	0.00	0.00	2.80	0.00	3.84	0.02	7.17	4.69	0.00	0.00	0.15
Oils and fats (vegetable)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.01
Oils and fats (animal)	1.61	23.44	0.00	0.00	0.12	0.00	0.08	0.00	0.04	0.00	0.00	0.00	0.00
Nonalcoholic beverages	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.26	0.00	0.00	0.00	0.00
Other urban areas													
Cereals	4.75	10.85	4.03	71.65	36.19	66.48	0.90	0.01	21.17	0.00	67.24	0.00	66.30
Roots and tubers	68.97	0.00	72.16	6.53	7.91	12.40	0.00	39.39	9.94	0.00	7.48	0.00	7.37
Sugars and syrups	0.00	0.00	0.00	0.00	16.07	0.00	0.00	0.04	0.19	0.00	0.30	0.00	0.30
Pulses	0.15	0.00	0.16	5.82	3.35	2.98	0.00	0.34	5.22	0.00	9.82	0.00	9.68
Tree nuts	0.00	0.00	0.00	0.03	0.15	0.01	0.00	0.00	0.27	0.00	0.07	0.00	0.07
Oil crops	0.00	0.00	0.00	1.26	0.53	1.51	0.00	0.51	3.68	0.00	4.35	0.00	4.29
Vegetables	16.70	0.00	17.46	9.10	23.36	4.70	0.00	26.12	11.61	0.00	5.09	0.00	5.02
Fruits	5.76	0.00	6.00	2.66	2.68	4.57	0.00	33.12	2.65	0.00	1.68	0.00	1.65

Table 6.7: Micronutrient Availability by Food Item

	Average micronutrient availability													
Food item	*Average edible quantity consumed (g/person/day)*	*RAE of vitamin A (mcg/person/day)*	*Retinol (mcg/person/day)*	*Beta-carotene (mcg/person/day)*	*Vitamin B1 (mg/person/day)*	*Vitamin B2 (mg/person/day)*	*Vitamin B6 (mg/person/day)*	*Vitamin B12 (mcg/person/day)*	*Vitamin C (mg/person/day)*	*Calcium (mg/person/day)*	*Animal iron (mg/person/day)*	*Nonanimal iron (mg/person/day)*	*Heme iron (mg/person/day)*	*Nonheme iron (mg/person/day)*
Rice paddy or rough	6.12	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.55	0.00	0.05	0.00	0.05
Rice husked	48.14	0.00	0.00	0.00	0.20	0.02	0.25	0.00	0.00	15.88	0.00	0.87	0.00	0.87
Maize cob fresh	9.29	0.65	0.00	7.80	0.01	0.00	0.00	0.00	0.28	0.09	0.00	0.03	0.00	0.03
Maize grain	72.72	8.00	0.00	70.54	0.28	0.15	0.45	0.00	0.00	5.09	0.00	1.97	0.00	1.97
Maize flour	163.94	18.03	0.00	216.40	0.66	0.33	0.49	0.00	0.00	9.84	0.00	5.74	0.00	5.74
Millet whole grain dried	1.68	0.34	0.00	4.04	0.01	0.00	0.01	0.00	0.00	0.71	0.00	0.13	0.00	0.13
Millet foxtail Italian whole grain	1.17	0.06	0.00	0.70	0.00	0.00	0.00	0.00	0.00	3.22	0.00	0.03	0.00	0.03
Sorghum whole grain brown	7.91	0.47	0.00	5.69	0.02	0.01	0.02	0.00	0.00	1.19	0.00	0.32	0.00	0.32
Sorghum average of all varieties	20.38	1.22	0.00	14.68	0.04	0.02	0.04	0.00	0.00	3.06	0.00	0.84	0.00	0.84
Wheat durum whole grain	0.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.16	0.00	0.02	0.00	0.02
Wheat meal or flour unspecied wheat	4.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.66	0.00	0.05	0.00	0.05
Wheat	1.30	0.00	0.00	0.00	0.02	0.01	0.02	0.00	0.00	0.51	0.00	0.08	0.00	0.08
Bread	3.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.31	0.00	0.02	0.00	0.02
Baby cereals	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.28	0.00	0.00	0.00	0.00
Biscuits wheat from Europe	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00	0.00	0.00
Buns cakes	3.25	1.11	1.11	0.00	0.00	0.00	0.00	0.01	0.00	1.24	0.00	0.01	0.00	0.01
Macaroni spaghetti	0.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00
Oats	2.31	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00	1.24	0.00	0.11	0.00	0.11
Cassava sweet roots raw	28.98	0.29	0.00	3.48	0.03	0.00	0.03	0.00	5.97	4.64	0.00	0.09	0.00	0.09
Cassava sweet roots dried	14.31	2.00	0.00	24.04	0.04	0.01	0.10	0.00	10.30	6.58	0.00	0.27	0.00	0.27
Cassava flour	35.67	4.99	0.00	59.93	0.11	0.04	0.25	0.00	25.68	16.41	0.00	0.68	0.00	0.68
Sweet potato	50.00	527.96	0.00	6335.53	0.05	0.05	0.10	0.00	9.00	10.00	0.00	0.20	0.00	0.20
Coco yam tuber	5.45	0.22	0.00	2.62	0.01	0.00	0.02	0.00	0.25	2.35	0.00	0.01	0.00	0.01

Table 6.8: Micronutrient Availability by Food Item and Area

	Average micronutrient availability													
	Average edible quantity consumed (g/person/day)	RAE of vitamin A (mcg/person/day)	Retinol (mcg/person/day)	Beta-carotene (mcg/person/day)	Vitamin B1 (mg/person/day)	Vitamin B2 (mg/person/day)	Vitamin B6 (mg/person/day)	Vitamin B12 (mcg/person/day)	Vitamin C (mg/person/day)	Calcium (mg/person/day)	Animal iron (mg/person/day)	Nonanimal iron (mg/person/day)	Heme iron (mg/person/day)	Nonheme iron (mg/person/day)
Food item/area														
Capital city														
Rice paddy or rough	0.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.00	0.00	0.00	0.00
Rice husked	108.34	0.00	0.00	0.00	0.45	0.05	0.55	0.00	0.00	35.75	0.00	1.95	0.00	1.95
Maize cob fresh	0.68	0.05	0.00	0.57	0.00	0.00	0.00	0.00	0.02	0.01	0.00	0.00	0.00	0.00
Maize grain	7.68	0.84	0.00	7.45	0.03	0.02	0.05	0.00	0.00	0.54	0.00	0.21	0.00	0.21
Maize flour	125.27	13.78	0.00	165.36	0.50	0.25	0.38	0.00	0.00	7.52	0.00	4.38	0.00	4.38
Millet whole grain dried	0.42	0.08	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.17	0.00	0.03	0.00	0.03
Millet foxtail Italian whole grain	0.71	0.04	0.00	0.42	0.00	0.00	0.00	0.00	0.00	1.94	0.00	0.02	0.00	0.02
Sorghum whole grain brown	0.13	0.01	0.00	0.09	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.01	0.00	0.01
Sorghum average of all varieties	0.04	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00
Wheat durum whole grain	0.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.01	0.00	0.01
Wheat meal or flour unspecified wheat	10.40	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	1.56	0.00	0.12	0.00	0.12
Wheat	0.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.01	0.00	0.01
Bread	16.83	0.00	0.00	0.00	0.02	0.02	0.00	0.00	0.00	1.68	0.00	0.08	0.00	0.08
Baby cereals	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.12	0.00	0.00	0.00	0.00
Biscuits wheat from Europe	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.60	0.00	0.01	0.00	0.01
Buns cakes	10.44	3.55	3.55	0.00	0.00	0.01	0.00	0.02	0.00	3.97	0.00	0.04	0.00	0.04
Macaroni spaghetti	2.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.33	0.00	0.03	0.00	0.03
Oats	10.69	0.00	0.00	0.00	0.08	0.01	0.01	0.00	0.00	5.77	0.00	0.50	0.00	0.50
Cassava sweet roots raw	13.06	0.13	0.00	1.57	0.01	0.00	0.01	0.00	2.69	2.09	0.00	0.04	0.00	0.04
Cassava sweet roots dried	1.09	0.15	0.00	1.83	0.00	0.00	0.01	0.00	0.78	0.50	0.00	0.02	0.00	0.02
Cassava flour	0.83	0.12	0.00	1.40	0.00	0.00	0.01	0.00	0.60	0.38	0.00	0.02	0.00	0.02
Sweet potato	13.23	139.66	0.00	1675.95	0.01	0.01	0.03	0.00	2.38	2.65	0.00	0.05	0.00	0.05
Coco yam tuber	2.45	0.10	0.00	1.18	0.00	0.00	0.01	0.00	0.11	1.06	0.00	0.00	0.00	0.00

Table 6.9: Micronutrient Availability by Food Item and Region

	Average micronutrient availability													
Food item/region	*Average edible quantity consumed (g/person/ day)*	*RAE of vitamin A (mcg/ person/ day)*	*Retinol (mcg/ person/ day)*	*Beta-carotene (mcg/ person/ day)*	*Vitamin B1 (mg/ person/ day)*	*Vitamin B2 (mg/ person/ day)*	*Vitamin B6 (mg/ person/ day)*	*Vitamin B12 (mcg/ person/ day)*	*Vitamin C (mg/ person/ day)*	*Calcium (mg/ person/ day)*	*Animal iron (mg/ person/ day)*	*Nonanimal iron (mg/ person/ day)*	*Heme iron (mg/ person/ day)*	*Nonheme iron (mg/ person/ day)*
Region 1														
Rice paddy or rough	0.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00
Rice husked	26.25	0.00	0.00	0.00	0.11	0.01	0.13	0.00	0.00	8.66	0.00	0.47	0.00	0.47
Maize cob fresh	11.73	0.82	0.00	9.85	0.01	0.00	0.00	0.00	0.35	0.12	0.00	0.04	0.00	0.04
Maize grain	50.15	5.52	0.00	48.64	0.19	0.10	0.31	0.00	0.00	3.51	0.00	1.36	0.00	1.36
Maize flour	240.10	26.41	0.00	316.93	0.96	0.48	0.72	0.00	0.00	14.41	0.00	8.40	0.00	8.40
Millet whole grain dried	1.68	0.34	0.00	4.03	0.01	0.00	0.01	0.00	0.00	0.71	0.00	0.13	0.00	0.13
Millet foxtail Italian whole grain	1.67	0.08	0.00	1.00	0.01	0.00	0.00	0.00	0.00	4.59	0.00	0.05	0.00	0.05
Sorghum whole grain brown	20.07	1.20	0.00	14.45	0.04	0.02	0.04	0.00	0.00	3.01	0.00	0.82	0.00	0.82
Sorghum average of all varieties	102.41	6.14	0.00	73.74	0.20	0.10	0.20	0.00	0.00	15.36	0.00	4.20	0.00	4.20
Wheat durum whole grain	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00
Wheat meal or flour unspecified wheat	3.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50	0.00	0.04	0.00	0.04
Wheat	1.40	0.00	0.00	0.00	0.03	0.01	0.02	0.00	0.00	0.55	0.00	0.09	0.00	0.09
Bread	1.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.00	0.01	0.00	0.01
Baby cereals	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.00	0.00	0.00	0.00
Biscuits wheat from Europe	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.00	0.00	0.00	0.00
Buns cakes	2.69	0.91	0.91	0.00	0.00	0.00	0.00	0.01	0.00	1.02	0.00	0.01	0.00	0.01
Macaroni spaghetti	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00
Oats	1.38	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.74	0.00	0.07	0.00	0.07
Cassava sweet roots raw	11.73	0.12	0.00	1.41	0.01	0.00	0.01	0.00	2.42	1.88	0.00	0.04	0.00	0.04
Cassava sweet roots dried	0.16	0.02	0.00	0.27	0.00	0.00	0.00	0.00	0.11	0.07	0.00	0.00	0.00	0.00
Cassava flour	0.29	0.04	0.00	0.48	0.00	0.00	0.00	0.00	0.21	0.13	0.00	0.01	0.00	0.01
Sweet potato	22.52	237.86	0.00	2854.31	0.02	0.02	0.05	0.00	4.05	4.50	0.00	0.09	0.00	0.09
Coco yam tuber	0.24	0.01	0.00	0.11	0.00	0.00	0.00	0.00	0.01	0.10	0.00	0.00	0.00	0.00

micronutrient availability for human consumption at the *regional level*. This table is useful to identify which food commodities are the main providers of micronutrients within each region and to detect differences across regions.

Availability of Amino Acids

Amino acids are the building blocks of proteins and have an important role in human bodies. Some of their functions include building cells, protecting the body from viruses or bacteria, repairing damaged tissue, providing nitrogen, and carrying oxygen throughout the body. They can be classified as dispensable or indispensable. The latter are also called essential amino acids and cannot be synthesized by the human body. Therefore, the indispensable amino acids should be supplied to the body through the consumption of proteins in food.

Please note that the statistics on amino acids shown in the tables exclude the food consumed away from home. Therefore, the total available amino acids are underestimated.

Disaggregated by Population Group: Tables 7.1 to 7.2

Table 7.1: Protein Consumption and Amino Acid Availability This table shows the availability of indispensable amino acids in terms of grams per person per day.

Table 7.2: Amino Acid Availability per Gram of Protein This table shows the availability of indispensable amino acids in terms of milligrams per gram of protein.

Disaggregated by Population Group: Tables 8.1 to 8.7

It is important to know that the statistics shown exclude the food consumed away from home. Therefore, the total available amino acids are underestimated.

Table 8.1: Availability of Amino Acids by Food Group This table shows the available grams of amino acids provided by each food commodity group at the *national level*. Each time N/A replaces a nutrient quantity, it means that the amount of nutrient available from the food commodity group is very low or null, or there was no acquisition of that food group.

Table 7.1: Protein Consumption and Amino Acid Availability

	Average amino acid availability (g/person/day)										
	Average food consumption in monetary value (LCU/ person/day)	*Average protein consumption (g/person/ day)*	*Lysine*	*Valine*	*Isoleucine*	*Leucine*	*Methio-nine and cystine*	*Threonine*	*Histidine*	*Phenyl-alanine and tyrosine*	*Tryptophan*
Total	211.61	46.64	1.83	1.73	1.39	2.88	1.20	1.28	0.95	2.67	0.61
Quintiles of income											
Lowest quintile	99.77	34.03	1.17	1.16	0.91	1.99	0.82	0.85	0.62	1.78	0.47
2	162.53	44.18	1.59	1.55	1.24	2.60	1.07	1.14	0.84	2.39	0.56
3	212.21	47.62	1.85	1.76	1.41	2.95	1.22	1.31	0.97	2.73	0.62
4	288.45	54.36	2.24	2.10	1.70	3.47	1.46	1.56	1.17	3.26	0.71
Highest quintile	413.21	64.14	2.94	2.63	2.16	4.22	1.83	1.97	1.46	4.05	0.82
Area											
Capital city	343.33	40.96	1.89	1.70	1.39	2.61	1.17	1.26	0.92	2.61	0.43
Other urban areas	277.28	45.48	1.94	1.85	1.49	3.12	1.29	1.38	1.04	2.90	0.53
Rural areas	190.86	47.25	1.80	1.71	1.37	2.85	1.19	1.27	0.93	2.64	0.64
Household size											
One person	421.07	67.97	3.20	2.69	2.24	4.28	1.86	2.05	1.50	4.13	0.83
Between 2 and 3 people	295.05	56.93	2.33	2.13	1.73	3.46	1.46	1.59	1.17	3.28	0.72
Between 4 and 5 people	228.21	48.36	1.90	1.79	1.44	2.96	1.24	1.33	0.99	2.77	0.61
Between 6 and 7 people	191.95	43.41	1.65	1.59	1.27	2.71	1.12	1.18	0.88	2.48	0.59
More than 7	165.59	42.15	1.61	1.56	1.24	2.61	1.09	1.15	0.84	2.41	0.57
Gender of the household head											
Male	209.51	46.58	1.83	1.73	1.39	2.89	1.20	1.28	0.95	2.68	0.61
Female	220.73	46.90	1.81	1.71	1.37	2.84	1.18	1.27	0.93	2.64	0.60
Age of the household head											
Less than 35	225.51	48.67	1.92	1.80	1.45	3.00	1.25	1.34	0.99	2.78	0.63
Between 35 and 45	213.95	45.43	1.80	1.70	1.36	2.83	1.18	1.26	0.93	2.63	0.60
Between 46 and 60	206.17	47.06	1.81	1.75	1.39	2.89	1.20	1.28	0.94	2.70	0.61
More than 60	192.61	44.84	1.76	1.65	1.32	2.72	1.15	1.22	0.89	2.53	0.61

Table 7.2: Amino Acid Availability per Gram of Protein

	Mg amino acid/g protein								
	Lysine	*Valine*	*Isoleucine*	*Leucine*	*Methionine and cystine*	*Threonine*	*Histidine*	*Phenylalanine and tyrosine*	*Tryptophan*
Total	39.2	37.1	29.7	61.7	25.7	27.5	20.3	57.3	13.1
Quintiles of income									
Lowest quintile	34.4	34.0	26.6	58.4	24.0	25.0	18.1	52.3	13.7
2	36.1	35.2	28.0	58.8	24.1	25.8	18.9	54.2	12.7
3	38.9	37.0	29.6	61.9	25.6	27.4	20.3	57.3	13.1
4	41.2	38.7	31.2	63.8	26.8	28.8	21.5	60.1	13.0
Highest quintile	45.8	40.9	33.6	65.8	28.5	30.7	22.8	63.1	12.8
Area									
Capital city	46.2	41.6	34.1	63.7	28.5	30.7	22.5	63.6	10.5
Other urban areas	42.6	40.7	32.8	68.6	28.3	30.4	22.8	63.8	11.6
Rural areas	38.2	36.2	29.0	60.4	25.1	26.8	19.7	55.9	13.5
Household size									
One person	47.1	39.6	32.9	63.0	27.3	30.2	22.1	60.7	12.2
Between 2 and 3 people	40.9	37.4	30.4	60.7	25.6	27.9	20.5	57.6	12.6
Between 4 and 5 people	39.4	37.1	29.9	61.2	25.6	27.5	20.4	57.3	12.7
Between 6 and 7 people	38.0	36.7	29.2	62.5	25.7	27.3	20.3	57.1	13.5
More than 7	38.3	37.0	29.4	61.9	25.8	27.2	19.9	57.1	13.4
Gender of the household head									
Male	39.3	37.2	29.8	61.9	25.9	27.6	20.4	57.5	13.1
Female	38.5	36.5	29.3	60.5	25.2	27.1	19.9	56.4	12.8
Age of the household head									
Less than 35	39.5	36.9	29.7	61.7	25.7	27.6	20.4	57.1	12.9
Between 35 and 45	39.6	37.4	30.0	62.2	26.0	27.8	20.6	57.9	13.1
Between 46 and 60	38.5	37.1	29.6	61.5	25.5	27.3	20.1	57.3	12.9
More than 60	39.3	36.9	29.4	60.7	25.6	27.2	19.9	56.5	13.6

Table 8.1: Availability of Amino Acids by Food Group

	Average amino acid availability (g/person/day)								
	Lysine	*Valine*	*Isoleucine*	*Leucine*	*Methionine and cystine*	*Threonine*	*Histidine*	*Phenylalanine and tyrosine*	*Tryptophan*
Food group									
Cereals	499.01	778.65	559.99	1419.81	526.21	526.08	384.94	1237.03	140.35
Roots and tubers	13.25	13.08	9.50	15.53	57.50	10.90	7.50	20.14	209.93
Sugars and syrups	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Pulses	215.13	152.61	131.89	248.01	78.85	121.70	96.17	281.47	37.57
Tree nuts	2.55	3.94	3.32	5.95	2.24	2.83	2.14	6.97	1.37
Oil crops	64.65	96.02	77.37	138.92	54.21	66.87	49.60	161.46	30.55
Vegetables	65.02	71.99	62.47	112.38	34.24	54.66	31.35	84.27	13.13
Fruits	8.26	5.17	3.31	5.40	14.16	3.28	27.98	6.14	43.48
Stimulants	2.76	2.19	1.93	2.75	1.13	1.79	0.88	3.21	0.47
Spices	2.09	2.53	1.87	2.71	1.26	1.57	0.97	2.73	0.62
Alcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Meat	419.64	245.40	230.07	392.42	187.63	217.53	168.80	363.80	55.49
Eggs	6.58	5.58	4.99	7.82	4.98	4.39	2.17	8.59	1.11
Fish	406.45	227.96	204.13	359.87	178.61	194.01	130.10	322.51	49.72
Milk and cheese	122.12	125.31	95.94	164.42	59.56	76.30	43.52	174.94	25.52
Oils and fats (vegetable)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Oils and fats (animal)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Nonalcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Table 8.2: Availability of Amino Acids by Food Group and Income Quintile This table shows the available grams of amino acids provided by each food commodity group at the *income quintile level*. Each time N/A replaces a nutrient quantity, it means that the amount of nutrient available from the food commodity group is very low or null, or there was no acquisition of that food group.

Table 8.3: Availability of Amino Acids by Food Group and Area This table shows the available grams of amino acids provided by each food commodity group at the *urban/rural level*. Each time N/A replaces a nutrient quantity, it means that the amount of nutrient available from the food commodity group is very low or null, or there was no acquisition of that food group.

Table 8.4: Availability of Amino Acids by Food Group and Region This table shows the available grams of amino acids provided by each food commodity group at the *regional level*. Each time N/A replaces a nutrient quantity, it means that the amount of nutrient available from the food commodity group is very low or null, or there was no acquisition of that food group.

Table 8.5: Contribution of Food Groups to Amino Acid Availability This table shows how much each food commodity group contributes, in percentage, to the total micronutrient availability at the *national level*. The total of each column is equal to 100 percent. This information is useful to identify the main food commodity groups that provide the available indispensable amino acids in the diet.

Table 8.6: Contribution of Food Groups to Amino Acid Availability by Area This table shows how much each food commodity group contributes, in percentage, to the total micronutrient availability at the *urban/rural level*. The total of each column is equal to 100 percent. This information is useful to identify the main food commodity groups that provide the available indispensable amino acids in the diet, and to highlight differences by area.

Table 8.2: Availability of Amino Acids by Food Group and Income Quintile

	Average amino acid availability (g/person/day)								
	Lysine	*Valine*	*Isoleucine*	*Leucine*	*Methionine and cystine*	*Threonine*	*Histidine*	*Phenylalanine and tyrosine*	*Tryptophan*
Quintiles of income									
Lowest quintile									
Cereals	0.37	0.58	0.41	1.10	0.39	0.40	0.29	0.91	0.10
Roots and tubers	0.01	0.01	0.00	0.01	0.06	0.01	0.00	0.01	0.23
Sugars and syrups	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Pulses	0.15	0.11	0.09	0.18	0.06	0.09	0.07	0.20	0.03
Tree nuts	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00
Oil crops	0.03	0.05	0.04	0.07	0.03	0.03	0.02	0.08	0.02
Vegetables	0.05	0.06	0.05	0.10	0.03	0.05	0.03	0.07	0.01
Fruits	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.00	0.02
Stimulants	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Spices	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Alcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Meat	0.18	0.11	0.10	0.17	0.08	0.10	0.07	0.16	0.02
Eggs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fish	0.26	0.15	0.13	0.23	0.12	0.13	0.08	0.21	0.03
Milk and cheese	0.09	0.09	0.07	0.12	0.04	0.05	0.03	0.12	0.02
Oils and fats (vegetable)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Oils and fats (animal)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Nonalcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Quintile 2									
Cereals	0.45	0.70	0.50	1.29	0.47	0.47	0.35	1.11	0.12
Roots and tubers	0.01	0.01	0.01	0.01	0.06	0.01	0.01	0.02	0.22
Sugars and syrups	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Pulses	0.19	0.14	0.12	0.22	0.07	0.11	0.09	0.25	0.03

Table 8.3: Availability of Amino Acids by Food Group and Area

	Average amino acid availability (g/person/day)								
	Lysine	*Valine*	*Isoleucine*	*Leucine*	*Methionine and cystine*	*Threonine*	*Histidine*	*Phenylalanine and tyrosine*	*Tryptophan*
Area/food group									
Capital city									
Cereals	0.54	0.82	0.61	1.25	0.54	0.53	0.37	1.28	0.18
Roots and tubers	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.03
Sugars and syrups	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Pulses	0.16	0.11	0.10	0.19	0.06	0.09	0.07	0.21	0.03
Tree nuts	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oil crops	0.04	0.07	0.05	0.08	0.04	0.04	0.03	0.10	0.02
Vegetables	0.04	0.04	0.04	0.05	0.02	0.03	0.02	0.05	0.01
Fruits	0.02	0.01	0.01	0.01	0.01	0.01	0.03	0.01	0.02
Stimulants	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Spices	0.01	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00
Alcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Meat	0.48	0.28	0.26	0.45	0.22	0.25	0.20	0.42	0.06
Eggs	0.02	0.01	0.01	0.02	0.01	0.01	0.01	0.02	0.00
Fish	0.54	0.30	0.27	0.48	0.24	0.26	0.17	0.43	0.07
Milk and cheese	0.03	0.04	0.03	0.05	0.02	0.03	0.01	0.05	0.01
Oils and fats (vegetable)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Oils and fats (animal)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Non alcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Other urban areas									
Cereals	0.59	0.93	0.67	1.70	0.63	0.63	0.47	1.50	0.18
Roots and tubers	0.02	0.02	0.01	0.02	0.03	0.01	0.01	0.02	0.11
Sugars and syrups	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Pulses	0.18	0.13	0.11	0.21	0.07	0.10	0.08	0.24	0.03

Table 8.4: Availability of Amino Acids by Food Group and Region

	Average amino acid availability (g/person/day)								
	Lysine	*Valine*	*Isoleucine*	*Leucine*	*Methionine and cystine*	*Threonine*	*Histidine*	*Phenylalanine and tyrosine*	*Tryptophan*
Region									
Region 1									
Cereals	0.59	0.92	0.65	1.56	0.62	0.60	0.42	1.40	0.16
Roots and tubers	0.01	0.01	0.01	0.01	0.01	0.01	0.00	0.01	0.03
Sugars and syrups	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Pulses	0.21	0.15	0.13	0.25	0.08	0.12	0.09	0.27	0.04
Tree nuts	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oil crops	0.12	0.18	0.15	0.27	0.10	0.13	0.10	0.32	0.06
Vegetables	0.10	0.12	0.10	0.18	0.06	0.09	0.05	0.15	0.03
Fruits	0.01	0.00	0.00	0.00	0.00	0.00	0.02	0.01	0.00
Stimulants	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Spices	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Alcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Meat	0.39	0.23	0.21	0.36	0.17	0.20	0.16	0.33	0.05
Eggs	0.01	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00
Fish	0.18	0.10	0.09	0.16	0.08	0.09	0.06	0.14	0.02
Milk and cheese	0.23	0.22	0.16	0.29	0.11	0.13	0.07	0.30	0.03
Oils and fats (vegetable)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Oils and fats (animal)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Nonalcoholic beverages	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Region 2									
Cereals	0.58	0.97	0.70	2.08	0.71	0.69	0.54	1.65	0.16
Roots and tubers	0.02	0.02	0.01	0.02	0.01	0.01	0.01	0.02	0.03
Sugars and syrups	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Pulses	0.21	0.15	0.13	0.24	0.08	0.12	0.10	0.28	0.04
Tree nuts	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oil crops	0.01	0.01	0.01	0.02	0.01	0.01	0.01	0.02	0.00

Table 8.5: Contribution of Food Groups to Amino Acid Availability

	Average amino acid availability, % of total availability								
	Lysine	*Valine*	*Isoleucine*	*Leucine*	*Methionine and cystine*	*Threonine*	*Histidine*	*Phenylalanine and tyrosine*	*Tryptophan*
Food group									
Cereals	27.31	45.00	40.38	49.37	43.83	41.04	40.69	46.27	23.03
Roots and tubers	0.73	0.76	0.69	0.54	4.79	0.85	0.79	0.75	34.45
Sugars and syrups	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pulses	11.77	8.82	9.51	8.62	6.57	9.49	10.17	10.53	6.17
Tree nuts	0.14	0.23	0.24	0.21	0.19	0.22	0.23	0.26	0.23
Oil crops	3.54	5.55	5.58	4.83	4.52	5.22	5.24	6.04	5.01
Vegetables	3.56	4.16	4.50	3.91	2.85	4.26	3.31	3.15	2.15
Fruits	0.45	0.30	0.24	0.19	1.18	0.26	2.96	0.23	7.14
Stimulants	0.15	0.13	0.14	0.10	0.09	0.14	0.09	0.12	0.08
Spices	0.11	0.15	0.14	0.09	0.11	0.12	0.10	0.10	0.10
Alcoholic beverages	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Meat	22.96	14.18	16.59	13.64	15.63	16.97	17.84	13.61	9.11
Eggs	0.36	0.32	0.36	0.27	0.41	0.34	0.23	0.32	0.18
Fish	22.24	13.17	14.72	12.51	14.88	15.13	13.75	12.06	8.16
Milk and cheese	6.68	7.24	6.92	5.72	4.96	5.95	4.60	6.54	4.19
Oils and fats (vegetable)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oils and fats (animal)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nonalcoholic beverages	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 8.6: Contribution of Food Groups to Amino Acid Availability by Area

	Average amino acid availability, % of total availability								
	Lysine	*Valine*	*Isoleucine*	*Leucine*	*Methionine and cystine*	*Threonine*	*Histidine*	*Phenylalanine and tyrosine*	*Tryptophan*
Area									
Capital city									
Cereals	28.71	48.03	43.76	48.10	46.26	41.94	40.60	49.23	42.14
Roots and tubers	0.70	0.75	0.66	0.55	1.19	0.70	0.60	0.73	7.62
Sugars and syrups	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pulses	8.53	6.73	7.07	7.11	5.10	7.26	7.92	8.19	6.66
Tree nuts	0.05	0.09	0.10	0.09	0.08	0.09	0.09	0.11	0.13
Oil crops	2.37	3.84	3.28	3.24	3.33	3.27	3.08	3.66	3.65
Vegetables	2.15	2.24	2.93	1.90	1.53	2.47	1.72	1.82	1.79
Fruits	0.88	0.68	0.53	0.39	0.95	0.48	3.65	0.53	3.84
Stimulants	0.06	0.06	0.06	0.04	0.04	0.06	0.04	0.06	0.05
Spices	0.29	0.39	0.35	0.27	0.29	0.33	0.27	0.27	0.38
Alcoholic beverages	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Meat	25.50	16.55	18.88	17.43	18.47	20.08	21.27	16.13	15.05
Eggs	0.88	0.83	0.91	0.76	1.08	0.89	0.60	0.84	0.66
Fish	28.34	17.68	19.32	18.22	20.20	20.41	18.68	16.34	15.25
Milk and cheese	1.55	2.13	2.16	1.89	1.49	2.03	1.49	2.10	2.79
Oils and fats (vegetable)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oils and fats (animal)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nonalcoholic beverages	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Other urban areas									
Cereals	30.23	49.97	45.06	54.42	49.14	45.33	44.86	51.66	33.30
Roots and tubers	0.87	0.87	0.79	0.58	2.57	0.85	0.74	0.83	20.01
Sugars and syrups	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pulses	9.36	6.94	7.40	6.67	5.19	7.39	7.88	8.25	6.09

Table 8.7: Contribution of Food Groups to Amino Acid Availability by Region This table shows how much each food commodity group contributes, in percentage, to the total micronutrient availability at the *regional level*. The total of each column is equal to 100 percent. This information is useful to identify the main food commodity groups that provide the available indispensable amino acids in the diet, and to highlight regional differences.

Disaggregated by Food Commodity: Tables 8.8 to 8.10

The food commodities analyzed are those collected in the survey excluding those consumed away from home. The food commodity quantities refer to edible portions, which mean they exclude the nonedible parts (peels, bones, etc.)

Table 8.8: Availability of Amino Acid by Food Item This table shows food commodity edible quantities and the available grams of amino acids provided by them at the *national level*. This table is useful to identify the main food commodities that provide indispensable amino acids at national level.

Table 8.9: Availability of Amino Acid by Food Item and Area This table shows food commodity edible quantities and the available grams of amino acids provided by them at the *urban/rural level*. This table is useful to identify the main food commodities that provide indispensable amino acids within rural and urban areas as well as to highlight differences between rural and urban patterns.

Table 8.10: Availability of Amino Acid by Food Item and Region This table shows food commodity edible quantities and the available grams of amino acids provided by them at the *regional level*. This table is useful to identify the main food commodity groups that provide the available of indispensable amino acids at the national level, and to highlight regional differences.

Table 8.7: Contribution of Food Groups to Amino Acid Availability by Region

	Average amino acid availability, % of total availability								
	Lysine	*Valine*	*Isoleucine*	*Leucine*	*Methionine and cystine*	*Threonine*	*Histidine*	*Phenylalanine and tyrosine*	*Tryptophan*
Region									
Region 1									
Cereals	31.89	47.31	43.04	50.26	50.11	43.73	42.95	47.38	37.70
Roots and tubers	0.45	0.40	0.38	0.27	0.91	0.39	0.37	0.39	7.40
Sugars and syrups	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pulses	11.48	7.79	8.71	8.02	6.19	8.81	9.36	9.15	8.23
Tree nuts	0.05	0.08	0.08	0.07	0.07	0.08	0.08	0.09	0.12
Oil crops	6.42	9.42	10.06	8.84	8.41	9.54	10.03	10.88	14.62
Vegetables	5.60	6.05	6.42	5.94	4.67	6.63	5.12	5.17	6.03
Fruits	0.34	0.22	0.19	0.14	0.23	0.20	2.45	0.17	0.79
Stimulants	0.21	0.16	0.18	0.13	0.13	0.18	0.13	0.15	0.15
Spices	0.08	0.09	0.08	0.06	0.07	0.08	0.07	0.06	0.10
Alcoholic beverages	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Meat	20.87	11.62	13.87	11.67	13.95	14.62	15.85	11.34	11.86
Eggs	0.28	0.22	0.26	0.20	0.31	0.25	0.17	0.23	0.20
Fish	9.76	5.22	5.96	5.15	6.43	6.29	5.88	4.85	5.12
Milk and cheese	12.56	11.43	10.77	9.25	8.53	9.21	7.55	10.14	7.68
Oils and fats (vegetable)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oils and fats (animal)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nonalcoholic beverages	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Region 2									
Cereals	32.94	52.72	47.93	61.11	55.97	49.94	51.90	55.59	37.04
Roots and tubers	1.03	0.90	0.83	0.53	1.13	0.80	0.68	0.82	6.57
Sugars and syrups	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pulses	11.90	7.97	8.61	6.98	6.15	8.48	9.18	9.38	8.66

Table 8.8: Availability of Amino Acid by Food Item

		Average amino acid availability (g/person/day)								
	Average edible quantity consumed (g/person/day)	*Lysine*	*Valine*	*Isoleucine*	*Leucine*	*Methio-nine and cystine*	*Threonine*	*Histidine*	*Phenylala-nine and tyrosine*	*Tryptophan*
Food item										
Rice paddy or rough	6.12	0.01	0.02	0.02	0.03	0.02	0.01	0.01	0.03	0.00
Rice husked	48.14	0.13	0.19	0.14	0.27	0.11	0.12	0.08	0.29	0.04
Maize cob fresh	9.29	0.02	0.03	0.02	0.05	0.01	0.02	0.01	0.02	0.00
Maize grain	72.72	0.13	0.24	0.17	0.59	0.19	0.18	0.15	0.43	0.03
Maize flour	163.94	0.11	0.14	0.10	0.27	0.07	0.10	0.07	0.21	0.02
Millet whole grain dried	1.68	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00
Millet foxtail Italian whole grain	1.17	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00
Sorghum whole grain brown	7.91	0.01	0.02	0.02	0.03	0.02	0.01	0.01	0.03	0.00
Sorghum average of all varieties	20.38	0.04	0.06	0.04	0.08	0.04	0.04	0.02	0.09	0.01
Wheat durum whole grain	0.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Wheat meal or flour unspecified wheat	4.41	0.02	0.03	0.02	0.04	0.02	0.02	0.01	0.04	0.01
Wheat	1.30	0.02	0.01	0.01	0.02	0.01	0.01	0.01	0.02	0.00
Bread	3.11	0.01	0.01	0.01	0.02	0.01	0.01	0.01	0.02	0.00
Baby cereals	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Biscuits wheat from Europe	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Buns cakes	3.25	0.01	0.01	0.01	0.01	0.01	0.00	0.00	0.01	0.00
Macaroni spaghetti	0.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oats	2.31	0.01	0.02	0.01	0.02	0.01	0.01	0.01	0.03	0.00
Cassava sweet roots raw	28.98	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.03
Cassava sweet roots dried	14.31	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.04
Cassava flour	35.67	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.10
Sweet potato	50.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.03
Coco yam tuber	5.45	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00
Potatoes tubers raw	9.00	0.01	0.01	0.01	0.01	0.00	0.01	0.00	0.01	0.00
Banana cooking	42.28	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.04
Starch	2.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 8.9: Availability of Amino Acid by Food Item and Area

	Average edible quantity consumed (g/person/day)	Average amino acid availability (g/person/day)								
		Lysine	*Valine*	*Isoleucine*	*Leucine*	*Methionine and cystine*	*Threonine*	*Histidine*	*Phenylalanine and tyrosine*	*Tryptophan*
Area/food item										
Capital city										
Rice paddy or rough	0.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rice husked	108.34	0.28	0.43	0.31	0.61	0.26	0.27	0.19	0.66	0.09
Maize cob fresh	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Maize grain	7.68	0.01	0.03	0.02	0.06	0.02	0.02	0.02	0.05	0.00
Maize flour	125.27	0.08	0.11	0.08	0.21	0.06	0.08	0.05	0.16	0.01
Millet whole grain dried	0.42	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Millet foxtail Italian whole grain	0.71	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00
Sorghum whole grain brown	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sorghum average of all varieties	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Wheat durum whole grain	0.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Wheat meal or flour unspecified wheat	10.40	0.04	0.06	0.05	0.09	0.05	0.04	0.03	0.11	0.02
Wheat	0.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bread	16.83	0.03	0.06	0.05	0.10	0.06	0.04	0.03	0.11	0.02
Baby cereals	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Biscuits wheat from Europe	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Buns cakes	10.44	0.02	0.02	0.02	0.03	0.02	0.01	0.01	0.04	0.01
Macaroni spaghetti	2.18	0.00	0.01	0.01	0.01	0.01	0.01	0.00	0.02	0.00
Oats	10.69	0.06	0.08	0.06	0.11	0.06	0.05	0.04	0.13	0.02
Cassava sweet roots raw	13.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02
Cassava sweet roots dried	1.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Cassava flour	0.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sweet potato	13.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Coco yam tuber	2.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Potatoes tubers raw	10.96	0.01	0.01	0.01	0.01	0.01	0.01	0.00	0.02	0.00

Table 8.10: Availability of Amino Acid by Food Item and Region

	Average edible quantity consumed	Average amino acid availability (g/person/day)								
	(g/person/day)	Lysine	Valine	Isoleucine	Leucine	Methionine and cystine	Threonine	Histidine	Phenylalanine and tyrosine	Tryptophan
Region/food item										
Region 1										
Rice paddy or rough	0.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rice husked	26.25	0.07	0.11	0.08	0.15	0.06	0.07	0.05	0.16	0.02
Maize cob fresh	11.73	0.03	0.04	0.02	0.07	0.02	0.02	0.02	0.03	0.00
Maize grain	50.15	0.09	0.17	0.12	0.41	0.13	0.12	0.10	0.30	0.02
Maize flour	240.10	0.16	0.21	0.15	0.40	0.11	0.15	0.10	0.31	0.03
Millet whole grain dried	1.68	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00
Millet foxtail Italian whole grain	1.67	0.00	0.01	0.01	0.02	0.01	0.01	0.00	0.01	0.00
Sorghum whole grain brown	20.07	0.04	0.06	0.04	0.08	0.04	0.03	0.02	0.08	0.01
Sorghum average of all varieties	102.41	0.18	0.30	0.22	0.41	0.22	0.18	0.12	0.43	0.06
Wheat durum whole grain	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Wheat meal or flour unspecified wheat	3.35	0.01	0.02	0.02	0.03	0.02	0.01	0.01	0.03	0.01
Wheat	1.40	0.02	0.02	0.01	0.02	0.01	0.01	0.01	0.02	0.00
Bread	1.07	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00
Baby cereals	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Biscuits wheat from Europe	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Buns cakes	2.69	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00
Macaroni spaghetti	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Oats	1.38	0.01	0.01	0.01	0.01	0.01	0.01	0.00	0.02	0.00
Cassava sweet roots raw	11.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Cassava sweet roots dried	0.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Cassava flour	0.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sweet potato	22.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Coco yam tuber	0.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Potatoes tubers raw	7.74	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.01	0.00

Glossary of Indicators

Overall Food Consumption Indicators

average carbohydrates consumption (g/person/day) Average quantity of available carbohydrates (excluding fiber) consumed by the household. See tables 1.9, 1.14, 2.1, 2.3, 2.4, 2.6, and 2.7.

average carbohydrates unit value (LCU/100 g) Measures the cost of 100 grams of carbohydrates by food groups. From this cost it is possible to identify among the food groups that provide carbohydrates those that provide low-cost carbohydrates. See table 2.8.

average dietary energy consumption (kcal/person/day) Measures the amount of calories consumed by the household. It is expressed in kilocalories per person per day. The dietary energy consumption is estimated from the food quantities collected in the survey. Food quantities that are collected "as purchased" (including bones, peels, etc.) first are transformed into edible quantities by taking into consideration the respective food item refuse factor and then are expressed in grams. Once all edible quantities are transformed into grams of nutrients, the nutrient densities (grams of nutrient per gram of food product) of each food item are used to estimate the amount of calories consumed. The dietary energy consumption should be within reasonable ranges from 800 to 4,000 kcal (whichever decile), and it tends to increase as income increases (although it is also possible that better-off households purchase more expensive and less energetic food). See tables 1.1, 1.3, 1.4, 1.9, 1.10, 1.11, 1.12, 2.1, 2.3, 2.4, 2.6, 2.7, 3.1, 3.3, 3.5, and 4.1.

average dietary energy requirement (kcal/person/day) Proper normative reference for adequate nutrition in the population. While it would be mistaken to take the average dietary energy requirement value as the cutoff point to determine the prevalence of undernourishment, its value is used to calculate

	the depth of the food deficit (FD), which is the amount of dietary energy needed to ensure that, if properly distributed, hunger would be eliminated. It is also used to estimate the nutrient recommended intake expressed in mg per 1000 kcal. See tables 1.1, 1.2, and 5.5.
average dietary energy unit value (LCU/1,000 kcals)	Measures the average cost of 1,000 kcal (in local currency). It usually increases as income increases because better-off households are more likely to buy food that is less caloric but more expensive (for instance meat instead of pulses) or to have meals in restaurants. Note that for all the tables presenting statistics by food commodity group or food commodity, the average dietary energy unit value refers to the median of the dietary energy value. See tables 1.3, 1.4, 2.8, 3.1, 3.3, 3.5, and 4.4.
average edible quantity consumed (g/person/day)	For each food item, provides the quantity of food consumed in grams per person per day after the nonedible portion has been removed. See tables 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 6.6, 6.8, 6.9, 8.8, 8.9, and 8.10.
average fat consumption (g/person/day)	Average quantity of fat consumed by the household (expressed in grams per person per day). See tables 1.9, 1.14, 2.1, 2.3, 2.4, 2.6, and 2.7.
average fat unit value (LCU/100 g)	Measures the cost of 100 grams of fat by food groups. From this cost it is possible to identify among the food groups that provide fat those that provide low-cost fat (for instance, to obtain 100 grams of fats from milk and cheese may cost more than to get 100 grams of fat from vegetable oil). See table 2.8.
average food consumption in monetary value (LCU/person/day)	Usually increases as income increases. It should be lower than the total consumption expenditure (which includes nonfood consumption expenditures such as education, health, transport, durable goods, etc.). See tables 1.3, 1.4, 1.9, 2.1, 2.3, 2.4, 2.6, 2.7, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 4.1, and 7.1.

average household size	Corresponds to the total number of household members, and usually wealthier households have fewer members than poor households. See tables 1.3, 1.4, 4.1, 4.2, and 4.3.
average income (LCU/person/day)	Average per person per day income, expressed in local unit of measurement. When the survey does not collect information on income or this information is not reliable, total expenditures corresponding to the sum of total consumption and nonconsumption is used as a proxy of income. See tables 1.4, 1.6, 1.8, 4.4, and 4.5.
average protein consumption (g/person/day)	Average quantity of proteins consumed. See tables 1.9, 1.14, 2.1, 2.3, 2.4, 2.6, 2.7, 3.2, 3.4, 3.6, and 7.1.
average protein unit value (LCU/100 g)	Measures the cost of 100 grams of proteins by food groups. From this cost it is possible to identify among the food groups that provide proteins those that provide low-cost proteins (for instance proteins from cereals may be less expensive than proteins coming from animal sources). See tables 2.8, 3.2, 3.4, and 3.6.
average total consumption in monetary value (LCU/person/day)	Usually increases as income increases and should be lower than total expenditures and income. See tables 1.3, 1.4, and 4.1.
coefficient of variation of dietary energy consumption (%)	Indicator of the dispersion of the dietary energy consumption within the general population. It should not be higher than 35 percent (maximum acceptable). A high CV should be corrected for excess variability. The CV should not be lower than 20 percent to account for at least the variability of dietary energy consumption due to factors other than income/household characteristics. See tables 1.1 and 1.2.

contribution of food groups to total nutrient consumption: share of total carbohydrates consumption (%)	Measures the contribution of each food group to the consumption of carbohydrates in percent (e.g., a share of 55 percent from cereal indicates that the group of cereals contributes 55 percent to the total carbohydrates consumption). See tables 2.2 and 2.5.
contribution of food groups to total nutrient consumption: share of total dietary energy consumption (%)	Measures the contribution of each food group to the total dietary energy consumption in percent (e.g., a share of 55 percent from cereals indicates that the group of cereals contributes 55 percent to the total dietary energy consumption). See tables 2.2, 2.5, and 2.9.
contribution of food groups to total nutrient consumption: share of total fat consumption (%)	Measures the contribution of each food group to the consumption of fats in percent (e.g., a share of 17 percent from vegetable oils indicates that the group of vegetable oils contributes 17 percent to the total fats consumption). See tables 2.2 and 2.5.
contribution of food groups to total nutrient consumption: share of total protein consumption (%)	Measures the contribution of each food group to the total consumption of proteins in percent (e.g., a share of 20 percent from the group meat and meat products indicates that the group of meat and products contributes 20 percent to the total protein consumption). See tables 2.2 and 2.5.
density—average carbohydrates consumption (g/1,000 kcal)	Measures the amount in grams of carbohydrates included in 1,000 kcal. See table 1.12.
density—average fat consumption (g/1,000 kcal)	Measures the amount in grams of fats contributing to 1,000 kcal. See table 1.12.
density—average protein consumption (g/1,000 kcal)	Measures the amount in grams of proteins contributing to 1,000 kcal. See table 1.12.

depth of food deficit (kcal/person/day) Indicates how many calories would be needed to lift the undernourished from its status, everything else being constant and considering food is equally distributed. The average intensity of food deprivation of the undernourished, estimated as the difference between the average dietary energy requirement and the average dietary energy consumption of the undernourished population (food-deprived), is multiplied by the number of undernourished to provide an estimate of the total food deficit in the country, and is then normalized by the total population. This usually is within the range of 100–400 kilocalories per day. When it is lower than 200 kcal, it is considered low; between 200 and 300, moderate; and above 300 kcal, high. See tables 1.1 and 1.2.

dietary energy supply adjusted for losses (kcal/person/day) Corresponds to the dietary energy supply as derived from the food balance sheets reduced by losses that occur at the retail level. It is expressed in kilocalories per person per day. It is used in the calculation of the prevalence of undernourishment indicator 1.9 of the MDG. See table 1.2.

estimated population Corresponds to the total number of people within a population group. It is calculated from the survey data as the sum of the product between household weight and number of household members. See table 1.4.

minimum dietary energy requirement (MDER) (kcal/person/day) Amount of energy needed for light activity and minimum acceptable body mass index (weight for attained height). MDER is the cutoff point, or threshold, used to estimate the prevalence (percentage) of the undernourished population in a country. Dietary energy requirements differ by gender and age, and for different levels of physical activity. As a result, minimum dietary energy requirements vary by country, and from year to year, depending on the sex and age structure of the population.

In countries with a high prevalence of undernourishment, a large proportion of the population typically consumes dietary energy levels close to the cutoff point, making the MDER a highly sensitive parameter. It is computed as a weighted average of the minimum energy requirements of different age/sex groups in the population.[25] See tables 1.1, 1.2, and 1.3.

The MDER shown at the national level in table 1.3 is calculated using the structure of the population as from the survey. Note that this MDER value is different from the MDER shown in the first row of tables 1.1 and 1.2. The MDER at the national level in tables 1.1 and 1.2 is based on the country population as published by the United Nations, biennially, and it is used to estimate the MDG 1.9 indicators. The difference in the values of these two MDERs at the national level can be due to differences in (1) the structure of the population by age and sex groups; (2) the heights used by age and sex groups; and (3) the birthrate used.

number of sampled households Total should be equal to the size of the survey sample: to obtain reliable estimates (at income decile levels) it is suggested to have more than 500 sampled households by category of analysis: region, household head's characteristics, etc. A statistic obtained with fewer than 30 households is considered not reliable. See tables 1.3, 1.4, 1.5, 1.6, 1.7, and 1.8.

other sources—proportion of households in total households (%) Measures the percentage of households whose consumption of a food item is coming from other sources (e.g., at national level, 5 percent associated to maize in grain means that 5 percent of households of this country have received maize in grain as a gift). See tables 3.7, 3.8, and 3.9.

own consumption—proportion of households in total households (%) Measures the percentage of households whose consumption of a food item is coming from their own production (e.g., at the national level 55 percent associated to wheat flour means that 55 percent of households of this country have consumed wheat flour coming from their own production). See tables 3.7, 3.8, and 3.9.

population ('000s) Corresponds to the total number of people (expressed in thousands) within a population group. It is calculated from the survey data as the sum of the product between household weight and number of household members. At the national level, the population estimates should be close to those published by the UN for the same year. See tables 1.1 and 1.2.

prevalence of undernourishment (%) Proportion of the population estimated to be at risk of caloric inadequacy. A value less than 5 percent is considered low, a value between 5 and 19 percent is considered moderate, and a value higher than 20 percent is considered high. For computing the official MDG 1.9 indicator FAO uses, as mean of the distribution, the dietary energy supply from food balance sheets minus waste of calories at the retail level. When the average dietary energy consumption from the national household surveys is used as the estimate of the mean of the distribution, the corresponding indicator should not be considered the official MDG 1.9. However, the two indicators should be compared and critically evaluated. See tables 1.1 and 1.2.

purchase—proportion of households in total households (%) Measures the percentage of households whose consumption of a food item is coming from purchases (e.g., at the national level, 55 percent associated to wheat flour means that 55 percent of households of this country have consumed wheat flour coming from purchases).

	Note that the sum of the proportion of households (HHs) that acquire the product through purchase, or received in kind, or from own consumption does not necessarily equal 100 percent because not all HH might have consumed the food. See tables 3.7, 3.8, and 3.9.
quantity as "produced" (g/person/day)	Measures the quantity of food product consumed coming from own production. It is expressed in grams per person per day. See tables 3.7, 3.8, and 3.9.
quantity as "purchased" (g/person/day)	Measures the quantity of food product consumed coming from purchases. It is expressed in grams per person per day. See tables 3.7, 3.8, and 3.9.
quantity as "received" from other sources (g/person/day)	Measures the quantity of food product consumed from other sources. It is expressed in grams per person per day. See tables 3.7, 3.8, and 3.9.
ratio to the first reference group of average dietary energy consumption	Measures the inequality in the average dietary energy consumption between the first quintile (used as the reference group) and the other quintiles. For instance, if the value of the ratio associated to quintile 4 is equal to 5, this means that the average dietary energy consumption in quintile 4 is five times higher than in quintile 1. It's a measure of inequality easier to interpret than the Gini coefficient or the coefficient of variation. This ratio is computed for the average dietary energy value, average food consumption in monetary value, average total consumption in monetary value, average income, and share of food source in dietary and monetary value. See table 4.1.
share of animal protein in total protein consumption (%)	Proportion of protein consumed from food of animal origin (meat [red and white], fish, eggs, milk, and cheese). See table 1.13.

share of dietary energy consumption from fats (%) Proportion of total calories from fats. The experts from WHO/FAO/UNU recommend a consumption of calories from fats between 15 and 30 percent of total calories consumed. See tables 1.10 and 1.11.

share of dietary energy consumption from protein (%) Proportion of total calories from proteins. The experts from WHO/FAO/UNU recommend a consumption of calories from proteins between 10 and 15 percent of total calories consumed. See tables 1.10 and 1.11.

share of dietary energy consumption from total carbohydrates and alcohol (%) Proportion of total calories from available carbohydrates and alcohol. The experts from WHO/FAO/UNU recommend a consumption of calories from available carbohydrates and alcohol between 55 and 75 percent of total calories consumed. See tables 1.10 and 1.11.

share of food consumed away from home in total food consumption (%) in dietary energy Share of dietary energy coming from the food eaten away from home (canteen at work, restaurants, bars, street food, etc.). Usually is greater for better-off households. Yet this rule widely depends on the eating habits of the country. Indeed, street food (sometimes cheap and highly caloric) may contribute significantly to the diet of poor people; whereby restaurants may provide expensive but not highly caloric food. See tables 1.5, 1.6, and 4.2.

share of food consumed away from home in total food consumption (%) in monetary value Contribution (expressed in monetary value) of food consumed away from home in total food monetary value. Usually is higher in urban areas and for higher income groups. Yet the rule widely depends on the eating habits of the country. See tables 1.7, 1.8, and 4.3.

share of food consumption in total income (%) (Engel ratio) According to Engel's law, the higher the income, the lower the proportion of income is spent on food. This ratio reflects the living standard of a population group and its vulnerability to food price increases. It can get close to 80 percent for low-income groups and 20 percent for high-income groups. See tables 1.7, 1.8, and 4.4.

share of food from other sources in total food consumption (%) in dietary energy Share of dietary energy coming from food received from other sources (e.g., received as payment, gift, aid, etc.). Usually is higher for low-income groups as they are more likely to receive food aid, gifts, etc. See tables 1.5, 1.6, and 4.2.

share of food from other sources in total food consumption (%) in monetary value Contribution (expressed in monetary value) of food received in kind to the total food monetary value. Usually it is higher for lower income deciles, which are mainly those receiving food in kind. See tables 1.7, 1.8, and 4.3.

share of own produced food in total food consumption (%) in dietary energy Share of dietary energy coming from own produced food. Should be higher in rural areas than urban areas, and it is usually higher for lower income groups. The greater the share is, the higher is the vulnerability to natural shocks affecting agricultural production. See tables 1.5, 1.6, and 4.2.

share of own produced food in total food consumption (%) in monetary value Contribution (expressed in monetary value) of food taken from own production to the total food monetary value. Should be higher in rural areas than urban areas and it is usually higher for lower income groups. The greater the share is, the higher is the vulnerability to natural shocks affecting agricultural production. Indeed, farming households will need to buy from the market the same amount of food they would have taken from their own production. See tables 1.7, 1.8, and 4.3.

share of purchased food in total food consumption (%) in dietary energy Share of dietary energy coming from food purchased from the market. Usually higher in urban areas. The greater the share is, the higher is the vulnerability to price increase. This share can be high for households living in urban areas and nonagricultural households. See tables 1.5, 1.6, and 4.2.

share of purchased food in total food consumption (%) in monetary value Contribution (expressed in monetary value) of purchased food to the total food monetary value. Usually higher in urban areas where most people get food from the market. The greater the share is, the higher is the vulnerability to price increase. See tables 1.7, 1.8, and 4.3.

skewness of dietary energy consumption Skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable. It measures the length of the tail of the distribution. In a lognormal distribution, the skewness is a function of the coefficient of variation: skewness = CV * (3 + CV2). In a skew(log) normal distribution, the skewness is independent from the CV's values. For this reason, the adoption of a skew(log)normal model allows for greater flexibility and for a truer representation of the consumption distribution. See tables 1.1 and 1.2.

within range of population fat intake goal: 15%–30% Indicates whether the proportion of total calories available from fats is within the range of 15–30 percent. See table 1.11.

within range of population protein intake goal: 10%–15% Indicates whether the proportion of total calories available from protein is within the range of 10–15 percent. See table 1.11.

within range of population total carbohydrates and alcohol intake goal: 55%–75% Indicates whether the proportion of total calories available from carbohydrates is within the range of 55–75 percent. See table 1.11.

Indicators on Micronutrients

95th percentile of the average absolute iron intakes required (mg/person/day)	Total absolute iron requirements depend on sex, age, and lactating and menopausal status (the latter two for women only). Values are those reported in FAO/WHO (2004, 196). See table 5.4.
average animal iron availability (mg/person/day)	Average amount of iron from animal sources available for consumption. Iron has several vital functions in the body. It serves as a carrier of oxygen to the tissues from the lungs by red blood cell hemoglobin. Iron deficiency (sideropenia or hypoferremia) is one of the most common nutritional deficiencies. Symptoms of iron deficiency include fatigue, dizziness, pallor, hair loss, twitches, irritability, weakness, pica, brittle or grooved nails, Plummer-Vinson syndrome, impaired immune function, pagophagia, and restless legs syndrome. See tables 5.4, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.
average availability of animal iron provided by each food group, out of total availability (%)	Relative contribution (percent) of each food group to total animal iron. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.
average availability of beta-carotene provided by each food group, out of total availability (%)	Relative contribution (percent) of each food group to total beta-carotene availability. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.
average availability of calcium provided by each food group, out of total availability (%)	Relative contribution (percent) of each food group to total calcium availability. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.
average availability of heme iron provided by each food group, out of total availability (%)	Relative contribution (percent) of each food group to total heme iron. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.

average availability of nonanimal iron provided by each food group, out of total availability (%) Relative contribution (percent) of each food group to total nonanimal iron. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.

average availability of nonheme iron provided by each food group, out of total availability (%) Relative contribution (percent) of each food group to total nonheme iron. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.

average availability of retinol provided by each food group, out of total availability (%) Relative contribution (percent) of each food group to total retinol availability. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.

average availability of vitamin B1 provided by each food group, out of total availability (%) Relative contribution (percent) of each food group to total vitamin B1 availability. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.

average availability of vitamin B12 provided by each food group, out of total availability (%) Relative contribution (percent) of each food group to total vitamin B12 availability. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.

average availability of vitamin B2 provided by each food group, out of total availability (%) Relative contribution (percent) of each food group to total vitamin B2 availability. Table 6.5 provides national values. Table 6.6 provides values disaggregated by area of residence.

average availability of vitamin B6 provided by each food group, out of total availability (%) Relative contribution (percent) of each food group to total vitamin B6 availability. Table 6.5 provides national level values. Table 6.6 provides values disaggregated by area of residence.

average availability of vitamin C provided by each food group, out of total availability (%)	Relative contribution (percent) of each food group to total vitamin C availability. Table 6.5 provides national level values. Table 6.6 provides values disaggregated by area of residence.
average beta-carotene availability (mcg/person/day)	Average amount of beta-carotene available, expressed in micrograms per person per day. Beta-carotene (β-Carotene) is a strongly colored red-orange pigment abundant in plants and fruits. Its absorption is enhanced if eaten with fats, as carotenes are fat-soluble. See tables 5.1, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.
average calcium availability (mg/person/day)	Average amount of calcium available for consumption. Calcium salts provide rigidity to the skeleton, and calcium ions play a role in many if not most metabolic processes. A positive calcium balance (i.e., net calcium retention) is required throughout growth, particularly during the first two years of life and during puberty and adolescence. See tables 5.3, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.
average calcium availability per 1,000 kcal	Average amount of calcium (expressed in milligrams) available in 1,000 kilocalories. Being a relative measure, we can talk about density of calcium per 1,000 kcal. See table 5.5.
average heme iron availability (mg/person/day)	Average amount of heme iron available for consumption. With respect to the mechanism of absorption, there are two kinds of dietary iron: heme iron and nonheme iron. Primary sources of heme iron are the hemoglobin and myoglobin from consumption of meat, poultry, and fish. Heme iron can be degraded and converted to nonheme iron if foods are cooked at a high temperature for a long time. See tables 5.4, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.

average nonanimal iron availability (mg/person/day)

Average amount of iron from nonanimal sources available for consumption. Iron has several vital functions in the body. It serves as a carrier of oxygen to the tissues from the lungs by red blood cell hemoglobin. Iron deficiency (sideropenia or hypoferremia) is one of the most common nutritional deficiencies. Symptoms of iron deficiency include fatigue, dizziness, pallor, hair loss, twitches, irritability, weakness, pica, brittle or grooved nails, Plummer-Vinson syndrome, impaired immune function, pagophagia, and restless legs syndrome. See tables 5.4, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.

average nonheme iron availability (mg/person/day)

Average amount of nonheme iron available for consumption. With respect to the mechanism of absorption, there are two kinds of dietary iron: heme iron and nonheme iron. Primary sources of nonheme iron are cereals, pulses, legumes, fruits, and vegetables. The absorption of nonheme iron is influenced by the individual's iron status and the presence of some food components such as ascorbic acid, polyphenols, and phytates. See tables 5.4, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.

average retinol activity equivalent of vitamin A availability (mcg/person/day)

Average amount of retinol activity equivalent of vitamin A available for consumption. There are two sources of vitamin A: one is food from animal origin, which includes retinol, and the second one is food from plant origin, which includes beta-carotene. One unit of retinol is equivalent to one unit of vitamin A; however, in the case of carotenoids, the body converts them to vitamin A as shown in this formula:

$$\begin{aligned}\text{Vitamin A} = {} & \text{mcg of retinol} \\ & + (\text{mcg of beta-carotene}/12) \\ & + (\text{mcg of other carotenoids})/24\end{aligned}$$

Vitamin A is an essential nutrient needed for the normal functioning of the visual system, growth and development, maintenance of epithelial cellular integrity, immune system functioning, and reproduction. The main consequence of vitamin A deficiencies is night blindness, which could develop into irreversible blindness. See tables 5.1, 5.6, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.

average retinol availability (mcg/person/day) Average amount of retinol available for consumption. See tables 5.1, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9

average vitamin A availability in 1,000 kcal Average amount of vitamin A (expressed in micrograms of retinol activity equivalents) available in 1,000 kcal. Being a relative measure, we can talk about density of calcium per 1,000 kcal. See table 5.6.

average vitamin B1 availability in 1,000 kcal Average amount of vitamin B1 (expressed in milligrams) available in 1,000 kcal. Being a relative measure, we can talk about density of vitamin B1 per 1,000 kcal. See table 5.7.

average vitamin B1 availability (mg/person/day) Average amount of vitamin B1 available for consumption. B1 (otherwise called thiamin) deficiency results in the disease called beriberi. Beriberi occurs in breastfed infants whose nursing mothers are deficient. It also occurs in adults with high carbohydrate intake mainly from milled rice and with intake of antithiamin factors. See tables 5.2, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.

average vitamin B12 availability in 1,000 kcal Average amount of vitamin B12 (expressed in micrograms) available in 1,000 kcal. Being a relative measure, we can talk about density of vitamin B12 per 1,000 kcal. See table 5.7.

average vitamin B12 availability (mcg/person/day) Average amount of vitamin B12 available for consumption. Vitamin B12 (otherwise called cobalamin) enters the human food chain through food of animal origin. Products from herbivorous animals, such as milk, meat, and eggs, constitute important dietary sources of vitamin B12. Vitamin B12 deficiency can cause permanent damage to nervous tissue if left untreated longer than six months. See tables 5.2, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.

average vitamin B2 availability in 1,000 kcal Average amount of vitamin B2 (expressed in milligrams) available in 1,000 kcal. Being a relative measure, we can talk about density of vitamin B2 per 1,000 kcal. See table 5.7.

average vitamin B2 availability (mg/person/day) Average amount of vitamin B2 available for consumption. B2 (otherwise called riboflavin) deficiency results into hypo- or ariboflavinosis, with sore throat, hyperemia, oedema of the pharyngeal and oral mucous membranes, cheilosis, angular stomatitis, glossitis, seborrheic dermatitis, and normochromic, normocytic bone marrow. The major cause of hyporiboflavinosis is inadequate dietary intake as a result of limited food supply, which is sometimes exacerbated by poor food storage or processing. See tables 5.2, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.

average vitamin B6 availability in 1,000 kcal Average amount of vitamin B6 (expressed in milligrams) available in 1,000 kcal. Being a relative measure, we can talk about density of vitamin B6 per 1,000 kcal. See table 5.7.

average vitamin B6 availability (mg/person/day) Average amount of vitamin B6 available for consumption. Vitamin B6 deficiency usually occurs in association with a deficit in other B-complex vitamins. Infants are especially susceptible to insufficient intakes, which can lead to epileptiform convulsions. Skin changes include dermatitis with cheilosis and glossitis. A decrease in the metabolism of glutamate in the brain, which is found in vitamin B6 insufficiency, reflects a nervous system dysfunction. As is the case with other micronutrient deficiencies, vitamin B6 deficiency results in an impairment of the immune system. See tables 5.2, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.

average vitamin C availability in 1,000 kcal Average amount of vitamin C (expressed in milligrams) available in 1,000 kcal. Being a relative measure, we can talk about density of vitamin C per 1,000 kcal. See table 5.6.

average vitamin C availability (mg/person/day) Average amount of vitamin C available for consumption. Vitamin C mainly works as an antioxidant. Therefore chronic lack of vitamin C in the diet can lead to a condition called scurvy (i.e., easy bruising, spontaneous bleeding, and the joint and muscle pains). The populations at risk of vitamin C deficiency are those for whom the fruit and vegetable supply is minimal. Epidemics of scurvy are associated with famine and war, when people are forced to become refugees and the food supply is small and irregular. In many developing countries, limitations in the supply of vitamin C are often determined by seasonal factors. See tables 5.3, 6.1, 6.2, 6.3, 6.4, 6.7, 6.8, and 6.9.

calcium recommended intake in 1,000 kcal Amount of recommended calcium intake per 1,000 kcal. See table 5.5.

calcium recommended intake (mg/person/day) Amount of recommended calcium intake to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. Values are those reported in FAO/WHO (2004, 162). See table 5.3.

median of the average absolute iron intake required (mg/person/day) Total absolute iron requirements depend on sex, age, and lactating and menopausal status (the latter two for women only). Values are those reported in FAO/WHO (2004, 196). See table 5.4.

ratio of calcium available to recommended (%) Indicates whether the amount of calcium available to the households is sufficient to meet the average daily nutrient intake needed by almost

	all apparently healthy individuals in the population group. When the amount of available calcium exceeds the recommended amount, the ratio is above 1. However, we cannot talk about population out of risk of calcium deficiency because we do not have information of the actual intake. See tables 5.3 and 5.5.
ratio of retinol available to vitamin A available (%)	Ratio between retinol and vitamin A available for consumption, or the percentage of vitamin A that is due to the presence of retinol. See table 5.1.
ratio of vitamin A available to recommended (%)	Indicates whether the amount of vitamin A available to the households is sufficient to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. When the amount of available vitamin A exceeds the recommended amount, the ratio is above 1. However, we cannot talk about population out of risk of vitamin A deficiency because we do not have information of the actual intake. See tables 5.1 and 5.6.
ratio of vitamin A available to required (%)	Indicates whether the amount of vitamin A available to the households is sufficient to meet the average daily nutrient intake needed by 50 percent of the "healthy" individuals in the population group. When the amount of available vitamin A exceeds the required amount, the ratio is above 1. However, we cannot talk about population out of risk of vitamin A deficiency because we do not have information of the actual intake. See tables 5.1 and 5.6.
ratio of vitamin B1 available to recommended (%)	Indicates whether the amount of vitamin B1 available to the households is sufficient to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. When the amount of available

vitamin B1 exceeds the recommended amount, the ratio is above 1. However, we cannot talk about population out of risk of vitamin B1 deficiency because we do not have information of the actual intake. See tables 5.2 and 5.7.

ratio vitamin B12 available to recommended (%) Indicates whether the amount of vitamin B12 available to the households is sufficient to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. When the amount of available vitamin B12 exceeds the recommended amount, the ratio is above 1. However, we cannot talk about population out of risk of vitamin B12 deficiency because we do not have information of the actual intake. See tables 5.2 and 5.7.

ratio vitamin B12 available to required (%) Indicates whether the amount of vitamin B12 available to the households is sufficient to meet the average daily nutrient intake needs by 50 percent of the "healthy" individuals in the population group. When the amount of available vitamin B12 exceeds the required amount, the ratio is above 1. However, we cannot talk about population out of risk of vitamin B12 deficiency because we do not have information of the actual intake. See tables 5.2 and 5.7.

ratio vitamin B2 available to recommended (%) Indicates whether the amount of vitamin B2 available to the households is sufficient to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. When the amount of available vitamin B2 exceeds the recommended amount, the ratio is above 1. However, we cannot talk about population out of risk of vitamin B2 deficiency because we do not have information of the actual intake. See tables 5.2 and 5.7.

ratio vitamin B6 available to recommended (%)	Indicates whether the amount of vitamin B6 available to the households is sufficient to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. When the amount of available vitamin B6 exceeds the recommended amount, the ratio is above 1. However, we cannot talk about population out of risk of vitamin B6 deficiency because we do not have information of the actual intake. See tables 5.2 and 5.7.
ratio vitamin C available to recommended (%)	Indicates whether the amount of vitamin C available to the households is sufficient to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. When the amount of available vitamin C exceeds the recommended amount, the ratio is above 1. However, we cannot talk about population out of risk of vitamin C deficiency because we do not have information of the actual intake. See tables 5.3 and 5.6.
vitamin A mean requirement in 1,000 kcal	Required amount of vitamin A (expressed in micrograms of retinol activity equivalent) per 1,000 kcalories. See table 5.6.
vitamin A mean requirement (mcg retinol activity equivalent/person/day)	Amount of required vitamin A intake to meet the average daily nutrient intake needed by 50 percent of the healthy individuals in the population group. Values are those reported in FAO/WHO (2004, 100). See table 5.1.
vitamin A recommended safe intake in 1,000 kcal	Recommended amount of vitamin A (expressed in micrograms of retino activity equivalent) per 1,000 kcal. The difference between vitamin A requirements and vitamin A recommended safe intake is reported in FAO/WHO (2004). See table 5.6.

vitamin A recommended safe intake (mcg retinol activity equivalent/ person/day)	Amount of recommended vitamin A intake to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. Values are those reported in FAO/WHO (2004, 100). See table 5.1.
vitamin B1 recommended intake (mg/person/day)	Amount of recommended vitamin B1 intake to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. Values are those reported in FAO/WHO (2004, 30). See table 5.2.
vitamin B1 recommended safe intake in 1,000 kcal	Recommended amount of vitamin B1 (expressed in milligrams) per 1,000 kcal. See table 5.7.
vitamin B12 average requirement in 1,000 kcal	Recommended amount of vitamin B12 (expressed in micrograms) per 1,000 kcal. See table 5.7.
vitamin B12 average requirement (mcg/person/day)	Amount of required vitamin B12 intake to meet the average daily nutrient intake needed by 50 percent of the healthy individuals in the population group. Values are those reported in FAO/ WHO (2004, 69). See table 5.2.
vitamin B12 recommended intake (mcg/person/day)	Amount of recommended vitamin B12 intake to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. Values are those reported in FAO/WHO (2004, 69). See table 5.2.
vitamin B12 recommended safe intake in 1,000 kcal	Recommended amount of vitamin B12 (expressed in micrograms) per 1,000 kcal. See table 5.7.
vitamin B2 recommended intake (mg/person/day)	Amount of recommended vitamin B2 intake to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. Values are those reported in FAO/WHO (2004, 33). See table 5.2.

vitamin B2 recommended safe intake in 1,000 kcal	Recommended amount of vitamin B2 (expressed in milligrams) per 1,000 kcal. See table 5.7.
vitamin B6 recommended intake (mg/person/day)	Amount of recommended vitamin B6 intake to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. Values are those reported in FAO/WHO (2004, 38). See table 5.2.
vitamin B6 recommended safe intake in 1,000 kcal	Recommended amount of vitamin B6 (expressed in milligrams) per 1,000 kcal. See table 5.7.
vitamin C recommended intake (mg/person/day)	Amount of recommended vitamin C intake to meet the average daily nutrient intake needed by almost all apparently healthy individuals in the population group. Values are those reported in FAO/WHO (2004, 79). See table 5.3.
vitamin C recommended safe intake in 1,000 kcal	Recommended amount of vitamin C (expressed in milligrams) per 1,000 kcal. See table 5.6.

Indicators on Amino Acids

amino acid availability as percentage of total availability (%)	Proportion of the essential amino acids (provided by a group of food items) in total availability of the same amino acid, after correcting for protein digestibility. See tables 8.5, 8.6, and 8.7.
amino acid availability per gram of protein (mg)	Amount of the essential amino acids available for consumption per gram of protein after correcting for protein digestibility. See table 7.2.
histidine—average amino acid availability (g/person/day)	Average amount of the essential amino acid histidine available for consumption after correcting for protein digestibility. Statistics shown by population group are expressed in daily milligrams per person. Statistics shown at food group or food item level are expressed in daily grams per person. Histidine belongs to the aromatic amino acids and was accepted as an

indispensable amino acid in human adults, despite controversy regarding its essentiality (WHO 2002). See tables 7.1, 8.1, 8.2, 8.3, 8.4, 8.8, 8.9, and 8.10.

isoleucine—average amino acid availability (g/person/day)
Average amount of the essential amino acid isoleucine available for consumption after correcting for protein digestibility. Statistics shown by population group are expressed in daily milligrams per person. Statistics shown at food group or food item level are expressed in daily grams per person. See tables 7.1, 8.1, 8.2, 8.3, 8.4, 8.8, 8.9, and 8.10.

leucine—average amino acid availability (g/person/day)
Average amount of the essential amino acid leucine available for consumption after correcting for protein digestibility. Statistics shown by population group are expressed in daily milligrams per person. Statistics shown at food group or food item level are expressed in daily grams per person. Leucine is the most abundant amino acid in tissue and food proteins (WHO 2002). See tables 7.1, 8.1, 8.2, 8.3, 8.4, 8.8, 8.9, and 8.10.

lysine—average amino acid availability (g/person/day)
Average amount of the essential amino acid lysine available for consumption after correcting for protein digestibility. Statistics shown by population group are expressed in daily milligrams per person. Statistics shown at food group or food item level are expressed in daily grams per person. Lysine is the likely limiting amino acid in cereals, especially wheat (WHO 2002). See tables 7.1, 8.1, 8.2, 8.3, 8.4, 8.8, 8.9, and 8.10.

methionine and cystine—average amino acid availability (g/person/day)
Average amount of the essential amino acids methionine and cystine available for consumption after correcting for protein digestibility. Statistics shown by population group are expressed in daily milligrams per person. Statistics shown at food group or food item level are expressed in

daily grams per person. Methionine and cystine are also called sulfur amino acids. The former is nutritionally indispensable while the latter, as a metabolic product of methionine catabolism, is dependent on there being sufficient methionine to supply the needs for both amino acids. Their concentrations are marginal in legume proteins, although they are equally abundant in cereal and animal proteins (WHO 2002). See tables 7.1, 8.1, 8.2, 8.3, 8.4, 8.8, 8.9, and 8.10.

phenylalanine and tyrosine—average amino acid availability (g/person/day)

Average amount of the essential amino acids phenylalanine and tyrosine available for consumption after correcting for protein digestibility. Statistics shown by population group are expressed in daily milligrams per person. Statistics shown at food group or food item level are expressed in daily grams per person. These two amino acids belong to the aromatic amino acids. Phenylalanine is nutritionally indispensable while tyrosine, as a metabolic product of phenylalanine catabolism, is dependent on there being sufficient phenylalanine to supply the needs for both amino acids (WHO 2002). See tables 7.1, 8.1, 8.2, 8.3, 8.4, 8.8, 8.9, and 8.10.

threonine—average amino acid availability (g/person/day)

Average amount of the essential amino acid threonine available for consumption after correcting for protein digestibility. Statistics shown by population group are expressed in daily milligrams per person. Statistics shown at food group or food item level are expressed in daily grams per person. Threonine is present at low concentrations in cereal proteins (WHO 2002). See tables 7.1, 8.1, 8.2, 8.3, 8.4, 8.8, 8.9, and 8.10.

tryptophan—average amino acid availability (g/person/day)	Average amount of the essential amino acid tryptophan available for consumption after correcting for protein digestibility. Statistics shown by population group are expressed in daily milligrams per person. Statistics shown at food group or food item level are expressed in daily grams per person. Tryptophan belongs to the aromatic amino acids. The occurrence of tryptophan in proteins is generally less than many other amino acids because its content is low in cereals, especially maize (WHO 2002). See tables 7.1, 8.1, 8.2, 8.3, 8.4, 8.8, 8.9, and 8.10.
valine—average amino acid availability (g/person/day)	Average amount of the essential amino acid valine available for consumption after correcting for protein digestibility. Statistics shown by population group are expressed in daily milligrams per person. Statistics shown at food group or food item level are expressed in daily grams per person. See tables 7.1, 8.1, 8.2, 8.3, 8.4, 8.8, 8.9, and 8.10.

Notes

1. Before executing ADePT-FSM, the user classifies the food commodities in different food groups. Further details can be found in chapter 2.
2. Further details can be found in chapter 2.
3. For the methodology applied at the subnational level, further details can be found in chapter 2.
4. Further details can be found in chapter 2.
5. See the following link: http://www.fao.org/economic/ess/ess-fs/fs-methods/adept-fsn/en/.
6. Further details can be found in chapter 2.
7. Further details can be found in chapter 2.
8. Available carbohydrates = total carbohydrates – fibers.

9. Available carbohydrates = total carbohydrates – fibers.
10. Available carbohydrates = total carbohydrates – fibers.
11. Available carbohydrates = total carbohydrates – fibers.
12. Available carbohydrates = total carbohydrates – fibers.
13. The food commodity quantities cannot be used for this comparison unless refuse factors and technical conversion factors for agricultural commodities (the same used in FBS) are applied to the food quantities consumed.
14. All food quantities include both the edible and the nonedible parts (i.e., peels, bones, spines, etc.).
15. All food quantities include both the edible and the nonedible parts (i.e., peels, bones, spines, etc.).
16. All food quantities include both the edible and the nonedible parts (i.e., peels, bones, spines, etc.).
17. EAR is the average daily nutrient intake level that meets the needs of 50 percent of the "healthy" individuals in a particular age and gender group. The RNI is the daily intake, set at the EAR plus 2 standard deviations, which meets the nutrient requirements of almost all apparently healthy individuals in an age- and sex-specific population group (FAO/WHO 2004). To express nutrient requirements and recommended intakes for population groups, the requirements by sex and age are applied to individuals and then summed for each population group of analysis. The individual requirements were defined for gender-age population groups by a FAO/WHO group of experts in 1998 (WHO 2004).
18. Iron deficiency is defined as a hemoglobin concentration below the optimum value in an *individual*, whereas iron deficiency anemia implies that the hemoglobin concentration is below the 95th percentile of the distribution of hemoglobin concentration in a *population* (disregarding effects of altitude, age, sex, etc., on hemoglobin concentration) (WHO 2004).
19. Further details can be found in chapter 2.
20. Further details can be found in chapter 2.
21. Further details can be found in chapter 2.
22. Further details can be found in chapter 2.
23. Further details can be found in chapter 2.
24. Further details can be found in chapter 2.
25. Further details can be found in chapter 2.

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